B.Tech.

IN

COMPUTER SCIENCE AND ENGINEERING
with Specialization in AI and Data Science

CURRICULUM
AND
SYLLABI

(Applicable from 2026 Admission onwards)

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY KOTTAYAM

Pala – 686635, KERALA, INDIA

Table of Contents

Programme Educational Objectives (PEOs) of B.Tech. in COMPUTER SCIENCE AND ENGINEERING

with Specialization in AI and Data Science

PEO1	Graduates shall apply the fundamental principles and techniques of Computer Science, Artificial Intelligence, and Data Science to pursue professional careers, higher studies, and research
PEO2	Graduates shall have the ability to design, develop, deploy, and maintain reliable and efficient intelligent systems and data-driven applications.
PEO3	Graduates shall have the necessary communication and management skills and ethical values to become competent professionals and responsible members of society.

Programme Outcomes (POs) and Programme Specific Outcomes (PSOs) of B.Tech. in COMPUTER SCIENCE AND ENGINEERING

with Specialization in AI and Data Science

PO1	**Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem analysis:** Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	**Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	**Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.
PSO1	Acquire strong fundamentals of computer science, artificial intelligence, and data science expressed through effective problem-solving techniques, algorithmic thinking, and intelligent system design.
PSO2	Exposure to modern computational platforms, emerging AI technologies, and interdisciplinary applications to fulfill and enhance the industry, research, innovation, and entrepreneurship ecosystem.

CURRICULUM

Total credits for completing B.Tech. in COMPUTER SCIENCE AND ENGINEERING

with Specialization in AI and Data Science is 160. (In addition, 10 mandatory Activity Credits must be earned; these are not counted towards the CGPA.)

COURSE CATEGORIES AND CREDIT REQUIREMENTS
The structure of the B.Tech. programme shall have the following course categories:

Sl. No.	Course Category	Number of Courses	Minimum Credits
1	Institute Core (IC)	17	51
2	Programme Core (PC)	19 Theory + 14 Lab	74
3	Programme Electives (PE)	4	12
4	Open Electives (OE)	4	11
5	Projects (P)	2	12
		Total credits (counted for CGPA)	**160**
6	Activity Credits (AC) – *mandatory, not counted for CGPA*	–	10

COURSE REQUIREMENTS. The effort to be put in by the student is indicated in the tables as follows:
L: Lecture (one unit is of 50 minutes) **T**: Tutorial (one unit is of 50 minutes) **P**: Practical (one unit is of one hour)

1. Institute Core (IC)

(a) Mathematics

Sl.	Code	Course Title	L	T	P	Credits
1	MAC111	Calculus and Linear Algebra	3	1	0	4
2	MAC121	Discrete Mathematics	3	1	0	4
3	MAC211	Probability, Statistics, and Random Processes	3	1	0	4
4	MACXXX	Computational Linear Algebra and Optimization Techniques	3	1	0	4
		Total	**12**	**4**	**0**	**16**

(b) Economics, Management & Entrepreneurship

Sl.	Code	Course Title	L	T	P	Credits
1	HMC211	Engg. Management, Entrepreneurship Development & Start-ups	2	0	0	2
2	HMC221	Engg. Economics, Finance & Business Analytics	2	0	0	2
		Total	**4**	**0**	**0**	**4**

(c) Communication Skills and Personality Development

Sl.	Code	Course Title	L	T	P	Credits
1	HMC111	Communication Skills for the Digital Age	3	0	0	3
2	HMC121	Personality Development & Soft Skills	2	1	0	3
		Total	**5**	**1**	**0**	**6**

(d) Engineering Foundation Courses

Sl.	Code	Course Title	L	T	P	Credits
1	CSC111	Computer Programming	3	1	0	4
2	ECC111	Electronic Devices and Digital Logic Design	3	1	0	4
3	CSL111	Computer Programming Lab	0	0	2	1
4	ECL111	Electronic Circuits and Digital Logic Design Lab	0	0	2	1
5	CSC121	Data Structures	3	1	0	4
6	CSC122	Computer Organization	3	1	0	4
7	CSC211	Computer Networks	2	1	0	3
8	CSL121	Data Structures Lab (C, C++)	0	0	2	1
9	CSC123	Artificial Intelligence	3	0	0	3
		Total	**17**	**5**	**6**	**25**

2. Programme Core (PC)

Sl.	Code	Course Title	L	T	P	Credits
1	CSC112	Foundations of Computing and Systems	2	0	0	2
2	CSL113	IT Workshop Lab	0	1	2	2
3	CSC123	Object Oriented Programming	2	1	0	3
4	CSL123	Object Oriented Programming Lab	0	0	2	1
5	CSC212	Design and Analysis of Algorithms	2	1	0	3
6	CSC213	Theory of Computation	3	1	0	4
7	CSC214	Operating Systems	3	0	0	3
8	CSL211	Computer Networks Lab	0	0	2	1
9	CSL212	Algorithm Design Lab	0	0	2	1
10	CSL214	OS Lab	0	0	2	1
11	CAC221	Machine Learning and Data Analysis	2	1	0	3
12	CSC222	Language Processors	2	1	0	3
13	CSC223	Database Management Systems	2	1	0	3
14	CSC224	Advanced Data Structures	2	1	0	3
15	CSL222	Language Processors Lab	0	0	2	1
16	CSL223	Database Management Systems Lab	0	0	2	1
17	CAL221	Machine Learning and Data Analysis Lab	0	0	2	1
18	CSC311	Software Engineering	2	1	0	3
19	CAC311	Data Engineering	3	0	0	3
20	CAC312	Parallel and Distributed Computing	3	0	0	3
21	CAC313	Advanced Machine Learning	3	0	0	3
22	CAL311	Data Engineering Lab	0	0	2	1
23	CSL312	Parallel and Distributed Computing Lab	0	0	2	1
24	CAL313	Advanced Machine Learning Lab	0	0	2	1
25	CAC321	AI Safety	2	0	0	2
26	CAC322	Deep Learning	3	0	0	3
27	CAC323	Big Data Analytics and Streaming Systems	3	1	0	4
28	CAC324	Number Theory and Cryptography	3	1	0	4
29	CAL325	Agentic AI Lab	0	0	2	1
30	CSL321	FOSS Project and Innovation Lab	0	1	2	2
31	CAC411	Machine Learning Operations (MLOps)	3	0	0	3
32	CAC412	Generative Artificial Intelligence Techniques	0	0	2	1
33	CAL411	MLOps Lab	0	0	2	1
		Total	**47**	**13**	**28**	**74**

3. List of Programme Electives (PE)

Sl.	Code	Course Title	L	T	P	Credits
1	CAE302	Pattern Recognition	3	0	0	3
2	CAE306	Artificial Intelligence: Knowledge Representation and Reasoning	3	0	0	3
3	CAE304	Computer Vision	3	0	0	3
4	CAE305	Data Handling, Analysis and Visualization	3	0	0	3
5	CSE302	Cloud Computing	3	0	0	3
6	CSE303	Computer Systems for Machine Learning	3	0	0	3
7	CSE304	Blockchain Technology and Applications	3	0	0	3
8	CSE309	Augmented & Virtual Reality Systems	3	0	0	3
9	CSE311	Internet of Things & Edge Intelligence	3	0	0	3
10	CSE312	Computational Complexity	3	0	0	3
11	CSE313	Online Algorithms	3	0	0	3
12	CSE315	Network Optimization	3	0	0	3
13	CSE316	Introduction to Quantum Machine Learning	3	0	0	3
14	CAE402	Dependable and Secure AI	3	0	0	3
15	CAE403	Explainable AI & Interpretable ML	3	0	0	3
16	CAE404	Reinforcement Learning	3	0	0	3
17	CAE405	Federated Learning and Privacy Preserving AI	3	0	0	3
18	CAE406	Graph Neural Network and Geometric Deep Learning	3	0	0	3
19	CAE407	Medical Image Analysis	3	0	0	3
20	CAE408	Applied Predictive Analytics	3	0	0	3
21	CAE409	Time Series Analytics	3	0	0	3
22	CAE410	Scientific Machine Learning	3	0	0	3
23	CAE411	Streaming Data Analytics	3	0	0	3
24	CAE412	Bioinformatics & Computational Biology	3	0	0	3
25	CAE413	Social Network Analysis & Graph Mining	3	0	0	3
26	CAE414	Responsible and Inclusive Computing	3	0	0	3
27	CAE415	Computational Neuroscience	3	0	0	3
28	CSE402	Embedded AI and TinyML	3	0	0	3
29	CSE403	Robotics and Autonomous Systems	3	0	0	3
30	CSE404	Quantum Computing & Post-Quantum Cryptography	3	0	0	3

Continued on next page

Sl.	Code	Course Title	L	T	P	Credits
31	CSE405	Malware Analysis & Reverse Engineering	3	0	0	3
32	CSE406	Multimedia Security & Forensics	3	0	0	3
33	CSE408	Approximation Algorithms	3	0	0	3
34	CSE409	Coding Theory	3	0	0	3
35	CSE410	Combinatorial Optimization	3	0	0	3
36	CSEXXX	AI for Climate Change and Sustainable Environment	3	0	0	3

4. List of Open Electives (OE)

Courses offered by other departments or approved online platforms (e.g. NPTEL), subject to the maximum permitted under the B.Tech. Ordinances and Regulations. PE courses offered by the parent department are also available to its own students under this category.

Sl.	Code	Course Title	L	T	P	Credits
1		Foreign Language (NPTEL)	2	0	0	2
2		Design Thinking & Innovation	3	0	0	3
3	CSO001	Human Computer Interaction	3	0	0	3
4	CSO002	Digital Signal Processing	3	0	0	3
5	CAO001	Soft Computing	3	0	0	3
6	CSO003	Quantum Computing for Engineers	2	1	0	3
7	CSO004	Introduction to DevOps and Microservices	3	0	0	3
8	CAO004	Large Language Models	3	0	0	3
9	CAO005	Prompt Engineering & LLM Application Development	3	0	0	3
10	CAO006	Introduction to AyurInformatics	3	0	0	3
11	CAO007	Introduction to Culture Computing	3	0	0	3
12	CAO008	Indian Theories of Meaning	3	0	0	3
13	CAO009	Health Informatics & Digital Health Systems	3	0	0	3
14	CSO006	Internet of Things: Concepts and Applications	3	0	0	3
15	CSO007	Project Management and Agile Practices	3	0	0	3
16	CSO008	Introduction to Indian Knowledge Systems	3	0	0	3
17	CSO009	Fundamentals of Sanskrit	3	0	0	3

5. Projects (P)

The major project represents the culmination of study towards the B.Tech. degree. It is offered in two parts – Project-1 and Project-2 – and gives students the opportunity to apply and extend the knowledge gained across the programme. The work may be analytical, simulation-based, hardware design, or a combination, in an emerging area of COMPUTER SCIENCE AND ENGINEERING

with Specialization in AI and Data Science, carried out under the supervision of a department faculty member. Project work may be carried out individually or in groups (size decided by the department) and may build on an industrial problem identified during an internship.

6. Activity Credits (AC)

A minimum of 40 Activity Points must be earned to obtain the 10 credits required in the curriculum; these credits are not counted towards CGPA. The scheme of activity points, as approved by the Senate, is announced at the start of the first semester. A minimum of 20 points must be earned by the end of the 4th semester and 40 points by the end of the 7th semester.

PROGRAMME STRUCTURE

Semester-wise Course Structure (I–VIII)

Semester I						Semester II					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
MAC111	Calculus and Linear Algebra	3	1	0	4	MAC121	Discrete Mathematics	3	1	0	4
CSC111	Computer Programming and Problem Solving	3	1	0	4	CSC121	Data Structures	3	1	0	4
ECC111	Electronic Devices and Digital Logic Design	3	1	0	4	CAC121	Artificial Intelligence	3	0	0	3
HMC111	Communication Skills for the Digital Age	3	0	0	3	HMC121	Personality Development and Soft Skills	2	1	0	3
CSC112	Foundations of Computing and Systems	2	0	0	2	CSC122	Computer Organization	3	1	0	4
CSL111	Computer Programming Lab	0	0	2	1	CSC123	Object Oriented Programming	2	1	0	3
CSL113	IT Workshop Lab	0	1	2	2	CSL121	Data Structures Lab (C, C++)	0	0	2	1
ECL111	Electronic Devices and Digital Logic Design Lab	0	0	2	1	CSL123	Object Oriented Programming Lab	0	0	2	1
Total		**14**	**4**	**6**	**21**	**Total**		**16**	**5**	**4**	**23**
Cumulative Credits at the End of First Year: 44											
Semester III						**Semester IV**					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
MAC211	Probability, Statistics, and Random Processes	3	1	0	4	MACXXX	Computational Linear Algebra and Optimization Techniques	3	1	0	4
HMC211	Engineering Management, Entrepreneurship Development and Start ups	2	0	0	2	HMC221	Engineering Economics, Finance and Business Analytics	2	0	0	2
CSC211	Computer Networks	2	1	0	3	CAC221	Machine Learning and Data Analysis	2	1	0	3
CSC212	Design and Analysis of Algorithms	2	1	0	3	CSC222	Language Processors	2	1	0	3
CSC213	Theory of Computation	3	1	0	4	CSC223	Database Management Systems	2	1	0	3
CSC214	Operating Systems	3	0	0	3	CSC224	Advanced Data Structures	2	1	0	3
CSL211	Computer Networks Lab	0	0	2	1	CAL221	Machine Learning and Data Analysis Lab	0	0	2	1
CSL212	Algorithm Design Lab	0	0	2	1	CSL222	Language Processors Lab	0	0	2	1
CSL214	Operating Systems Lab	0	0	2	1	CSL223	Database Management Systems Lab	0	0	2	1
Total		15	4	6	**22**	**Total**		13	5	6	**21**
Cumulative Credits at the End of Second Year: 87											
Semester V						**Semester VI**					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
CSC311	Software Engineering	2	1	0	3	CAC321	AI Safety	2	0	0	2
CAC311	Data Engineering	3	0	0	3	CAC322	Deep Learning	3	0	0	3
CSC312	Parallel and Distributed Computing	3	0	0	3	CAC323	Big Data Analytics and Streaming Systems	3	1	0	4
CAC313	Advanced Machine Learning	3	0	0	3	CAC324	Number Theory and Cryptography	3	1	0	4
CXOXXX	Open Elective-1	2	1	0	3	CXEXXX	Stream Elective-2	2	0	2	3
CXEXXX	Stream Elective-1	3	0	0	3	CXOXXX	OE 2 Foreign Language	2	0	0	2
CAL311	Data Engineering Lab	0	0	2	1	CXOXXX	OE 3 Design Thinking and Innovation	3	0	0	3
CSL312	Parallel and Distributed Computing Lab	0	0	2	1	CAL325	Agentic AI Lab	0	0	2	1
CAL313	Advanced Machine Learning Lab	0	0	2	1	CSL321	FOSS Project and Innovation Lab	0	1	2	2
							Hons. Course-1	3	0	0	3
Total		**16**	**2**	**6**	**21**	**Total**		**18**	**3**	**6**	**24**
Cumulative Credits at the End of Third Year: 132 (B.Tech.); $132 + 3 = 135$ (B.Tech.(Hon.))											
Semester VII						**Semester VIII**					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
CAC411	Machine Learning Operations (MLOps)	3	0	0	3	CAP421	B.Tech Project Phase II	0	0	18	9

CXOXXX	Open Elective-4	3	0	0	3	CHP421	Hons. Project Phase II	0	0	22	11
CAC412	Generative Artificial Intelligence Techniques	3	0	0	3						
CXEXXX	Stream Elective-3	3	0	0	3						
CXEXXX	Stream Elective-4	3	0	0	3						
CAL411	Machine Learning Operations (MLOps) Lab	0	0	2	1						
CAP411	B.Tech Project Phase I	0	0	6	3						
	Hons. Course-2	3	0	0	3						
CHP411	Hons.Project Phase-I	0	0	6	3						
Total		**15**	**0**	**8**	**19**	**Total**		**0**	**0**	**18**	**9**
Total Credits at the End of Fourth Year:**160** (excluding Activity Credits) (B.Tech.); 135 + 28 + 17 = 180 (B.Tech.(Hon.))											
Semester IX						**Semester X**					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
IRP 511	Research Project-1	0	0	16	8	IRP521	Research Project-2	0	0	24	12
Total		**0**	**0**	**16**	**8**	**Total**		**0**	**0**	**24**	**12**
Total Credits at the End of Fifth Year:**180+20=200** (**BTech-MS**)											

Detailed Semester-wise Structure

Semester I

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC111	Calculus and Linear Algebra	3	1	0	4	IC
2	CSC111	Computer Programming and Problem Solving	3	1	0	4	IC
3	ECC111	Electronic Devices and Digital Logic Design	3	1	0	4	IC
4	HMC111	Communication Skills for the Digital Age	3	0	0	3	IC
5	CSC112	Foundations of Computing and Systems	2	0	0	2	PC
6	CSL111	Computer Programming Lab	0	0	2	1	IC
7	CSL113	IT Workshop Lab	0	1	2	2	PC
8	ECL111	Digital Systems and Electronics Foundations Lab	0	0	2	1	IC
		Total	14	4	6	21	

Semester II

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC121	Discrete Mathematics	3	1	0	4	IC
2	CSC121	Data Structures	3	1	0	4	IC
3	CAC121	Artificial Intelligence	3	0	0	3	IC
4	HMC121	Personality Development and Soft Skills	2	1	0	3	IC
5	CSC122	Computer Organization	3	1	0	4	IC
6	CSC123	Object Oriented Programming	2	1	0	3	PC
7	CSL121	Data Structures Lab (C, C++)	0	0	2	1	IC
8	CSL123	Object Oriented Programming Lab	0	0	2	1	PC
		Total	16	5	4	23	

Semester III

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC211	Probability, Statistics, and Random Processes	3	1	0	4	IC
2	HMC211	Engineering Management, Entrepreneurship Development and Start ups	2	0	0	2	IC
3	CSC211	Computer Networks	2	1	0	3	IC
4	CSC212	Design and Analysis of Algorithms	2	1	0	3	PC
5	CSC213	Theory of Computation	3	1	0	4	PC
6	CSC214	Operating Systems	3	0	0	3	PC
7	CSL211	Computer Networks Lab	0	0	2	1	PC
8	CSL212	Algorithm Design Lab	0	0	2	1	PC
9	CSL214	Operating Systems Lab	0	0	2	1	PC
		Total	15	4	6	22	

Semester IV

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MACXXX	Computational Linear Algebra and Optimization Techniques	3	1	0	4	IC
2	HMC221	Engineering Economics, Finance and Business Analytics	2	0	0	2	IC
3	CAC221	Machine Learning and Data Analysis	2	1	0	3	PC
4	CSC222	Language Processors	2	1	0	3	PC
5	CSC223	Database Management Systems	2	1	0	3	PC
6	CSC224	Advanced Data Structures	2	1	0	3	PC
7	CAL221	Machine Learning and Data Analysis Lab	0	0	2	1	PC
8	CSL222	Language Processors Lab	0	0	2	1	PC
9	CSL223	Database Management Systems Lab	0	0	2	1	PC
		Total	13	5	6	21	

Semester V

Sl.	Code	Course Title	L	T	P	Credits	Category
1	CSC311	Software Engineering	2	1	0	3	PC
2	CAC311	Data Engineering	3	0	0	3	PC
3	CSC312	Parallel and Distributed Computing	3	0	0	3	PC
4	CAC313	Advanced Machine Learning	3	0	0	3	PC
5	CXOXXX	Open Elective-1	2	1	0	3	OE
6	CXEXXX	Stream Elective-1	3	0	0	3	PE
7	CAL311	Data Engineering Lab	0	0	2	1	PC
8	CSL312	Parallel and Distributed Computing Lab	0	0	2	1	PC
9	CAL313	Advanced Machine Learning Lab	0	0	2	1	PC
		Total	16	2	6	21	

Semester VI

Sl.	Code	Course Title	L	T	P	Credits	Category
1	CAC321	AI Safety	2	0	0	2	PC
2	CAC322	Deep Learning	3	0	0	3	PC
3	CAC323	Big Data Analytics and Streaming Systems	3	1	0	4	PC
4	CAC324	Number Theory and Cryptography	3	1	0	4	PC
5	CXEXXX	Stream Elective-2	2	0	2	3	PE
6	CXOXXX	Open Elective-2 Foreign Language	2	0	0	2	OE
7	CXOXXX	Open Elective-3 Design Thinking and Innovation	3	0	0	3	OE
8	CAL325	Agentic AI Lab	0	0	2	1	PC
9	CSL321	FOSS Project and Innovation Lab	0	1	2	2	PC
10		Hons. Course-1	3	0	0	3	
		Total	18	3	6	24	

Semester VII

Sl.	Code	Course Title	L	T	P	Credits	Category
1	CAC411	Machine Learning Operations (MLOps)	3	0	0	3	PC
2	CXOXXX	Open Elective-4	3	0	0	3	OE
3	CAC412	Generative Artificial Intelligence Techniques	3	0	0	3	PC
4	CXEXXX	Stream Elective-3	3	0	0	3	PE
5	CXEXXX	Stream Elective-4	3	0	0	3	PE
6	CAL411	Machine Learning Operations (MLOps) Lab	0	0	2	1	PC
7	CAP411	B.Tech Project Phase-I	0	0	6	3	P
8		Hons. Course-2	3	0	0	3	
9	CHP411	Hons.Project Phase- 1	0	0	6	3	
		Total	15	0	8	19	

Semester VIII

Sl.	Code	Course Title	L	T	P	Credits	Category
1	CAP421	B.Tech Project Phase-II	0	0	18	9	P
2	CHP421	Hons. Project Phase-II	0	0	22	11	
		Total	0	0	18	9	

DETAILED SYLLABUS

MAC111 Calculus and Linear Algebra [3-1-0-4]

Course Objectives

- To study the basic topological properties of the real numbers.
- To develop knowledge of sequences of real numbers and their convergence.
- To study the notion of continuous functions and their properties.
- To gain an understanding of systems of linear equations.
- To introduce the fundamental concepts of vector spaces.
- To impart the basics of linear transformations, orthogonalization, basis, dimension, and eigenvalues.
- To provide the knowledge required to apply concepts of linear algebra in engineering applications.

Course Outcomes

- CO1: Understand the concepts of limits, continuity, differentiability, and classical theorems (Rolle's, Mean Value, Fundamental Theorem of Calculus) to model and solve real-world problems in science and engineering.
- CO2: Analyze convergence of sequences and series, and optimize multivariable functions using Hessian matrices to address practical problems in data modelling, optimization and artificial intelligence.
- CO3: Formulate and solve systems of linear equations using matrix methods, and interpret solutions in applied contexts such as network flows, resource allocation, and computational simulations.
- CO4: Apply vector space concepts, orthogonalization techniques, and inner product structures to design efficient representations in signal processing, machine learning, and numerical computation.
- CO5: Compute eigenvalues and eigenvectors, perform matrix diagonalization, and apply these techniques in stability analysis, principal component analysis, engineering design etc.

Syllabus

Calculus: Real Numbers, Properties of Real Numbers, Limit, Continuity, Discontinuity, Uniform continuity, Differentiability, Rolle's theorem, Mean Value theorem, Reimann Integration, fundamental theorem of calculus. Sequences and Series: Convergence and limit laws, Finite and infinite series, Sums of non-negative numbers, Absolute and conditional convergence of an infinite series, tests of convergence; Function of several variables: Limit, Continuity, Partial derivatives, Local maxima and local minima, Saddle point, Hessian Matrix.

Linear Algebra: System of linear equations, Row reduced echelon matrices, Rank of a matrix. Definition of a linear vector space and examples; linear independence of vectors, basis and dimension, Subspaces; Linear transformations, Matrix representation of Linear transformation, Inner product, norms, orthogonal basis, Gram-Schmidt orthogonalization process, Eigen vectors of a matrix and matrix diagonalization, Applications.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1										2	
CO2	3	3	1	2									3	1
CO3	3	3	2	2	1								3	1
CO4	3	2	2	1	2								3	2
CO5	3	2	2	2	2							1	3	2

Textbooks / References

1. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th Edition, Wiley, 2011.
2. T. M. Apostol, *Calculus*, Volume I, 2nd Edition, Wiley, 2007.
3. G. Strang, *Linear Algebra and Its Applications*, 5th Edition, Wellesley–Cambridge Press/SIAM, 2016.
4. K. Hoffman and R. Kunze, *Linear Algebra*, 2nd Edition, PHI, 2009.
5. E. Kreyszig, *Advanced Engineering Mathematics*, 10th Edition, Wiley, 2015.
6. Thomas and Finney, *Calculus*, 9th Edition, Addison Wesley Publishing, 1998.
7. Anton, Biven, *Calculus*, 11th Edition, Wiley, 2015
8. Gilbert Strang, *Linear Algebra and Learning from Data*, Wellesley-Cambridge Press, 2019.

CSC111 Computer Programming & Problem Solving [3-1-0-4]

Course Objectives

- To introduce problem solving techniques and their representation using algorithms, flowcharts, and pseudocode.
- To teach core concepts of C programming including data types, operators, control flow constructs, and input/output mechanisms.
- To familiarize students with arrays, character arrays, and user-defined functions with parameter passing mechanisms in C.
- To introduce the concepts of recursion and pointer-based programming in C.
- To familiarize students with derived data types such as typedef, enumerations, unions, and structures in C.

Course Outcomes

- CO1: Gain knowledge of problem-solving techniques using algorithmic thinking, flowcharts, and pseudocode.
- CO2: Students learn to apply core C programming concepts including data types, control flow, arrays, functions, recursion, and pointers in problem solving
- CO3: Students learn to use derived data types and structured programming constructs for developing efficient solutions in C

Syllabus

Introduction to Problem Solving and Computational Thinking: Problem decomposition and abstraction, Algorithm design-Pseudocode and flowcharts. Problem-solving strategies (brute force, divide-and-conquer, iterative and recursive thinking), Algorithm verification and testing

Introduction to C Programming: Introduction to algorithm, flowchart, pseudocode. Introduction to C programming - Character set of C, basic data types, operators, scope of variables, formatted input and output using scanf and printf. Writing and compiling simple C programs.

Control Flow Constructs: Conditional statements, if-else and switch constructs, nested conditions. Entry tested loops, exit tested loops, nested loop programs.

Arrays and Functions: Single-dimensional arrays, multidimensional arrays, character arrays. User-defined functions, function declaration, definition, call, return types, actual and formal parameters, parameter passing by value.

Recursion and Pointers: Recursive problem definitions, base case and recursive case. Example programs: factorial using recursion, Fibonacci series, Printing patterns. Introduction to pointers, address and dereferencing, pass-by-reference using pointers.

Derived Data Types: Use of typedef for user-defined type names, enumeration, unions and structures in C. Defining and accessing union members and structure members.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	1								1	3	
CO2	3	3	3	2	2							1	3	1
CO3	3	3	3	2	2							1	3	2

Textbooks / References

1. V. Anton Spraul, Think Like a Programmer: An Introduction to Creative Problem Solving is a programming, No Starch Press, Inc., 2012
2. Pólya, George. How to Solve It: A New Aspect of Mathematical Method. Princeton University Press, 2004. (Princeton Science Library Edition)
3. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Second Edition, Prentice Hall, 1988.
4. E. Balagurusamy, Programming in ANSI C, Ninth Edition, McGraw Hill, 2024.
5. Paul Deitel and Harvey Deitel, C: How to Program, Ninth Edition, Pearson Education, 2022.
6. V. Rajaraman, Computer Programming in C, Second Edition, Prentice Hall India, 2004.
7. R. G. Dromey, How to Solve It by Computer, Pearson Education India, 2007.
8. Reema Thareja, Computer Fundamentals and Programming in C, Second Edition, Oxford University Press, 2016.

ECC111 Electronic Devices and Digital Logic Design [3-1-0-4]

Course Objectives

- To impart the basic concepts of semiconductor devices.
- To analyse the operation of fundamental diode circuits and transistor switches.
- To develop a strong foundation in number systems, Boolean algebra, logic gates, and simplification techniques used in digital electronic systems.
- To analyze and design combinational and sequential logic circuits using standard digital building blocks such as multiplexers, counters, flip-flops, and shift registers.

Course Outcomes

- CO1: Analyse pn-junction behaviour under various operating conditions and design diode circuits.
- CO2:Understand the working of MOSFETs as switching devices.
- CO3: Apply number system conversions, Boolean algebra, Karnaugh maps, and logic gate principles to analyze and simplify digital circuits.
- CO4: Design and evaluate combinational and sequential circuits including adders, multiplexers, counters, flip-flops, and shift registers with appropriate timing analysis.

Syllabus

Introduction to Semiconductors: Intrinsic and extrinsic semiconductors, pn junction diode, diode equation (derivation not required), diode characteristics, ideal and practical diode (Silicon), diode applications: clipping and clamping circuits. Field Effect Transistors: voltage-current relation in MOSFET (derivation not required), MOSFETs as switches.

Number Systems and base conversion, Binary Arithmetic, Representation of Negative Numbers: sign and magnitude, 2's complement, 1's complement forms, shifts, and overflow, Binary Coded Decimal (BCD). Boolean algebra, De Morgan's laws, Basic Theorems, truthtables, SOP/POS. Minterm and Maxterm Expansions, Incompletely Specified Functions, don't–care conditions, Karnaugh map, Karnaugh map SOP and POS minimizations (Two, Three, Four and Five-variable Karnaugh maps), NAND/NOR realization.

Combinational Logic Circuits–Half and Full Adders, Parallel Binary Adders, Binary Subtractor, Magnitude Comparators, Decoders, Encoders, Code Converters, Multiplexers, Demultiplexers.

Sequential Logic Circuits – Latches and Flip–Flops, SR Latch, Gated D Latch, Edge–Triggered D Flip-Flop, SR Flip–Flop, JK Flip-Flop, T Flip–Flop, Flip-Flops with Asynchronous Clear and Preset inputs, Master-Slave Flip–Flops, Excitation table, Flip-Flop conversions, Counters – Asynchronous Counters, Synchronous Counters, Up/Down Synchronous Counters, Design of Synchronous Counters. Shift Registers – data movements in shift registers, SISO, SIPO, PISO, PIPO shift registers, Design of sequence detector circuits.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2					1		2	3	2
CO2	3	3	3	2	3					1		2	3	3
CO3	3	3	2	2	3					1		2	3	3
CO4	3	3	3	3	3				2	2	1	3	3	3

Textbooks / References

1. Sedra A. and Smith K. C., Microelectronic Circuits, Oxford University Press, Seventh Edition, 2015.
2. Robert L. Boylestad and Louis Nashelsky, Electronic Devices and Circuit Theory, Pearson Education, Eleventh Edition, 2015.
3. Ben G. Streetman and Sanjay Kumar Banerjee, Solid State Electronic Devices, Seventh Edition, Pearson Education, 2016.
4. M. Morris Mano and Michael D. Ciletti, Digital Design, 6th Edition, Pearson Education, 2018.
5. Thomas L. Floyd, Digital Fundamentals, 11th Edition, Pearson Education, 2015.
6. Charles H. Roth Jr. and Larry L. Kinney, Fundamentals of Logic Design, 7th Edition, Cengage Learning, 2013.

HMC111 Communication Skills for the Digital Age [3-0-0-3]

Course Objectives

- To understand the fundamentals of communication and technical communication.
- To develop effective listening, speaking, reading, and writing skills for academic and professional contexts.
- To apply grammatical structures and vocabulary appropriately in oral and written communication.
- To analyze communication barriers, audience needs, and communication styles in different contexts.
- To create professional documents, presentations, and technical content using appropriate language and format.
- To understand the impact of Artificial Intelligence and Digital Humanities on contemporary communication practices.
- To practice ethical, responsible, and inclusive communication in digital and multilingual environments.

Course Outcomes

- CO1: Apply effective listening, speaking, reading, and writing strategies in academic and professional contexts
- CO2: Analyze communication barriers, audience needs, and communication styles in different contexts
- CO3: Create professional documents, presentations, and technical content using appropriate language and format
- CO4: Evaluate the role of AI and digital technologies in communication with ethical awareness
- CO5: Demonstrate responsible and effective communication in multilingual and digital environments

Syllabus

Fundamentals of Communication: Process and types of communication- Technical communication and its importance - Barriers to communication - Non-verbal communication -Communication in academic and professional contexts -Shifts in communication in the digital age.

Listening and Speaking Skills: Listening process and active listening - Barriers to listening and strategies to overcome them - Fluency and accuracy in speech - Self-expression and tonal variations - Group discussion and interpersonal communication - Public speaking and presentation skills - Interviews and professional interaction - Responsible use of AI in communication.

Reading, Writing and Digital Communication: Reading comprehension and critical reading - Elements of effective writing - Professional emails and formal letters - Technical reports and presentations - Introduction to academic writing - AI in multilingual communication and translation - Ethical and legal considerations in AI-mediated communication - Introduction to Digital Humanities and digital communication practices.

Grammar and Vocabulary for Communication: Parts of speech and sentence structures - Tenses and Subject–Verb Agreement - Reported Speech and Active–Passive Voice - Common errors in English usage - Punctuation and mechanics of writing - Vocabulary enhancement in context.

Practical / Language Laboratory Component (1 Hour per Week): Listening to recorded talks and comprehension activities - Active listening exercises - Group discussion and interpersonal conversation - Pronunciation and tonal variation practice - Extempore speech and presentation activities -Mock interviews and persuasive speaking exercises -Speech assessment and peer feedback - Technology-assisted listening and speaking practice.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1					1			1	2	3		1		
CO2		1		1		1		2	2	3		1		
CO3			1		2			1	2	3		1		1
CO4		1		1	2	1		3		2		2		2
CO5					2	1		3	2	3		2		2

Textbooks / References

1. Anderson, Paul V., and Kerry Surman. Technical communication: A reader-centered approach. Thomson/Wadsworth, 2007.
2. Berry, David M. "Introduction: Understanding the digital humanities." Understanding digital humanities. London: Palgrave Macmillan UK, 2012. 1-20.
3. Davis, Marjorie T. "Assessing technical communication within engineering contexts tutorial." IEEE Transactions on professional communication 53.1 (2010): 33-45.
4. Duin, Ann Hill, and Isabel Pedersen. Augmentation technologies and artificial intelligence in technical communication: Designing ethical futures. Routledge, 2023.
5. Durack, K. T. (1997). Gender, Technology, and the History of Technical Communication. Technical Communication Quarterly, 6(3), 249–260.
6. Eggleston, A. G., & Rabb, R. J. (2018, June), Technical Communication for Engineers: Improving Professional and Technical Skills Paper presented at 2018 ASEE Annual Conference & Exposition , Salt Lake City, Utah. 10.18260/1-2—31068
7. Hart-Davidson, Bill, et al. "The history of technical communication and the future of generative AI." Proceedings of the 42nd ACM International Conference on Design of Communication. 2024.
8. Raman, Meenakshi, and Sangeeta Sharma. Technical communication: Principles and practice. Oxford University Press, 2004.
9. Ranade, Nupoor, and Marly Saravia. "Teaching AI ethics in technical and professional communication: A systematic review." IEEE Transactions on Professional Communication 67.4 (2024): 422-436.
10. Reeves, Carol, and J. J. Sylvia IV. "Generative AI in technical communication: A review of research from 2023 to 2024." Journal of technical writing and communication 54.4 (2024): 439-462.
11. Thomson, Audrey Jean, and Agnes V. Martinet. A practical English grammar. New York: Oxford University Press, 2015.
12. Verma, Shalini. Technical communication for engineers. Vikas Publishing House, 2015.
13. Yule, George. Oxford practice grammar advanced. Oxford University Press, 2015.

CSC112 Foundations of Computing and Systems [2-0-0-2]

Course Objectives

- Develop a foundational understanding of computing systems by examining computation as information processing, computational models, and methods of organizing and representing data.
- Establish the mathematical and logical fundamentals governing digital data representation, reliability, and physical hardware execution.
- Provide a unified view of computer systems by bridging the structural hierarchy from physical logic gates up through software layers and networking frameworks.
- Introduce the mechanics of distributed execution, concurrency, and real-world system modeling.
- Evaluate the physical and mathematical limits of classical computing while exploring the architecture of emerging, next-generation paradigms.

Course Outcomes

- CO1: Explain computation as information processing, distinguish between major models of computation, and describe how information is organized using fundamental logical structures and elementary data structures.
- CO2: Apply number system conversions and explain how digital information is represented, encoded, compressed, and protected against transmission errors.
- CO3: Map the abstraction layers of a computing system, linking high-level software execution down to machine code, processor cycles, and physical hardware switches.
- CO4: Analyze data flow and structural topologies by mapping algorithmic design strategies (such as graphs or recursion) onto packet routing mechanisms and network layered frameworks.
- CO5: Construct conceptual models for distributed and cloud-based applications, evaluating how communication topologies and architectural frameworks influence systemic performance.
- CO6: Formulate simulation workflows for complex, real-world processes while appraising basic computational complexity (P vs. NP), the necessity of algorithmic ethics, and adaptability to non-classical hardware.

Syllabus

Foundations of Computational Systems: Computing as information processing, Models of computation, Data Organization and logical structures. Introduction to Linear Data Structures: conceptual overview of storing relationships using Arrays, lists, matrices, and basic graphs.

The mathematical foundations of digital data storage and transmission. Number Systems: positional notation, binary, octal, decimal, and hexadecimal bases; standard conversion techniques. Data Encoding & Multimedia: text representation (ASCII, Unicode), and the digitization of continuous media (image pixels, color depth, audio sampling). Introduction to Compression & Reliability: conceptual difference between lossy and lossless compression; basic error detection (parity bits).

The basic structural hierarchy of physical computer hardware and system software. The Hardware Layer: transistors as binary switches, basic logic gates (AND, OR, NOT, XOR), and simple Boolean expressions. Digital Logic Circuits: introductory overview of combinational logic (adders) and sequential logic components (registers). Processor Architecture: the classic Von Neumann model framework (ALU, Control Unit, Memory) and a conceptual overview of the Fetch-Decode-Execute instruction cycle. Software Layer Abstraction: high-level languages vs. machine code, and the basic role of an Operating System.

Introduction to interconnected processing units and network communication. Networking Foundations: physical topologies (star, mesh), and layered abstraction frameworks (basic concepts of the OSI and TCP/IP models). The Internet Stack: packet routing, logical addressing (IP), and physical addressing (MAC). Distributed Frameworks: conceptual introduction to Distributed Systems and message passing / Remote Procedure Calls (RPC), foundational overview of Cloud Computing models (SaaS, PaaS, IaaS) and client-server architectures.

System Modeling, Complexity Horizons, and Computing Ethics. Computational approaches to mimicking reality, future paradigms, and societal impact. System Modeling & Simulation: introduction to using computers to simulate real-world processes (stochastic vs. deterministic context, queuing systems). Complexity Horizons: Basic notions of computational complexity (time and space trade-offs, scalability), conceptual overview of what makes a problem computationally hard (P vs. NP mapping). Emerging Paradigms: qualitative overview of Quantum Computing (qubits, superposition), neuromorphic architectures, and the Internet of Things (IoT). Responsible Computing and Ethics: introduction to algorithmic bias, data privacy, user confidentiality, and computing sustainability (carbon footprint of large-scale systems).

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	1	1							2	3	2
CO2	3	3	2	1	2							2	3	2
CO3	3	3	2	2	2							2	3	3
CO4	3	3	2	2	3				1			2	3	3
CO5	3	3	2	2	3			1	2	1		2	3	3
CO6	3	3	2	2	2		1	3				3	3	3

Textbooks / References

1. Brookshear, J. G., & Brylow, D. Computer Science: An Overview. 13th Edition, Pearson, 2019.
2. Patterson, D. A., & Hennessy, J. L. Computer Organization and Design: The Hardware/Software Interface. 6th Edition, Morgan Kaufmann, 2020.
3. Kurose, J. F., & Ross, K. W. Computer Networking: A Top-Down Approach. 8th Edition, Pearson, 2020.
4. Patt, Yale N., and Sanjay J. Patel. Introduction to Computing Systems: From Bits & Gates to C/C++ & Beyond. 3rd ed., McGraw-Hill Education, 2020.
5. Tanenbaum, A. S., & Austin, T. Structured Computer Organization. 6th Edition, Prentice Hall, 2012.
6. George Coulouris, Jean Dollimore, and Tim Kindberg. (2011). Distributed Systems: Concepts and Design (5th ed.). Addison-Wesley.
7. Buyya, R., Vecchiola, C., & Selvi, S. T. (2013). Mastering Cloud Computing: Foundations and Applications Programming. McGraw-Hill Education.
8. Sipser, Michael. Introduction to the Theory of Computation. 3rd ed., Cengage Learning, 2012.
9. Quinn, M. J. (2020). Ethics for the Information Age (8th/9th Edition). Pearson.
10. Zeigler, Bernard P., et al. Theory of Modeling and Simulation: Discrete Event & Iterative System Computational Foundations. 3rd ed., Academic Press, 2018.

CSL111 Computer Programming Lab [0-0-2-1]

Course Objectives

- To enable students to write, compile, and execute C programs systematically
- To familiarize students with core C programming concepts including data types, operators, control flow constructs, and input/output mechanisms
- To develop programming skills using arrays, character arrays, and user-defined functions with parameter passing mechanisms.
- To introduce recursive and pointer-based programming through hands-on implementation in C.
- To enable students to implement programs using derived data types such as typedef, enumerations, unions, and structures in C.

Course Outcomes

- CO1: Students learn to translate programming concepts into C programs using structured programming techniques.
- CO2: Students learn to develop recursive and pointer-based solutions for computational problems.
- CO3: Students learn to use C programs for solving real world problems.

Syllabus

Introduction to C Programming: Basic data types, operators, scope of variables, formatted input and output using scanf and printf. Writing and executing simple C programs.

Control Flow Constructs: Implementation of programs using conditional statements and loop constructs.

Arrays and Functions: Implementation of programs using arrays and character arrays. Programs involving user-defined functions with parameter passing by value.

Recursion and Pointers: Implementation of recursive programs. Implementation of programs using pointers and pass-by-reference mechanisms.

Derived Data Types: Implementation of programs using typedef, enumerations, unions, and structures in C. Defining and accessing union members and structure members.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	2	2							1	3	1
CO2	3	3	3	2	2							1	3	1
CO3	3	3	3	2	2							1	3	2

Textbooks / References

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Second Edition, Prentice Hall, 1988.
2. E. Balagurusamy, Programming in ANSI C, Ninth Edition, McGraw Hill, 2024.
3. Paul Deitel and Harvey Deitel, C: How to Program, Ninth Edition, Pearson Education, 2022.
4. Ashok N. Kamthane, Programming in C, Third Edition, Pearson India, 2015

CSL113 IT Workshop Lab [0-1-2-2]

Course Objectives

- To introduce students to modern software development environments, workflows, and collaborative programming practices.
- To develop proficiency in Linux command-line tools, file system management, shell scripting, and developer productivity utilities.
- To enable effective use of Git and GitHub for version control, repository management, collaborative development, and open-source contribution workflows.
- To train students in the use of integrated development environments, debugging techniques, and software documentation standards for professional software development.
- To introduce foundational web programming concepts and modern front-end development using HTML, CSS, JavaScript, and basic React JS for building interactive web applications.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Demonstrate proficiency in Linux environments, command-line tools, shell scripting, and software development utilities.
- CO2: Apply Git and GitHub workflows for repository creation, version control, collaboration, branching, merging, and remote repository management.
- CO3: Develop, debug, and document software projects using modern IDEs, debugging tools, and software documentation practices.
- CO4: Design and implement basic web pages using HTML, CSS, and JavaScript while understanding web application architecture.

Lab Practice

1. **Linux Environment Familiarization and Shell Scripting:** Linux file system, terminal basics, directory navigation, file manipulation commands, package management overview, environment variables, basic shell commands, simple shell scripts, and automation concepts.
2. **Git Fundamentals, Branching, and GitHub Repository Management:** Git installation, repository initialization, staging and committing, viewing history, undoing changes, branch creation, feature branch workflow, merge operations, conflict resolution, introduction to rebase, remote repositories, clone, push, pull, fetch, SSH and HTTPS authentication, repository hosting and management, and forking repositories.
3. **Collaborative Development and Technical Documentation:** Pull requests, issues and discussions, code reviews, open-source contribution workflow, README creation, Markdown syntax, code documentation, and documentation standards.
4. **IDE Setup, Debugging, and Project Management:** Modern IDE features, extensions and plugins, integrated terminal, workspace management, breakpoints, stepping through code, variable inspection, error diagnosis, debugging best practices, and basic project organization.
5. **Introduction to Web Programming:** World Wide Web, client-server architecture, static and dynamic websites, browser developer tools, HTML page structure, forms, tables, lists, semantic elements, CSS selectors, layouts, responsive design, JavaScript variables, functions, events, DOM manipulation, and browser developer tools.
6. **React JS Fundamentals:** Setting up a React application using Vite, understanding components, JSX fundamentals, conceptual introduction to props and state, event handling in React, and development of a simple interactive component.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	1		3							2	2	2
CO2	2	2	2	1	3				2	2		2	2	3
CO3	2	2	2	2	3				2	2	1	2	3	3
CO4	2	2	3	2	3				1	2		2	3	3

Textbooks / References

1. Dean, J., *Web Programming with HTML5, CSS, and JavaScript*, Jones & Bartlett Learning, 2018.
2. Robbins, J., *Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics*, 6th

Edition, O'Reilly Media, 2024.
3. Banks, A., and Porcello, E., *Learning React: Modern Patterns for Developing React Apps*, 3rd Edition, O'Reilly Media, 2024.

ECL111 Electronic Devices and Digital Logic Design Lab [0-0-2-1]

Course Objectives

- To introduce the fundamental concepts and characteristics of semiconductor devices.
- To analyse the operation and applications of fundamental electronic devices in various circuits.
- To develop a strong foundation in number systems, Boolean algebra, logic gates, and logic simplification techniques used in digital electronic systems.
- To enable the analysis and design of combinational and sequential logic circuits using standard digital building blocks such as multiplexers, counters, flip-flops, and shift registers.

Course Outcomes

- CO1 Identify and operate basic electronic laboratory equipment, and analyse the characteristics and applications of diode-based circuits such as clippers and clampers.
- CO2 Design, implement, and verify combinational logic circuits including adders, multiplexers, and demultiplexers using digital electronic components.
- CO3 Analyse the operation of sequential logic circuits by implementing various flip-flops, latches, and asynchronous counters.
- CO4 Design, implement, and evaluate synchronous counters and shift registers for digital system applications.

Experiments

1. Familiarization of basic electronic lab equipment and components.
2. Diode Clippers.
3. Diode Clampers.
4. Design of Combinational Logic Circuits - Adders, Multiplexers, Demultiplexers
5. Familiarization of Flip-Flops and Latches. SR, D and JK.
6. Design of asynchronous counters.
7. Design of synchronous counters
8. Shift registers.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3					1		2	3	2
CO2	3	3	2	2	3					1		2	3	3
CO3	3	3	3	3	3				2	2	1	2	3	3
CO4	3	2	3	2	3				2	2		2	3	3

Textbooks / References

1. Sedra A. and Smith K. C., Microelectronic Circuits, Oxford University Press, Seventh Edition, 2015.
2. Robert L. Boylestad and Louis Nashelsky, Electronic Devices and Circuit Theory, Pearson Education, Eleventh Edition, 2015.
3. M. Morris Mano and Michael D. Ciletti, Digital Design, 6th Edition, Pearson Education, 2018.
4. Thomas L. Floyd, Digital Fundamentals, 11th Edition, Pearson Education, 2015.
5. Charles H. Roth Jr. and Larry L. Kinney, Fundamentals of Logic Design, 7th Edition, Cengage Learning, 2013.

MAC121 Discrete Mathematics [3-1-0-4]

Course Objectives

- To extend students' logical and mathematical maturity and their ability to deal with abstraction.
- To introduce the fundamental terminologies and concepts used in electronics, communication engineering, and computer science.
- To explain and apply the basic methods of discrete mathematics in engineering and computing applications.
- To develop the ability to write clear, concise, and correct mathematical proofs.
- To solve counting problems involving permutations, combinations, and the Pigeonhole Principle.
- To understand the fundamental concepts of graph theory and abstract algebra.

Course Outcomes

- CO1: Apply propositional and predicate logic, rules of inference, and proof techniques to analyze mathematical statements and algorithms.
- CO2: Analyze sets, relations, functions, and recursive structures used in discrete mathematical modeling.
- CO3: Apply counting principles, recurrence relations, and generating functions to solve combinatorial problems.
- CO4: Solve problems involving graphs, trees, groups, rings, fields, and lattices in engineering and computing contexts.

Syllabus

Logic: Propositions, negation, conjunction, disjunction, implication and equivalence, truth tables, predicates, quantifiers, rules of inference, methods of proof.

Set Theory: Definitions and simple proofs in set theory, inductive definition of sets and proof by induction, inclusion and exclusion principle, relations, representation of relations by graphs, properties of relations, equivalence relations and partitions, partial orderings.

Functions: Mappings, injections and surjections, composition of functions, inverse functions, special functions, recursive function theory.

Elementary Combinatorics: Counting techniques, Pigeonhole Principle, recurrence relations, generating functions.

Graph Theory: Elements of graph theory, Euler graphs, Hamiltonian paths, trees, tree traversals, spanning trees.

Algebra: Groups, Lagrange's theorem, homomorphism theorem, rings and fields, structure of the ring $\mathbb{Z}_n$ and the unit group $\mathbb{Z}_n$, lattices.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	1							2	3	1
CO2	3	3	2	2	1							2	3	1
CO3	3	3	3	2	1							2	3	2
CO4	3	3	3	2	2							2	3	2

Textbooks / References

1. Kenneth H. Rosen, *Discrete Mathematics and Its Applications*, 7th Edition, McGraw-Hill, 2017.
2. Norman L. Biggs, *Discrete Mathematics*, 2nd Edition, Oxford University Press, 2003.
3. J. P. Tremblay and R. Manohar, *Discrete Mathematical Structures with Applications to Computer Science*, McGraw-Hill, 2017.
4. K. A. Ross and C. R. B. Wright, *Discrete Mathematics*, 5th Edition, Pearson, 2003.
5. P. B. Bhattacharya, S. K. Jain, and S. R. Nagpaul, *Basic Abstract Algebra*, 2nd Edition, Cambridge University Press, 2003.
6. J. A. Gallian, *Contemporary Abstract Algebra*, 9th Edition, Cengage Learning, 2017.

CSC121 Data Structures [3-1-0-4]

Course Objectives

- Define and describe simple data structures like arrays, linked lists, trees and graphs.
- Design and specify algorithms for searching and sorting, and those associated with the above data structures .
- Analyse simple algorithms, like sorting and searching using mathematical tools, like formulation and solving of recurrences.
- Analyse application problems and abstract them to formulate solutions involving data structures and algorithms.

Course Outcomes

- CO1: Students learn to define operations of data structures like arrays, linked lists, trees and graphs.
- CO2: Students learn to design and specify algorithms involving above types of data structures.
- CO3: Students learn to analyze simple algorithms and solve recurrences, asymptotic analysis and probabilistic analysis.
- CO4: Students learn to analyze application problems and abstract them to formulate solutions involving data structures and algorithms

Syllabus

Introduction- Algorithm Analysis, Finding Complexity. Fundamental data structures - List, Sorted Lists, Single and Double Linked Lists, Skip list. Dynamic Arrays, Circular Linked Lists, Applications of Lists in Browser History and Text Editors.

Stack & Queue, variants of queues and applications.

Binary Trees- Insertion and Deletion of nodes, Tree Traversals, Polish Notations, AVL Trees, Heaps, Priority Queues. Optimal binary search tree, Application problems on Optimal binary search Tree.

Sorting -Bubble, Selection, Insertion, Merge Sort, Quick Sort, Radix Sort, Heap sort. Searching. Hashing- Application problems on hashing.

Graphs- Representations, types and properties of graphs, Shortest path algorithms, Minimum Spanning Trees, BFS, DFS.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2							1	3	1
CO2	3	3	3	2	2							1	3	2
CO3	3	3	2	3	2							2	3	2
CO4	3	3	3	2	2							2	3	2

Textbooks / References

1. Clifford A Shaffer, Data Structures and Algorithm Analysis, Edition 3.2 (Java Version), 2011.
2. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser. Data Structures And Algorithms In Java TM Sixth Edition, Wiley Publishers, 2014.
3. Mark Allen Weiss Data Structures And Algorithm Analysis In Java, Third Edition, 2012.
4. Robert L. Kruse, Data Structures And Program Design In C++, Pearson Education, Second Edition, 2006.
5. Ellis Horowitz, Fundamentals of Data Structures in C++, University Press, 2015.
6. Ajay Agarwal, Data Structure through C, A Complete Reference Guide, Cyber Tech Publications, 2005.

CAC121 Artificial Intelligence [3-0-0-3]

Course Objectives

- To understand the foundations of Artificial Intelligence and intelligent agents, environments, and rational decision-making processes in AI systems.
- To apply problem-solving techniques in uninformed, informed, adversarial, and constraint search space.
- To understand knowledge representation, reasoning and planning in applications

- To explore advanced AI concepts including natural language processing,expert systems, fuzzy logic, and hybrid intelligent systems.

Course Outcomes

- CO1: Explain the core principles and architectures of Artificial Intelligence systems.
- CO2:Implement search algorithms and reasoning techniques to solve AI problems.
- CO3: Apply knowledge representation and probabilistic reasoning methods in intelligent applications.
- CO4: Analyze the applicability of advanced AI techniques such as NLP, fuzzy systems, expert systems in real-world applications.

Syllabus

Artificial Intelligence & Problem Solving: Introduction to AI: What is AI? Foundation, History, State of the art, Risks and Benefits. Intelligent agent: Agents and environments, concept of rationality, nature of environment, structure of agents. Problem solving: Search space problem, uninformed search strategies, informed search strategies, local search algorithms, constraint satisfaction problems, adversarial search and games.

Knowledge, reasoning and planning: Logical agents: Knowledge-Based Agents, Propositional logic, propositional theorem proving. First order logic: syntax and semantics, knowledge engineering, inference: unification, forward and backward chaining, resolution. Knowledge representation: Ontology, categories, objects, events. Planning: classical planning, heuristics of planning, hierarchical planning.

Uncertain Knowledge and reasoning: Quantifying uncertainty: acting under uncertainty, Bayes' rule, Naïve Bayes model, semantics of Bayes networks, inference in temporal models, combining beliefs and desires under uncertainty, decision networks, sequential decision problems, relational probability models. Fuzzy logic: Crisp logic, Fuzzy logic, Membership function, Membership function, Fuzzy logic Applications.

Expert systems and Advanced AI: Expert Systems: What is Expert system, Conventional systems vs. Expert systems, Basic Concepts, Human Expert Behaviors, Knowledge Types, Inferencing, Rules, Structure of Expert Systems, ES Components, Knowledge Engineer, Expert Systems Working, Problem Areas Addressed by Expert Systems, benefits limitations- Applications of expert systems. Advanced AI: Natural language processing, Hybrid intelligent systems, Introduction to: Ethics of AI, Responsible AI, Generative AI, Explainable AI, Agentic AI.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2							2	3	2
CO2	3	3	3	2	2							2	3	2
CO3	3	3	3	2	2							2	3	3
CO4	3	3	2	2	2	2	2	2			1	2	3	3

Textbooks / References

1. Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, Fourth edition, Pearson Education, 2021
2. Elaine rich and kevin knight, introduction to artificial intelligence, mcgraw hill, third edition, 2017.
3. Michael Negnevitsley, artificial intelligence: a guide to intelligent systems, addison wesley, third edition, 2017
4. David L. Poole and Alan K. Mackworth, Artificial Intelligence: Foundations of Computational Agents, 3rd edition,2023. (https://artint.info/3e/html/ArtInt3e.html)
5. Deepak Khemani. A First Course in Artificial Intelligence. McGraw Hill Education (India), 2013.
6. Denis Rothman. Artificial Intelligence by Example, Packt, 2020, 2nd Edition.
7. Explainable AI for Practitioners by Michael Munn, David Pitman, O'Reilly Media 2022, ISBN: 9781098119133
8. Web Resources:
   1.Stanford Encyclopedia of Artificial Intelligence
   2. IBM AI Learning Hub
   3. NPTEL Artificial Intelligence Course

HMC121 Personality Development and Soft Skills [2-1-0-3]

Course Objectives

- To understand the significance of soft skills, professional etiquette, self-awareness, and ethical behaviour in personal and professional development
- To help students understand and explain individual behaviour and personality in interpersonal and professional contexts.
- To apply communication, leadership, teamwork, presentation, and interpersonal skills in academic and workplace contexts
- To create awareness regarding common psychological concerns and the importance of psychological wellbeing in academic and personal life.
- To analyze workplace situations and develop problem-solving, emotional intelligence, adaptability, and stress management skills for professional
- To familiarize students with strategies for emotional regulation, coping, and healthy personality development.

Course Outcomes

- CO1: Apply self-assessment and goal-setting techniques for personal and professional growth

- CO2: Understand one's own personality and that of others, appreciate individual differences, and adapt effectively to people and situations.
- CO3:Analyze workplace situations using communication, leadership, and emotional intelligence skills
- CO4: Develop the ability to cope with personal, academic, and professional challenges through an understanding of human behaviour.
- CO5: Demonstrate effective speaking, presentation, and interpersonal communication skills
- CO6: Evaluate professional challenges related to stress, ethics, teamwork, and work–life balance

Syllabus

Foundations of soft skills and personality development: Introduction to soft skills - Importance and aspects of soft skills - Self-esteem and self-awareness - Professional etiquette and social etiquette - Professional ethics and workplace behavior - Professional body language - SMART goals and personal goal setting - SWOT analysis for personality development.

From Campus to Corporate: Leadership skills and teamwork - Group discussion and meeting management - Adaptability and work ethics - Time management and stress management - Advanced speaking skills - Oral presentations, speeches, and debates - Combating nervousness and stage fear - Planning and preparation of presentations - Making effective presentations - Speeches for various occasions - Interviews and facing job interviews - Emotional intelligence and critical thinking -Work–life balance - Applied grammar for professional communication.

Understanding Personality and Behaviour: Personality: Meaning & Assessment. Psychoanalytic & Neo-Psychoanalytic Approach; Behavioural Approach; Cognitive Approach; Social- Cognitive Approach; Humanistic Approach; The Traits Approach; Models of healthy personality: the notion of the mature person, the self-actualizing personality; Indian perspective on personality.

Mental Health, Institutions, and Social Contexts: Mental health in social, cultural, and institutional contexts; campus and workplace mental health. Institutional violence, bullying, harassment, and discrimination based on caste, religion, gender, sexuality, disability, body image, and other social locations. Structural inequalities, stigma, inclusion, and access to mental health support. Student mental health and suicide prevention, including institutional accountability and recommendations of the Supreme Court's National Task Force on Student Suicides. Critical perspectives on positive psychology and mainstream mental health interventions; indigenous and community-based approaches to wellbeing. Help-seeking, collective care, peer support, and professional mental health services.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1						1		1	1	2	1	3		
CO2						2	1	2	2	2		2		
CO3						2	1	2	3	3	2	2		
CO4						3	2	2	2	2		3		
CO5						1		1	2	3	1	2		
CO6						3	2	3	2	2	2	3		

Textbooks / References

1. Carnegie, Dale. How to win friends and influence people. John Wiley & Sons, 2026.
2. Covey, Stephen R. "The 7 habits of highly effective people." Personal Finance Magazine 2024.527 (2024): 8-13.
3. Ghai, A. (2015). Rethinking disability in India. Routledge.
4. Guru, G., & Sarukkai, S. (2012). The cracked mirror: An Indian debate on experience and theory. Oxford University Press.
5. Goleman, Daniel. "Emotional Intelligence Bloomsbury Publishing." (2020).
6. Kumar, N. (2023). Mental health and well-being: An Indian psychology perspective. Routledge.
7. Medden, Stephanie, et al. "From Public Speaking to Multimodal Communication: A Reevaluation of the Basic Communication Course." Basic Communication Course Annual 37.1 (2025): 4.
8. Patel, V. (2014). Where there is no psychiatrist: A mental health care manual (2nd ed.). Royal College of Psychiatrists.
9. Patel, V., Saxena, S., Lund, C., Thornicroft, G., Baingana, F., Bolton, P., ... Unützer, J. (2018). The Lancet Commission on global mental health and sustainable development. The Lancet, 392(10157), 1553–1598.
10. Pease, Barbara, and Allan Pease. The definitive book of body language: The hidden meaning behind people's gestures and expressions. Bantam, 2008.
11. Rao, Manchanahalli Satyaranayana. Soft skills-enhancing employability: connecting campus with corporate. IK International Pvt Ltd, 2010.
12. Tracy, Brian. Eat that frog!: 21 great ways to stop procrastinating and get more done in less time. Berrett-Koehler Publishers, 2025.
13. Nevid, J. S., Rathus, S. A., & Greene, B. (2022). Abnormal psychology in a changing world (11th ed.). Pearson.
14. Snyder, C. R., & Lopez, S. J. (Eds.). (2021). Positive psychology: The scientific and practical explorations of human strengths (5th ed.).
15. Carr, A. (2022). Positive psychology: The science of happiness and human strengths (4th ed.).
16. Kabat-Zinn, J. (2018). Mindfulness for beginners: Reclaiming the present moment—and your life.
17. Ryckman, R. M. (2017). Theories of personality (10th ed.).

18. Schultz, D. P., & Schultz, S. E. (2020). Theories of personality (12th ed.).

CSC122 Computer Organization [3-1-0-4]

Course Objectives

- This course will introduce to students the fundamental concepts underlying modern computer organization and architecture.
- Students should be able to know the overall working of a computer.
- Students should be able to get a detailed understanding of the design principles involved in developing a computer.
- They should know the representation of data, how programs are represented, executed and how programs manipulate and operate on data.
- They should also be able to appreciate how the memory organization is done and how to organize memory for faster execution of programs.
- Students should also be able to appreciate the concepts in pipelining.

Course Outcomes

- CO1: Understand the basics of computer hardware and how software interacts with computer Hardware.
- CO2:Analyse and evaluate the performance of computers.
- CO3: Understand basics of Instruction Set Architectures and their impact on processor design
- CO4: Understand how computers, represent and manipulate data.
- CO5: Understand how computer, perform arithmetic operations, how they are optimized and made to run faster.
- CO6: Design a pipeline for consistent execution of instructions with minimum hazards.
- CO7: Understand how the memory management takes place in a computer system
- CO8: Design a simple computer with hardware design including data format, instruction format, instruction set, addressing modes, bus structure, input/output, memory, Arithmetic/Logic unit, control unit, and data, instruction and address flow.

Syllabus

Basic principles, hardware components Measuring performance: evaluating, comparing and summarizing performance.

Instructions: operations and operands of the computer hardware, representing instructions, making decision, supporting procedures, character manipulation, styles of addressing, starting a program.

Computer Arithmetic: signed and unsigned numbers, addition and subtraction, logical operations, constructing an ALU, multiplication and division, floating point representation and arithmetic, Parallelism and computer arithmetic.

The processor: building a data path, simple and multicycle implementations, microprogramming, exceptions, Pipelining, pipeline Data path and Control, Hazards in pipelined processors

Memory hierarchy: caches, cache performance, virtual memory, common framework for memory hierarchies

Input/output: I/O performance measures, types and characteristics of I/O devices, buses, interfaces in I/O devices, design of an I/O system, parallelism and I/O. Introduction to multicores and multiprocessors.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							1	3	1
CO2	3	3	2	3	2							2	3	2
CO3	3	3	2	2	2							1	3	2
CO4	3	2	2	2	2							1	3	1
CO5	3	3	2	2	2							1	3	2
CO6	3	3	3	2	2							2	3	2
CO7	3	3	2	2	2							1	3	2
CO8	3	3	3	2	3						1	2	3	2

Textbooks / References

1. D. A. Pattersen and J. L. Hennesy, Computer Organisation and Design: The Hardware/ Software Interface, Fourth Edition, Morgan Kaufman, 2009.
2. V. P. Heuring and H. F. Jordan, Computer System Design and Architecture, Prentice Hall, 2003.
3. Computer Architecture: A Quantitative Approach, Fifth Edition, Morgan Kaufman, 2011.
4. Carl Hamazher, ZvonkoVranesic and SafwatZaky, Computer Organization, Fifth Edition, McGraw Hill, 2002.

CSC123 Object Oriented Programming [2-1-0-3]

Course Objectives

- Understand the fundamentals of object-oriented programming and the Java programming environment.
- Design and implement programs using classes, objects, constructors, and access control mechanisms.
- Apply inheritance, polymorphism, abstraction, and interfaces to develop reusable and extensible software components.
- Organize programs using packages and manage program robustness through exception handling mechanisms.
- Develop Java applications involving arrays, strings, and object-oriented design principles for real-world problem solving.

Course Outcomes

- CO1: Explain the concepts of object-oriented programming and the architecture of the Java platform, including JVM, JRE, and JDK.

- CO2: Design and implement Java classes using constructors, access modifiers, encapsulation, static members, method overloading, and object management techniques.
- CO3: Develop applications using inheritance, polymorphism, abstract classes, interfaces, and packages to create reusable and modular software systems.
- CO4: Implement exception handling mechanisms, including custom exceptions, to build reliable and fault-tolerant Java applications.
- CO5: Develop Java programs using arrays, strings, string manipulation techniques, and related programming constructs to solve computational problems effectively.

Syllabus

Introduction to OOP and Java Fundamentals: Procedural vs. object-oriented thinking; OOPs Principles, advantages of OOPs, Java ecosystem: Bytecode, JDK, JVM, JRE; compiling and running Java programs, Language Fundamentals

Class and Objects: Introducing Class Fundamentals - Object and Object reference, Defining classes: fields, methods, constructors (default, parameterised, copy), Access modifiers: private, protected, public, package-private, Encapsulation: getters, setters, and data hiding principles, 'this' keyword; static members, method overloading, constructor overloading, Inner Class and Anonymous Class, Object Lifetime and Garbage Collection.

Extending Classes and Inheritance: Use and Benefits of Inheritance in OOP, Types of Inheritance in Java, Role of Constructors in inheritance, Overriding Super Class Methods, use of "super", constructor chaining, Polymorphism in inheritance, dynamic method dispatch, final classes, methods, and variables, Abstract classes and abstract methods; Interfaces: defining contracts, multiple interface implementation, default methods, static methods, functional interfaces, Packages and import statements; Package as access protection.

Exception Handling: Throwable, Error, Exception, RuntimeException, try-catch-finally blocks; multi-catch; Checked vs unchecked exceptions; Creating custom exception classes.

Array & Strings: Defining an Array, Initializing & Accessing Array, Multi –Dimensional Array, Operation on String, Mutable & Immutable String, Using Collection Bases Loop for String, Tokenizing a String, Creating Strings using String Buffer.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							1	3	1
CO2	3	3	3	2	2							1	3	2
CO3	3	3	3	2	2							2	3	2
CO4	3	2	2	2	2			1				1	3	2
CO5	3	3	2	2	2							1	3	2

Textbooks / References

1. Introduction to Java Programming and Data Structures (12th Ed.), Y. Daniel Liang, Pearson
2. Thinking in Java (4th Ed.), Bruce Eckel, Prentice Hall
3. Core Java Volume I – Fundamentals (12th Ed.), Cay S. Horstmann, Oracle Press / Pearson
4. Head First Java (3rd Ed.), Kathy Sierra & Bert Bates, O'Reilly Media
5. Clean Code: A Handbook of Agile Software Craftsmanship, Robert C. Martin, Prentice Hall
6. C. Thomas Wu, An Introduction to Object Oriented Programming with Java, Fifth Edition, 2009.
7. Ken Arnold, James Gosling and David Holmes, The Java Programming Language, Fourth Edition, 2005.
8. Herbert Schildt, Java: The Complete Reference, McGraw Hill Education(India), 11th Edition 2018. 4. 1. Steven Holzner, PHP: The Complete Reference, McGraw Hill Education, 2017.

CSL121 Data Structures lab (C, C++) [0-0-2-1]

Course Objectives

- Develop proficiency in implementing fundamental data structures and algorithms.
- Analyze the time and space complexity of data structure operations.
- Apply appropriate data structures to solve computational problems efficiently.
- Gain practical experience with tree, graph, sorting, searching, and hashing techniques.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Implement and evaluate linear and non-linear data structures.
- CO2: Analyze the performance of data structure operations and associated algorithms.
- CO3: Apply trees, graphs, hashing, sorting, and searching techniques to solve computational problems.
- CO4: Design efficient solutions using suitable data structures and justify their selection.

Syllabus

Linear Data Structures: Implementation and analysis of Lists, Sorted Lists, Singly Linked Lists, Doubly Linked Lists, Circular Linked Lists, Dynamic Arrays, and Skip Lists. Applications of lists in Browser History Management and Text Editors.

Stacks and Queues: Implementation using arrays and linked lists. Applications including Expression Evaluation, Celebrity Problem, Largest Rectangular Area in a Histogram, and Queue-based Scheduling Problems.

Trees: Binary Tree Operations, Insertion and Deletion of Nodes, Tree Traversals, Polish Notations, AVL Trees, Heaps, Priority Queues, Optimal Binary Search Trees, and Application Problems on Optimal Binary Search Trees.

Sorting and Searching: Implementation and performance analysis of Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort, Radix Sort, Heap Sort, Linear Search, and Binary Search.

Hashing: Hash Functions, Collision Resolution Techniques, and Application Problems on Hashing.

Graphs: Graph Representations, Types and Properties of Graphs, Breadth First Search (BFS), Depth First Search (DFS), Shortest Path Algorithms, and Minimum Spanning Tree Algorithms.

Mini Project / Case Study (optional): Design, implementation, analysis, and presentation of a data structure-based solution to a real-world problem.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	2	3							1	3	2
CO2	3	3	2	3	2							2	3	2
CO3	3	3	3	2	3							2	3	2
CO4	3	3	3	3	2							2	3	2

Textbooks / References

1. Clifford A Shaffer, Data Structures and Algorithm Analysis, Edition 3.2 (Java Version), 2011.
2. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser. Data Structures And Algorithms In Java TM Sixth Edition, Wiley Publishers, 2014.
3. Mark Allen Weiss Data Structures And Algorithm Analysis In Java, Third Edition, 2012.
4. Robert L. Kruse, Data Structures And Program Design In C++, Pearson Education, Second Edition, 2006.
5. Ellis Horowitz, Fundamentals of Data Structures in C++, University Press, 2015.
6. Ajay Agarwal, Data Structure through C, A Complete Reference Guide, Cyber Tech Publications, 2005.

CSL123 Object Oriented Programming Lab [0-0-2-1]

Course Description

This laboratory course provides hands-on experience in developing Java applications using object-oriented programming principles. Students gain practical knowledge of the Java programming environment and learn to design and implement programs using classes, objects, constructors, and access control mechanisms. The course emphasizes software development through inheritance, polymorphism, abstraction, interfaces, packages, and exception handling. Students also develop applications involving arrays, strings, and object-oriented design principles to solve real-world problems.

Course Objectives

- To enable students to write, compile, and execute Java programs using the Java programming environment.
- To familiarize students with object-oriented programming concepts including classes, objects, constructors, and access control mechanisms.
- To develop programming skills using inheritance, polymorphism, abstraction, interfaces, and packages for modular software development.
- To introduce exception handling techniques for developing robust Java applications.
- To enable students to implement object-oriented applications using arrays, strings, and Java class libraries.

Course Outcomes

- CO1:Students learn to design and implement Java programs using classes, objects, constructors, and object-oriented programming principles.
- CO2: Students learn to develop reusable and extensible applications using inheritance, polymorphism, abstraction, interfaces, and packages.
- CO3: Students learn to develop robust Java applications using exception handling, arrays, strings, and object-oriented design principles for solving real-world problems.

Syllabus

Introduction to Java Programming: Writing, compiling, and executing Java programs. Basic language constructs, data types, operators, input/output, and program structure. Classes and Objects: Implementation of programs using classes, objects, constructors, access modifiers, encapsulation, static members, method overloading, constructor overloading, inner classes, and anonymous classes.

Inheritance and Polymorphism: Implementation of programs using inheritance, constructor chaining, method overriding, dynamic method dispatch, abstract classes, interfaces, and packages.

Exception Handling: Implementation of programs using exception handling mechanisms including try-catch-finally blocks, multiple catch blocks, checked and unchecked exceptions, and user-defined exceptions.

Arrays and Strings: Implementation of programs using one-dimensional and multidimensional arrays, string manipulation, String, StringBuffer, StringBuilder, tokenizing strings.

Object-Oriented Applications: Development of Java applications integrating object-oriented concepts, exception handling, arrays, strings, and packages to solve real-world problems.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3					1		2	3	3
CO2	3	3	3	2	3			1	2	1	2	2	3	3
CO3	3	3	3	3	3		1	2	2	2	2	3	3	3

Textbooks / References

1. Introduction to Java Programming and Data Structures (12th Ed.), Y. Daniel Liang, Pearson
2. Thinking in Java (4th Ed.), Bruce Eckel, Prentice Hall
3. David A. Bell,
4. Core Java Volume I – Fundamentals (12th Ed.), Cay S. Horstmann, Oracle Press / Pearson
5. C. Thomas Wu, An Introduction to Object Oriented Programming with Java, Fifth Edition, 2009.
6. Ken Arnold, James Gosling and David Holmes, The Java Programming Language, Fourth Edition, 2005.

MAC211 Probability, Statistics and Random Processes [3-1-0-4]

Course Objectives

- To expose the students to the modern theory of probability, concept of random variables and their expectations.
- To introduce various discrete and continuous distributions and concept of estimation theory, confidence interval.
- To develop the ability to perform statistical inference and data analysis using estimation, confidence intervals, and hypothesis testing techniques.
- To illustrate the concept of hypothesis testing, tests for means and variances, Goodness-of-fit tests.
- To introduce the concept of random processes, Markov chains, and Brownian Motion.
- To introduce stochastic modeling techniques and their applications in data-driven systems.

Course Outcomes

- CO1: Define and apply the concepts of probability and conditional probability.
- CO2: Define and illustrate discrete and continuous random variables, their probability mass functions and probability density functions.
- CO3: Understand the concepts and significance of estimation theory and hypothesis testing.
- CO4: Apply estimation and hypothesis testing techniques to construct confidence intervals for means and variances and conduct goodness-of-fit tests for statistical inference and data-driven decision making.
- CO5: Analyze random processes, Markov chains, and Brownian motion and explain their applications in stochastic modelling and computing systems.

Syllabus

Basics of Probability: Axiomatic construction of the theory of probability, independence, conditional probability, and basic formulas.

Random Variables and Distributions: Univariate, Bivariate and multivariate random variables, cumulative and marginal distribution functions, conditional and multivariate distributions, functions of random variables: sum, product, ratio, change of variables, mathematical expectations, moments, moment generating function, characteristic functions

Special Probability Distributions: discrete and continuous distributions, Limit theorems, Binomial distribution, Geometric distribution, Poisson distribution, Normal distribution, Exponential distribution, Gamma distribution, Beta distribution, Central Limit Theorem, Tchebeychev's inequality, Law of Large Numbers.

Estimation Theory: Bias of estimates, confidence intervals, minimum variance unbiased estimation, Bayes' estimators, moment estimators, maximum likelihood estimators, Chi-square distribution, confidence intervals for parameters of normal distribution.

Hypothesis Testing: Tests for means and variances, hypothesis testing and confidence intervals, Bayes' decision rules, power of tests, Goodness-of-fit tests, Kolmogorov-Smirnov Goodness-of-fit test.

Random Processes: Definition and classification of random processes, discrete-time Markov chains, Poisson process, continuous-time Markov chains, stationary processes, Gaussian process, Brownian motion.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2		1	1							1	2	1
CO2	3	3	1	2	2							1	2	2
CO3	3	3	2	2	2							1	3	2
CO4	3	3	2	3	2							2	3	3
CO5	3	3	2	3	2	1						2	3	3

Textbooks / References

1. S. Ross, *Introduction to Probability and Statistics for Engineers and Scientists*, 3rd ed., Elsevier, 2004.
2. P. G. Hoel, S. C. Port, and C. J. Stone, *Introduction to Probability Theory*, Universal Book Stall, 2000.
3. S. M. Ross, *Introductory Statistics*, 2nd ed., Academic Press, 2009.
4. J. Medhi, *Stochastic Processes*, 3rd ed., New Age International, 2009.
5. V. K. Rohatgi and A. K. Saleh, *An Introduction to Probability and Statistics*, 3rd ed., Wiley Student Edition, 2006.
6. G. R. Grimmett and D. R. Stirzaker, *Probability and Random Processes*, Oxford Univ. Press, 2001.
7. W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1, 3rd ed., Wiley, 1968.
8. S. M. Ross, *Stochastic Processes*, 2nd ed., Wiley, 1996.
9. C. M. Grinstead and J. L. Snell, *Introduction to Probability*, 2nd ed., Universities Press India, 2009.
10. S. Ross, *A First Course in Probability*, 10th ed., Pearson Education, Delhi, 2018.

11. K. P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022.
12. G. Casella and R. L. Berger, *Statistical Inference*, Chapman and Hall/CRC, 2024.
13. S. Ross, *Introduction to Probability Models*, 13th ed., Elsevier, 2023.
14. R. V. Hogg, J. W. McKean, and A. T. Craig, *Introduction to Mathematical Statistics*, 8th ed., Pearson, 2019.

HMC211 Engineering Management, Entrepreneurship Development and start ups [2-0-0-2]

Course Objectives

- To introduce the principles and practices of engineering management and enable students to understand managerial functions in technology-oriented organizations.
- To provide students with tools and techniques for planning, organizing, leading, and controlling engineering projects and resources.
- To develop an entrepreneurial mindset by exposing students to innovation, opportunity identification, startup ecosystems, and venture creation.
- To equip students with the knowledge required to evaluate business opportunities and prepare sustainable technology-based business models and ventures.

Course Outcomes

- CO1: Understand the functions, responsibilities, and practices of managers in engineering and technology organizations.
- CO2: Apply basic management, project management, and decision-making tools for effective management of engineering projects and teams.
- CO3: Analyze entrepreneurial opportunities, innovation processes, and startup ecosystems for technology-driven ventures.
- CO4: Develop a preliminary business plan incorporating market, financial, legal, and sustainability considerations.

Syllabus

Engineering Management: Meaning, definition, scope and significance of management in engineering organizations; Evolution of management thought; Contributions of F.W. Taylor, Henry Fayol, Elton Mayo and Behavioral School; Functions of management; Levels of management; Managerial roles and skills; Strategic thinking; Management lessons from Indian ethos and philosophy; Teamwork, productivity, effectiveness and efficiency.

Planning and Organizing: Planning process, objectives, policies, strategies and decision making; Management by Objectives (MBO); Organizing principles; Organizational structures and design; Departmentation; Centralization and decentralization; Delegation of authority; Span of control; Staffing and human resource management; Team building; Communication and coordination in engineering organizations.

Leadership, Operations and Project Management: Leadership concepts, leadership styles and motivation theories; Directing and controlling functions; Operations management fundamentals; Productivity and quality management; Lean concepts and continuous improvement; Engineering project management; Project life cycle; Work Breakdown Structure (WBS); PERT and CPM techniques; Resource allocation; Cost estimation; Risk management; Project monitoring and control.

Entrepreneurship and Venture Creation: Entrepreneurship concepts and characteristics; Entrepreneurial competencies; Creativity, innovation and design thinking; Opportunity identification and feasibility analysis; Startup ecosystem in India; Technology entrepreneurship; Incubation and acceleration; Business Model Canvas; Customer discovery and value proposition; Marketing fundamentals for startups; Sources of finance including bootstrapping, angel investment, venture capital and crowdfunding; Intellectual Property Rights (IPR); Business planning; Sustainable and social entrepreneurship; Case studies of successful technology ventures.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2				2	2	2	2	2	1	1		
CO2	2	3	2	2	2	2	2	2	3	2	2	1	1	1
CO3	1	2	3	2	2	2	3	2	3	3	2	2	1	1
CO4	1	2	3	2	2	2	3	3	3	3	2	2	1	1

Textbooks / References

1. Engineering Management, Addison-Wesley, 1998.
2. Essentials of Management, Tata McGraw-Hill, 1998.
3. Management, Pearson Education, 6th Edition, 2004.
4. Management, Pearson Education, 15th Edition, 2022.
5. Principles of Management, Tata McGraw-Hill, 2017.
6. Fundamentals of Business Organisation and Management, Sultan Chand & Co., 2019.
7. Entrepreneurship, McGraw-Hill Education, 2023.
8. Entrepreneurship Development, S. Chand Publishing, 2022.
9. Business Model Generation, Wiley, 2010.
10. Technology Ventures: From Idea to Enterprise, McGraw-Hill Education, 2020.
11. Innovation and Entrepreneurship, Harper Business, 2006.
12. The Lean Startup, Crown Business, 2011.

CSC211 Computer Networks [2-1-0-3]

Course Objectives

- To understand the principles, architecture, and operation of modern computer networks.
- To study networking protocols and services across different layers of the TCP/IP model.

- To understand routing, switching, and transport mechanisms in computer networks.
- To explore wireless networking technologies and emerging networking paradigms.
- To analyze network performance, reliability, and security mechanisms.
- To provide the foundation required for network design, administration, and advanced networking studies.

Course Outcomes

- CO1: Explain networking architectures, communication models, and protocol layering principles.
- CO2: Analyze data transmission, error control, flow control, and medium access mechanisms in communication networks.
- CO3: Apply Internet addressing, routing, and transport protocols for network communication and management.
- CO4: Explain the operation of common application-layer protocols and network services.
- CO5: Assess wireless networking technologies, network security threats, and emerging networking paradigms.

Syllabus

Introduction to Computer Networks: Evolution of computer networks, network components and network types, network topologies, layered network architecture, OSI reference model, TCP/IP protocol suite, Internet standards and RFCs.

Data Communication and Data Link Layer: Data transmission fundamentals, transmission media, encoding techniques, framing, error detection and correction, flow control, sliding window protocols, HDLC and PPP protocols, medium access control protocols, switching concepts.

Network Layer and Routing: Internet Protocol (IP), addressing and subnetting, ARP, ICMP, DHCP, CIDR and NAT, routing principles, distance vector and link state routing, RIP, OSPF and BGP, introduction to Software Defined Networking (SDN).

Transport and Application Layers: UDP and TCP, reliable data transfer, flow control, congestion control, Quality of Service (QoS), DNS, HTTP and HTTPS, electronic mail protocols, multimedia networking basics.

Wireless, Network Security and Emerging Networks: Wireless LAN and Wi-Fi standards, network security fundamentals, firewalls and intrusion detection concepts, Internet of Things (IoT) networking, edge computing, and future Internet architectures.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2			1					1			2	1
CO2	3	3	2	2	2					1			3	2
CO3	3	3	3	2	3				1	1			3	3
CO4	2	2	2	1	2				1	2			2	2
CO5	2	3	2	2	3	2	2	1		1	1	2	3	3

Textbooks / References

1. Kurose, J. F. and Ross, K. W., *Computer Networking: A Top-Down Approach*, 8th Edition, Pearson, 2021.
2. Tanenbaum, A. S. and Wetherall, D. J., *Computer Networks*, 6th Edition, Pearson, 2021.
3. Peterson, L. L. and Davie, B. S., *Computer Networks: A Systems Approach*, 6th Edition, Morgan Kaufmann, 2021.
4. Forouzan, B. A., *Data Communications and Networking*, 5th Edition, McGraw-Hill Education, 2013.
5. Comer, D. E., *Internetworking with TCP/IP: Principles, Protocols, and Architecture*, 6th Edition, Pearson, 2013.

CSC212 Design and Analysis of Algorithms [2-1-0-3]

Course Objectives

- Understand the principles of algorithm design and analysis.
- Analyze the correctness and efficiency of algorithms using appropriate mathematical techniques.
- Understand fundamental algorithmic paradigms such as divide-and-conquer, greedy methods, and dynamic programming to solve computational problems.
- Develop solutions using graph algorithms.
- Understand computational complexity, NP-completeness, and approaches for solving intractable problems.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Analyze the time and space complexity of algorithms using asymptotic and amortized analysis techniques.
- CO2: Design and implement algorithms using divide-and-conquer, greedy, and dynamic programming paradigms.
- CO3: Apply graph algorithms to solve real-world computational problems.
- CO4: Classify computational problems based on complexity classes and explain concepts of NP-completeness and approximation algorithms.

Syllabus

Asymptotic Analysis of algorithms: Asymptotic notations, Insertion Sort- Proof of Correctness and Complexity.

Amortized Analysis of algorithms: Aggregate Method, Accounting Method, Potential Method, Dynamic Tables–Balanced Trees.

Graph Algorithms: Topological Sorting, Strongly Connected Components, Minimum Spanning Tree, Shortest Path.

Introduction to algorithm design paradigms: Branch and Bound, Backtracking.

Divide and Conquer: Analysis by Recurrence Relations Methods for solving recurrence: Recursion Tree, Substitution Method, Master Theorem. Applications of the Master Theorem.

Greedy Algorithms: Optimization problems, optimal substructures, greedy choice, Applications of advanced data structures - Union-Find, Fibonacci heap.

Dynamic Programming: Reusing work across sub computations, optimal substructures, overlapping subproblems, ordering of subproblems and memoization.

Intractable Problems: Polynomial Time– class P, Polynomial Time Verifiable Algorithms– class NP, NP completeness and reducibility, NP Hard Problems, NP completeness proofs.

Introduction to Approximation Algorithms.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2			1					1			2	1
CO2	3	3	2	2	2					1			3	2
CO3	3	3	3	2	3				1	1			3	3
CO4	2	2	2	1	2				1	2			2	2
CO5	2	3	2	2	3	2	2	1		1	1	2	3	3

Textbooks / References

1. Introduction to Algorithms, T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, 3rd ed. Cambridge, MA: MIT Press, 2009.
2. Algorithms, 1st Edition, Sanjoy Dasgupta, Christos Papadimitriou, Umesh Vazirani, 2024.
3. The Design and Analysis of Algorithms, Dexter C. Kozen, Monographs in Computer Science, Springer New York, NY, 2011.
4. Algorithm Design, Kleinberg, J., & Tardos, E., Pearson/ Addison-Wesley, 2006.

CSC213 Theory of Computation [3-1-0-4]

Course Objectives

The primary objectives of this course are to enable students to:

- Understand the abstract mathematical models of computation and how they relate to modern computer hardware and software design (such as compilers).
- Analyze and Design various types of automata, including Deterministic Finite Automata (DFA), Nondeterministic Finite Automata (NFA), Pushdown Automata (PDA), and Turing Machines (TM) for specified languages.
- Master the Relationship between formal languages, grammars, and abstract machines across the Chomsky hierarchy.
- Apply Mathematical Tools, such as the Pumping Lemma, to rigorously prove the limitations of specific computational models.
- Distinguish between decidable and undecidable problems, and gain a foundational understanding of computational complexity classes.

Course Outcomes

By the end of this course, learners will be able to:

- CO1: Design, minimize, and interconvert Finite State Machines, regular expressions, and regular grammars.
- CO2: Apply formal mathematical tools, such as the Pumping Lemma, to prove the limitations of regular and context-free languages.
- CO3: Construct, simplify, and transform Context-Free Grammars and design equivalent Pushdown Automata models.
- CO4: Synthesize standard and variant Turing Machines to compute functions and classify complex languages within the Chomsky Hierarchy.
- CO5: Evaluate the decidability of computational problems using reductions and analyze foundational principles of Complexity Theory.

Syllabus

Introduction: Notion of formal language- Strings, Alphabet, Language, Operations, Finite State Machine, definitions, finite automaton model, acceptance of strings and languages.

Finite Automata – deterministic and nondeterministic, Regular operations, Regular Expression, equivalence between DFA, NFA and REs, Interconversions of DFA, NFA and REs, minimization of FSM, Non regular languages and pumping lemma, Moore and Mealy machines. Regular grammars, right linear and left linear grammars equivalence between regular linear grammar and FA, inter conversion between RE and RG.

Context free grammars: Derivation, parse trees. Language generated by a CFG. Eliminating useless symbols, productions, and unit productions, Chomsky Normal Form, Non CFLs and pumping lemma for CFLs.

Pushdown automata: Definition, instantaneous description as a snapshot of PDA computation, notion of acceptance for PDAs, Equivalence of PDA and CFG, Properties of CFLs, DCFLs.

Turing machine: definition, model, design of TM, Computable Functions, recursive enumerable language, Church's Hypothesis, Counter machine, types of TM's, RAM machine, Configuration graph, closure properties of decidable languages, decidability properties of regular languages and CFLs.

Undecidability and classes of problems: Reductions, Chomsky hierarchy of languages, Rice's Theorem, Universal Turing Machine, un-decidability of post's correspondence problem, introduction to complexity theory.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	1	2					1			3	2
CO2	3	3	2	2	1					1			3	2
CO3	3	3	3	2	2					1			3	3
CO4	3	3	3	2	2					1	1		3	3
CO5	3	3	2	3	2					2	1		3	3

Textbooks / References

1. MSipser, Introduction to the Theory of Computation, Second Edition, Thomson, 2005.
2. Lewis H.P. and Papadimition C.H. Elements of Theory of Computation, Prentice Hall of India, Fourth Edition, 2007.
3. S. Arora and B. Barak, Computational Complexity: A Modern Approach, Cambridge University Press, 2009.
4. C. H. Papadimitriou, Computational Complexity, Addison-Wesley Publishing Company, 1994.
5. D. C. Kozen, Theory of Computation, Springer, 2006.
6. D. S. Garey and G. Johnson, Computers and Intractability: A Guide to the Theory of NP- Completeness, Freeman, New York, 1979.
7. J Hopcroft, JD Ullman and R Motwani, Introduction to Automata Theory, Languages and Computation, Third Edition, Pearson, 2008.

CSC214 Operating Systems [3-0-0-3]

Course Objectives

- To understand the architecture, structures, and services of operating systems.
- To study process management, thread management, CPU scheduling, synchronization, and deadlock handling techniques.
- To understand memory management and virtual memory mechanisms.
- To analyze file systems, disk management, and I/O management techniques.
- To introduce Linux-based systems, multicore systems, and virtualization concepts.
- To develop analytical and problem-solving skills related to operating system resource management and performance.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Explain the structure, services, and operating modes of modern operating systems.
- CO2: Analyze process management, thread management, CPU scheduling, and inter-process communication mechanisms.
- CO3: Apply synchronization and deadlock handling techniques in concurrent systems.
- CO4: Evaluate memory management and virtual memory management strategies used in operating systems.
- CO5: Analyze file systems, disk management, I/O management, and resource management techniques.
- CO6: Understand Linux-based operating system environments, virtualization concepts, containers, and contemporary operating system technologies.

Syllabus

Introduction and Operating System Structures: Evolution and functions of operating systems, Operating system services, system calls, and command interpreter (shell). Modes: User mode and kernel mode. Operating system structures: Monolithic, Layered, Microkernel, and Hybrid architectures. Brief introduction to Linux/xv6 operating systems.

Processes, Threads, and CPU Scheduling: Processes and Process Control Block (PCB), Process states and context switching, Threads and multithreading models, and Inter-process communication (IPC). CPU Scheduling: Scheduling Criteria, Scheduling Algorithms, Multiple-Processor Scheduling, Algorithm Evaluation, and Process Scheduling Models. Introduction to Linux scheduling.

Process Synchronization and Deadlocks: Race conditions and critical section problem, Hardware support for synchronization. Mutexes, semaphores, monitors, condition variables, and classical synchronization problems. **Deadlocks:** System Model, Characterization, Prevention. Avoidance, Detection, and Recovery.

Memory Management: Memory Management Background, Swapping, Contiguous and non-contiguous memory allocation, Paging and segmentation, Virtual memory concepts, and Demand paging. Page Replacement Algorithms, Allocation of Frames, Thrashing, and Overview of Memory management in Linux/Windows systems.

File Systems and I/O Management: File concepts and access methods, directory structures, file sharing, and protection mechanisms. File-system structure and implementation, file allocation methods, and free-space management.

Disk structure and disk scheduling concepts: buffering and caching; RAID concepts; application I/O interface; kernel I/O subsystem; and an overview of journaling file systems.

Modern Operating System Concepts Introduction to virtualization, Basics of containers, Protection and security mechanisms, and Overview of mobile and real-time operating systems.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2			2					1			2	1
CO2	3	3	2	2	2				1	1			3	2
CO3	3	3	3	2	2				1	1			3	3
CO4	3	3	3	2	2					1			3	3
CO5	3	3	2	2	3				1	1			3	3
CO6	2	2	2	2	3	1	1			2	1	1	3	3

Textbooks / References

1. Abraham Silberschatz, Peter B. Galvin, Greg Gagne, Operating System Concepts, Wiley.
2. Andrew S. Tanenbaum and Herbert Bos, Modern Operating Systems, Pearson.
3. Robert Love, Linux System Programming, O'Reilly.
4. W. Richard Stevens and Stephen A. Rago, Advanced Programming in the UNIX Environment, Addison-Wesley.
5. Thomas Anderson and Michael Dahlin, Operating Systems: Principles and Practice.

CSL211 Computer Networks Lab [0-0-2-1]

Course Objectives

- To provide practical exposure to networking protocols, services, and communication mechanisms.
- To develop skills in network programming, protocol analysis, and network simulation.
- To enable students to analyze, configure, and troubleshoot networked systems using modern networking tools.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Use networking tools to analyze network traffic and protocol behavior.
- CO2: Develop client-server applications using socket programming.
- CO3: Simulate and evaluate networking protocols using network simulators.
- CO4: Configure, troubleshoot, and evaluate network communication scenarios.

Syllabus

Network Programming

1. Introduction to Linux Networking Commands
2. Socket Programming – TCP Client/Server
3. Socket Programming – UDP Client/Server
4. Multi-client Communication Applications
5. Remote Procedure Call (RPC) Concepts
6. Implementation of Routing Algorithms

Network Analysis

7. Packet Capture and Analysis using Wireshark
8. Measurement of Network Performance Metrics

Network Simulation

10. Introduction to NS3 Simulator
11. Point-to-Point Network Simulation
12. Routing Protocol Simulation
13. Mini Project on Network Design and Performance Evaluation

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3		3	3			1		1			3	2
CO2	3	3	3	2	3			1	1	1			3	3
CO3	2	3	3	3	3			1	1	1			3	3
CO4	3	3	3	3	3	1	1	2	1	2		1	3	3

Textbooks / References

1. Panwar, S., Mao, S., Ryoo, J.-D., and Li, Y., *Computer Networking Fundamentals*, Cambridge University Press, 2024.
2. Stevens, W. R., *UNIX Network Programming*.
3. Wireshark Documentation.
4. NS-3 Documentation.

CSL212 Algorithm Design Lab [0-0-2-1]

Course Objectives

- Analyze and compare the performance of algorithms through experimentation.
- Apply algorithm design paradigms to solve computational problems.
- Gain experience with graph algorithms.
- Strengthen problem-solving and programming skills through algorithmic projects and case studies.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Implement and evaluate the performance of fundamental algorithms and data structures.
- CO2: Apply divide-and-conquer, greedy, dynamic programming, and graph-based techniques to solve computational problems.
- CO3: Design efficient algorithmic solutions and analyze their correctness and complexity.

Syllabus

Implementation and Analysis of Basic Algorithms: Insertion Sort, Merge Sort, Binary Search,

Asymptotic and Recurrence Analysis: Experimental verification of asymptotic growth, Solving recurrence relations and complexity comparison

Graph Algorithms like Topological Sorting, Strongly Connected Components, Minimum Spanning Tree (Prim's and Kruskal's Algorithms), Shortest Path Algorithms

Greedy Algorithms like Interval Scheduling, Fractional Knapsack, Huffman Coding, Minimum Spanning Tree Applications

Dynamic Programming algorithms like Rod Cutting Problem, Matrix Chain Multiplication, Longest Common Subsequence, Machine Scheduling Problems, Bellman-Ford and Floyd-Warshall Algorithms

Complexity and Computational Problems: Problem classification (P, NP, NP-Complete, NP-Hard), Simple reductions and complexity analysis

Approximation: Implementation of basic approximation algorithms Mini Project / Case Study (optional): Design, implementation, analysis, and presentation of an algorithmic solution to a real-world problem.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3			1		1			3	2
CO2	3	3	3	2	3			1	1	1			3	3
CO3	3	3	3	3	2			2	1	2			3	3

Textbooks / References

1. The algorithm design manual, Skiena, S. S. (2nd ed.). Springer (2008).
2. Algorithms, 1st Edition, Sanjoy Dasgupta, Christos Papadimitriou, Umesh Vazirani, 2024.

3. Algorithm Design, Kleinberg, J., & Tardos, E., Pearson/ Addison-Wesley, 2006.

CSL214 Operating Systems Lab [0-0-2-1]

Course Objectives

- To provide practical understanding of operating system concepts.
- To develop shell scripting and Linux system programming skills.
- To implement process management, thread synchronization, and inter-process communication techniques.
- To simulate CPU scheduling, deadlock handling, and memory management algorithms.
- To familiarize students with Linux utilities for file handling, disk management, and process/resource monitoring.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Use Linux commands and shell scripting for system operations.
- CO2:Develop programs using process creation and Inter Process Communication mechanisms.
- CO3:Implement synchronization techniques using mutexes and semaphores.
- CO4: Simulate CPU scheduling, deadlock handling, and memory management algorithms.
- CO5: Perform file handling, directory operations, and disk scheduling in Linux environments.
- CO6: Analyze process behavior and operating system resource utilization using Linux monitoring tools.

Syllabus

1. Familiarization with Linux environment and shell commands.
2. Shell scripting: File handling, Pattern matching, and Process automation.
3. Process creation using fork (), exec (), and wait ().
4. Inter-process communication using Pipes, Shared memory, and Message queues.
5. Simulation of CPU scheduling algorithms, including FCFS, SJF, Priority Scheduling, and Round Robin.
6. Thread creation using POSIX threads
7. Synchronization using Mutexes and Semaphores
8. Implementation of Producer–Consumer and Reader–Writer problems using synchronization primitives.
9. Deadlock avoidance using Banker's Algorithm.
10. Simulation of contiguous and non-contiguous memory allocation, and Page replacement algorithms: FIFO, LRU, and Optimal.
11. File handling, process monitoring, and directory operations in Linux,
12. Disk scheduling simulations.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	1		3							2	2	1
CO2	3	3	2	1	3							2	3	2
CO3	3	3	3	2	2				2			2	3	2
CO4	3	3	3	2	2				1			2	3	2
CO5	2	2	2	1	3							2	3	2
CO6	2	3	1	2	3				1			3	2	2

Textbooks / References

1. Abraham Silberschatz, Peter B. Galvin, and Greg Gagne, Operating System Concepts, Wiley.
2. Robert Love, Linux System Programming, O'Reilly.
3. Andrew S. Tanenbaum and Herbert Bos, Modern Operating Systems, Pearson.
4. W. Richard Stevens and Stephen A. Rago, Advanced Programming in the UNIX Environment, Addison-Wesley.
5. Michael Kerrisk, The Linux Programming Interface, No Starch Press.

MACXXX Computational Linear Algebra and Optimization Techniques [3-1-0-4]

Course Objectives

- To develop a strong foundation in modern numerical linear algebra, including matrix norms, conditioning, matrix factorizations, and singular value decomposition.
- To introduce spectral methods, eigenvalue problems, graph linear algebra, and Perron–Frobenius theory for analyzing networks and complex systems.
- To provide mathematical foundations of optimization and machine learning, including gradient-based, stochastic, and second-order optimization methods.
- To develop the ability to formulate and solve regression, classification, clustering, and regularized optimization problems arising in data science and machine learning.
- To understand computational graphs and automatic differentiation as essential tools for modern machine learning algorithms.
- To emphasize computational efficiency, numerical stability, and scalable algorithms for large-scale scientific computing and data-driven applications.

Course Outcomes

- CO1: Apply computational linear algebra techniques including matrix norms, QR decomposition, SVD, Fourier transforms, and least-squares methods to solve computational and data-driven problems.
- CO2: Analyze eigenvalue problems and graph structures using spectral methods, iterative eigenvalue algorithms, and Perron–Frobenius theory for network analysis and ranking applications.
- CO3: Formulate and solve optimization problems using gradient-based, stochastic, coordinate descent, and second-order optimization techniques.

- CO4: Employ matrix calculus, regularization methods, and advanced optimization algorithms for developing efficient machine learning models.
- CO5: Integrate linear algebra, optimization, and computational graph concepts to design and implement machine learning algorithms for regression, classification, clustering, and neural networks.

Syllabus

Norms, Matrix Factorizations and Numerical Linear Algebra

Vector and matrix norms, conditioning of linear systems, sensitivity analysis, condition numbers, geometric interpretation, orthogonality and orthogonal projections, QR decomposition, Gram–Schmidt orthogonalization, Householder transformations, applications of QR factorization, Singular Value Decomposition (SVD), low-rank approximations, dimensionality reduction, complex vector spaces, Discrete Fourier Transform (DFT), matrix representation of DFT, least-squares problems, normal equations, QR- and SVD-based least-squares solutions, numerical stability, applications in regression and data fitting.

Eigenvalue Problems, Spectral Methods and Graph Linear Algebra

Eigenvalues and eigenvectors, characteristic polynomial, algebraic and geometric multiplicities, diagonalization, spectral decomposition, iterative eigenvalue algorithms including Power Method, Inverse Power Method, Shifted Power Method and Rayleigh Quotient Iteration, Hermitian eigenvalue problems, spectral theorem, graph matrices, adjacency matrix, degree matrix, graph Laplacian, spectral graph theory, Perron–Frobenius theorem, positive and non-negative matrices, network analysis, ranking algorithms, graph centrality measures, spectral clustering and applications in machine learning.

Optimization and Machine Learning Foundations

Optimization problems and objective functions, optimization landscapes, gradient descent, convex optimization, gradient computation and verification, learning-rate selection, line search techniques, batch and stochastic optimization, Stochastic Gradient Descent (SGD), matrix calculus, derivatives of vectors and matrices, quadratic optimization, linear regression, ridge regularization, pseudoinverse, support vector machines, logistic regression, softmax regression, coordinate descent, block coordinate descent and K-means clustering.

Advanced Optimization and Computational Graphs for Machine Learning

Optimization challenges in machine learning, local optima, saddle points, ill-conditioning, momentum methods, adaptive optimization methods including AdaGrad, RMSProp and Adam, Newton's method, Hessian matrices, quasi-Newton methods, BFGS and L-BFGS algorithms, conjugate gradient methods, Krylov subspace methods, non-smooth optimization, subgradient methods, computational graphs, directed acyclic graphs (DAGs), automatic differentiation, backpropagation, neural networks and efficient gradient computation.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3							2	3	2
CO2	3	3	2	3	2							2	3	2
CO3	3	3	3	2	3							2	3	3
CO4	3	3	3	2	3							2	3	3
CO5	3	3	3	3	3	1			1	1		3	3	3

Textbooks / References

1. Robert A. van de Geijn and Margaret Myers, *Advanced Linear Algebra: Foundations to Frontiers*, Open Textbook, 2020.
2. Charu C. Aggarwal, *Linear Algebra and Optimization for Machine Learning*, Springer, 2020.
3. Howard Anton and Chris Rorres, *Elementary Linear Algebra*, 9th Edition, John Wiley & Sons, 2000.
4. D. Poole, *Linear Algebra: A Modern Introduction*, 2nd Edition, Brooks/Cole, 2005.
5. Gilbert Strang, *Linear Algebra and Its Applications*, 3rd Edition, Harcourt College Publishers, 1988.
6. S. S. Rao, *Optimization Theory and Applications*, 2nd Edition, New Age International, 1995.
7. Edwin K. P. Chong and Stanislaw H. Zak, *An Introduction to Optimization*, 2nd Edition, Wiley-Interscience, 2004.
8. Kalyanmoy Deb, *Optimization for Engineering Design: Algorithms and Examples*, Prentice Hall of India, 2004.
9. M. Asghar Bhatti, *Practical Optimization Methods with Mathematics Applications*, Springer, 2000.

HMC221 Engineering Economics, Finance and Business Analytics [2-0-0-2]

Course Objectives

- To introduce the fundamental concepts of engineering economics and their application in engineering decision-making.
- To provide students with knowledge of financial management principles, financial markets, and investment evaluation techniques.
- To familiarize students with business analytics concepts, data-driven decision making, and analytical tools used in modern organizations.
- To develop the ability to evaluate economic, financial, and business alternatives for effective managerial and entrepreneurial decision making.

Course Outcomes

- CO1: Understand the principles of engineering economics, cost analysis, and economic decision-making relevant to engineering and technology sectors.
- CO2: Apply financial concepts and techniques for budgeting, investment appraisal, and financial planning.

- CO3: Analyze business data using descriptive, predictive, and prescriptive analytics approaches to support managerial decisions.
- CO4: Evaluate business opportunities and organizational performance using economic, financial, and analytical tools.

Syllabus

Engineering Economics: Nature and scope of engineering economics; Economic decision making in engineering; Demand, supply and market equilibrium; Cost concepts and cost analysis; Fixed and variable costs; Break-even analysis; Time value of money; Simple and compound interest; Cash flow diagrams; Economic comparison of alternatives; Present worth, annual worth and future worth methods; Depreciation concepts and methods; Inflation and its impact on economic decisions; Replacement and maintenance analysis.

Financial Management and Financial Markets: Introduction to finance and financial management; Objectives and functions of financial management; Financial statements and financial analysis; Ratio analysis; Working capital management; Capital budgeting; Sources of finance; Cost of capital; Risk and return concepts; Financial planning and budgeting; Overview of financial markets and institutions; Equity, debt and mutual funds; Basics of investment management and portfolio concepts; Venture capital and startup financing.

Business Analytics Fundamentals: Introduction to business analytics; Role of analytics in business decision making; Types of analytics—descriptive, diagnostic, predictive and prescriptive analytics; Data collection and data visualization; Measures of central tendency and dispersion; Probability concepts; Hypothesis Testing; Correlation and regression analysis; Forecasting techniques; Data-driven decision making; Introduction to spreadsheet-based analytics and business intelligence tools.

Applications of Analytics in Business and Engineering: Analytics for operations, marketing, finance and human resources; Key Performance Indicators (KPIs); Dashboard reporting; Decision trees and risk analysis; Data ethics and governance; Artificial Intelligence and analytics in business; Case studies in engineering economics, financial decision making and business analytics; Applications in startups, technology ventures and digital enterprises.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	2	1	1	2	1	1			3	2	1	
CO2	2	2	2	1	2	2	1	1			3	2	1	1
CO3	2	3	2	2	3	1	1	1		1	2	2	1	2
CO4	2	3	3	2	2	2	2	2	1	1	3	3	1	2

Textbooks / References

1. Engineering Economy, McGraw-Hill Education, 2018.
2. Engineering Economy, Pearson Education, 2020.
3. Principles of Engineering Economic Analysis, Wiley, 2012.
4. Fundamentals of Financial Management, Cengage Learning, 2021.
5. Financial Management, Vikas Publishing House, 2021.
6. Financial Management: Theory and Practice, McGraw-Hill Education, 2023.
7. Business Analytics, Pearson Education, 2020.
8. Business Analytics: Data Analysis and Decision Making, Cengage Learning, 2019.
9. Data Science for Business, O'Reilly Media, 2013.
10. Business Intelligence, Analytics, and Data Science, Pearson Education, 2018.
11. Business Analytics: The Science of Data-Driven Decision Making, Wiley India, 2017.
12. Managerial Economics, Oxford University Press, 2020.
13. Managerial Economics, McGraw-Hill Education, 2022.
14. Statistics for Business and Economics, Pearson Education, 2020.
15. Competing on Analytics, Harvard Business Review Press, 2017.

CAC221 Machine Learning and Data Analysis [2-1-0-3]

Course Objectives

- To introduce the mathematical and statistical foundations required for machine learning and data analysis.
- To develop skills in exploratory data analysis, preprocessing, and visualization techniques.
- To provide knowledge of supervised, unsupervised, and reinforcement learning methods.
- To enable students to design, implement, and evaluate machine learning models.
- To develop analytical and problem-solving skills for intelligent data-driven applications.

Course Outcomes

- CO1: Understand the mathematical and statistical foundations of machine learning.
- CO2: Perform exploratory data analysis and preprocessing on datasets.
- CO3: Apply supervised and unsupervised learning algorithms for predictive analytics.
- CO4: Evaluate and optimize machine learning models using appropriate metrics.
- CO5:Develop intelligent data-driven solutions for real-world applications.

Syllabus

Mathematical Foundations and Exploratory Data Analysis Overview of Linear Algebra, Overview of Probability and Statistics, Overview of Optimization Techniques.Exploratory Data Analysis (EDA): Data collection and preprocessing, Handling missing values, Data transformation, Outlier detection, Correlation analysis, Data visualization techniques.

Statistical Learning and Model Selection Introduction to Machine Learning: Definitions and goals of machine learning, History and applications of machine learning, Types of machine learning. Statistical Learning: Linear classification, Classification errors, Regression techniques, Predictive analytics concepts. Model Selection Techniques: Empirical Risk Minimization (ERM), Structural Risk Minimization (SRM), Cross-validation techniques, Regularization methods, Bias-variance tradeoff, Performance evaluation metrics.

Mining Frequent Patterns, Associations, and Correlations Association Rule Mining: Frequent itemset mining, Apriori algorithm, FP-growth concepts, Association rules, Support and confidence measures, Market basket analysis applications.

Supervised Learning Fundamentals: Supervised learning concepts, Generative and discriminative learning, Parametric and non-parametric learning. Regression Techniques: Linear regression, Model fitting and prediction. Classification Techniques: Decision trees, k-Nearest Neighbors (KNN), Naïve Bayes classifier with Laplace smoothing, Logistic regression with Maximum Likelihood Estimation Model Evaluation:Regression Metrics and Confusion matrix, Precision, Recall, F1-score, ROC curve.

Unsupervised Learning: Unsupervised learning concepts, Clustering techniques, K-Means clustering, Hierarchical clustering. Dimensionality Reduction: Principal Component Analysis (PCA), Feature selection and feature extraction techniques.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	2	2							2	3	2
CO2	3	3	2	2	3							2	3	3
CO3	3	3	3	2	3							2	3	3
CO4	3	3	2	3	3							2	3	3
CO5	3	3	3	3	3	1		1	1	1		3	3	3

Textbooks / References

1. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer.
2. Tom M. Mitchell, Machine Learning, McGraw-Hill.
3. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow, O'Reilly Media.
4. Gareth James et al., An Introduction to Statistical Learning, Springer.
5. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, Springer.
6. Gilbert Strang, Introduction to Linear Algebra, Wellesley-Cambridge Press.
7. 3. Stephen Boyd and Lieven Vandenberghe, Convex Optimization, Cambridge University Press.

CSC222 Language Processors [2-1-0-3]

Course Objectives

- To introduce the major concept areas in language processors and know the various types of Language processors
- To analyze lexical analysis techniques and parsing methods used in compiler construction.
- To understand the various parsing algorithms and comparison of the same.
- To gain knowledge about the various code generation principles.
- To understand the necessity for code optimization

Course Outcomes

- CO1:Apply the knowledge of LEX, YACC, ANTLR tools to develop a scanner and parser.
- CO2: Suggest the necessity for appropriate code optimization techniques.
- CO3: Design a Translator for any programming language
- CO4: Design compiler components and recommend suitable code optimization techniques for efficient program execution.
- CO5: Design a machine learning translators for real-time applications

Syllabus

Introduction to Language Processing: Translators, Assemblers, Compilers, and Interpreters. Phases of a Compiler, System Software, Linkers, Loaders, and macro processors. Scanners: Tokens, lexemes, and patterns, Input Buffering Strategies, Scanner generations tools like Lex or Flex or Antlr.

Parser: CFG, Derivations, parse trees, and ambiguity, Top-Down Parsing: Recursive Descent Parsing, Predictive Parsing, and LL(1) grammars, Bottom-Up Parsing: Shift-Reduce Parsing, Operator Precedence Parsing, Different LR Parsers construction, Parser Generators: Using tools Yacc or Bison.4, or Antlr.

Syntax-Directed Translation (SDT) Syntax-Directed Definitions: Inherited and synthesized attributes, S-Attributed and L-Attributed Definitions. Translation Schemes: Construction of syntax trees. Intermediate Code Generation: Intermediate Representations (IR), Declarations and Assignment Statements, Boolean Expressions and Case Statements.

Code Optimization: Sources of optimization, Optimization Techniques, and data-flow analysis. Code Generation and Runtime Environment Runtime Storage Management: Activation trees, stack allocation, and heap management. Target Machine Code: Instruction selection, register allocation, and basic block generation.

Understanding AI specific compiler Architectures: Machine learning compilers, Multi-Level Intermediate Representations, Quantum Compilers.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3					1		2	3	2
CO2	3	3	3	2	3			1		1		2	3	3
CO3	3	3	3	3	3					2	1	2	3	3
CO4	3	3	3	3	3			1		2	2	3	3	3
CO5	3	3	3	2	3	1		2	1	2	2	3	3	3

Textbooks / References

1. Mössenböck, Hanspeter. Compiler Construction: Fundamentals and Applications. Springer Nature, 2025.
2. Cooper, K. D., &; Torczon, L. (2022). Engineering a compiler (3rd ed.). Morgan Kaufmann.
3. Thain, Douglas. Introduction to compilers and language design. Lulu. com, 2016.
4. Aho A.V., Lam M. S., Sethi R., and Ullman J. D., Compilers: Principles, Techniques and Tools, Pearson Education, 2007.

Web References

1. Machine Learning Compilation, https://book.mlc.ai/
2. Bettina Heim, Mathias Soeken, Sarah Marshall, Chris Granade, Martin Roetteler, Alan Geller, Matthias Troyer & Krysta Svore. Quantum programming languages. Nat Rev Phys 2, 709–722 (2020). https://doi.org/10.1038/s42254-020-00245-7

CSC223 Database Management Systems [2-1-0-3]

Course Objectives

- To understand the fundamentals of database systems and data modeling techniques.
- To design and implement relational databases using SQL and advanced SQL features.
- To apply normalization techniques for efficient database design.
- To understand indexing, query processing, transaction management, and database security mechanisms.
- To introduce modern database technologies including NoSQL, cloud databases, and vector databases.

Course Outcomes

- CO1: Explain database concepts, architectures, and data modeling techniques.
- CO2: Design relational database schemas and write SQL queries for data management and retrieval.
- CO3: Apply normalization techniques to develop efficient database designs.
- CO4: Analyze indexing, query processing, transaction management, and security mechanisms in database systems.
- CO5: Identify and apply modern database technologies such as NoSQL, cloud databases, and vector databases for emerging applications.

Syllabus

Database Modelling: Database system concepts and architecture, Data models, Schemas and instances, Database system environment, Data modelling using Entity Relationship (ER) model, Enhanced ER model, Specialization, Generalization.

Relational Databases: Relational Model, Relational database design using ER to relational mapping, Constraints, Relational algebra, Relational calculus, SQL, Advanced SQL (Views, Stored Procedures, Triggers, CTEs, Window Functions), PL/SQL.

Database Design: Database design theory and methodology, Functional dependencies and normalization of relations, Normal Forms, Properties of relational decomposition, Algorithms for relational database schema design, Denormalization and scalable schema design.

Indexing and Query Processing in DBMS: Single level and multi-level indexing, Dynamic Multi-level Indexing using B Trees and B+ Trees, Hash-Based Indexing (Static and Dynamic Hashing), Query processing, Query evaluation plans, Query optimization.

Database Transactions and Security: Transaction processing concepts, ACID properties, Schedules and serializability, Concurrency control, Two Phase Locking Techniques, Deadlock, Database recovery concepts and techniques. Database Security: Introduction to database security, Authentication and authorization, Access control mechanisms, Privileges using GRANT and REVOKE.

Advanced Database Systems and Emerging Database Technologies: Big Data overview and NoSQL database concepts, NewSQL Databases, Cloud Databases, Vector Databases for AI Applications, Emerging trends in database technologies.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							2	3	2
CO2	3	3	3	2	3							2	3	2
CO3	3	3	3	2	2							2	3	2
CO4	3	3	2	3	3			1				2	3	2
CO5	3	3	3	2	3	1		1	1	1		3	3	3

Textbooks / References

1. R. Elmasri and S. B. Navathe, *Fundamentals of Database Systems*, 7th Edition, Pearson, 2022.
2. R. Ramakrishnan and J. Gehrke, *Database Management Systems*, 3rd Edition, McGraw-Hill Higher Education, 2003.
3. M. Kleppmann, *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*, 1st Edition, O'Reilly Media, 2017.

CSC224 Advanced Data Structures [2-1-0-3]

Course Objectives

The course aims to:

- Master Advanced Data Structures.
- Optimize Information Retrieval.
- Solve Network Connectivity.
- Design Probabilistic Solutions.

Course Outcomes

After successful completion of the course, students will be able to:

- CO1: Analyze and Implement Self-Balancing Search Structures: Demonstrate the ability to implement and evaluate balanced search trees to optimize disk access and dynamic data retrieval.
- CO2: Evaluate Advanced Heap Architectures: Compare the structural and amortized performance of Binomial and Fibonacci Heaps to optimize priority queue operations in network and graph algorithms.
- CO3: Design Efficient Pattern Matching and String Processing Systems: Apply advanced string algorithms (KMP, Rabin-Karp) and prefix/suffix structures (Tries, Suffix Trees/Arrays) to solve complex text-processing and biological data problems.
- CO4: Formulate Graph Connectivity Solutions: Deploy advanced graph algorithms to identify strongly connected components, detect cycles, and map topology in complex networks.
- CO5: Synthesize and Apply Probabilistic Algorithmic Strategies.

Syllabus

Self-Balancing Trees & Extensions but not limited to: Red-Black Trees, B-Trees & B+ Trees.

Advanced Heaps & Priority Queues but not limited to: Fibonacci Heaps , Binomial Heaps.

String and Prefix Processing but not limited to: Tries (Prefix Trees), Suffix Trees & Suffix Arrays, KMP Algorithm, Rabin Karp algorithm.

Algorithms on Graph connectivity but not limited to: Tarjan's algorithm and Kosaraju's algorithm for detection of strongly connected components, Detect cycles in an undirected graph.

Network Flow: Max Flow min cut theorem, Ford Fulkerson algorithm, Bipartite matching using flows.

Succinct data structures, Dynamic Graphs, Probabilistic data structures.

Introduction to Randomized Algorithms but not limited to: Las Vegas Algorithms -Quick Sort, Monte Carlo Algorithms- Miller-Rabin primality test.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3							2	3	2
CO2	3	3	2	2	3							2	3	2
CO3	3	3	3	2	3							2	3	3
CO4	3	3	3	3	3							2	3	3
CO5	3	3	3	3	3				1	1		3	3	3

Textbooks / References

1. Introduction to Algorithms (4th Edition) by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein.
2. Algorithms (4th Edition) by Robert Sedgewick and Kevin Wayne.
3. Advanced Data Structures by Peter Brass.
4. Randomized Algorithms by Rajeev Motwani and Prabhakar Raghavan.

CAL221 Machine Learning and Data Analysis Lab [0-0-2-1]

Course Overview

This laboratory course provides hands-on experience in Python programming, data preprocessing, exploratory data analysis, visualization, and machine learning techniques. Students implement regression, classification, clustering, association rule mining, and model evaluation methods using Python libraries to analyze real-world datasets and develop practical data-driven solutions.

Course Objectives

- Recall the core terms, concepts, and mathematical foundations used in data analysis and machine learning.
- Interpret and understand statistical methods, learning paradigms, and visualization principles.
- Apply and analyse data preprocessing, statistical, mining, and machine learning techniques on datasets.
- Synthesize and create a complete data analysis and machine learning workflow for a real problem.

Course Outcomes

- CO1: Identify the essential concepts, tools, and mathematical ideas used in data analysis and machine learning.
- CO2: Explain statistical procedures, learning methods, and visualization principles in practical contexts.
- CO3: Implement and evaluate preprocessing, prediction, clustering, mining, and visualization techniques on datasets.
- CO4: Design and develop an end-to-end analytical solution using machine learning and visualization tools.

Syllabus

1. **Introduction to Python Programming:** Installation, development environments, Python packages, and

scientific computing libraries for machine learning and data analysis.

2. **Mathematical Foundations using Python:** Implementation of matrix operations, eigenvalues and eigenvectors, gradients, and probability distributions.
3. **Data Preprocessing and Feature Engineering:** Data cleaning, transformation, normalization, feature selection, and dimensionality reduction techniques.
4. **Exploratory Data Analysis (EDA):** Computation of measures of central tendency and dispersion, and exploratory analysis using Python.
5. **Regression Analysis:** Implementation of simple and multiple linear regression models with performance evaluation and error analysis.
6. **Model Selection and Generalization:** Application of cross-validation, regularization techniques, and analysis of the bias–variance trade-off.
7. **Supervised Learning:** Implementation and evaluation of parametric and non-parametric classification algorithms such as Decision Trees, Naïve Bayes, and k-Nearest Neighbors (k-NN).
8. **Unsupervised Learning:** Implementation of clustering, dimensionality reduction, and outlier detection techniques.
9. **Association Rule Mining:** Implementation of frequent pattern mining algorithms such as Apriori and FP-Growth for association rule discovery.
10. **Data Visualization:** Generation of line plots, bar charts, scatter plots, histograms, box plots, heat maps, and treemaps using Matplotlib, Seaborn, and Plotly.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	1	1	2						2	3	2
CO2	3	2	2	2	3						2	3	2
CO3	3	3	3	3	3						2	3	3
CO4	3	3	3	3	3	1	1	1	1		3	3	3

Textbooks / References

1. Hastie, T., Tibshirani, R., and Friedman, J. H. The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd Edition, Springer Series in Statistics, Springer, New York, USA, 2009. ISBN: 978-0-387-84857-0 (Hardcover), ISBN: 978-0-387-84858-7 (eBook), 745 pages.
2. Bishop, C. M. Pattern Recognition and Machine Learning, Information Science and Statistics Series, Springer, New York, USA, 2006. ISBN: 978-0-387-31073-2 (Hardcover), ISBN: 978-0-387-45528-0 (eBook), 738 pages.
3. Han, J., Kamber, M., and Pei, J. Data Mining: Concepts and Techniques, 3rd Edition, Morgan Kaufmann (Elsevier), Waltham, MA, USA, 2011. ISBN: 978-0-12-381479-1, 744 pages.
4. Murphy, K. P. Machine Learning: A Probabilistic Perspective, Adaptive Computation and Machine Learning Series, The MIT Press, Cambridge, MA, USA, 2012. ISBN: 978-0-262-01802-9 (Hardcover), ISBN: 978-0-262-30432-0 (eBook), 1104 pages.
5. Géron, A. Hands-On Machine Learning with Scikit-Learn and PyTorch: Concepts, Tools, and Techniques to Build Intelligent Systems, 1st Edition, O'Reilly Media, Sebastopol, CA, USA, 2025. ISBN: 979-8341607989.

CSL222 Language Processors Lab [0-0-2-1]

Course Objectives

- To provide practical exposure to the implementation of various language processing components.
- To develop lexical, syntax, semantic analysis, and intermediate code generation modules using compiler construction tools.
- To familiarize students with runtime support mechanisms, code optimization techniques, and interpreter design.
- To introduce modern language processing frameworks including parser generators, intermediate representations, and AI-oriented compiler infrastructures.
- To enable students to design simple domain-specific language (DSL) translators and machine learning-assisted language processing applications.

Course Outcomes

- CO1: Develop lexical analyzers, parsers, and syntax-directed translation modules using compiler construction tools.
- CO2: Implement semantic analysis, symbol table management, intermediate code generation, and runtime support mechanisms.
- CO3: Design translators, interpreters, and simple domain-specific language processors.
- CO4: Apply code optimization techniques and experiment with modern compiler infrastructures for AI and machine learning applications.

Syllabus

- Design and implementation of lexical analyzers using LEX/FLEX or ANTLR.
- Construction of LL(1) and LR parsers using YACC/Bison or ANTLR.
- Development of Recursive Descent Parser for arithmetic expressions.
- Construction of Abstract Syntax Trees (ASTs) and Parse Trees.
- Implementation of Symbol Table Management and Semantic Analysis.
- Syntax Directed Translation for arithmetic expressions and assignment statements.
- Generation of Intermediate Code (Three Address Code, Quadruples and Triples).
- Implementation of Runtime Environment concepts including stack-based activation records and memory allocation simulation.

- Development of a simple Interpreter for arithmetic or expression-based languages.
- Design of a simple Domain Specific Language (DSL) using ANTLR.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3					1		2	3	3
CO2	3	3	3	3	3			1		2		2	3	3
CO3	3	3	3	3	3				2	2	2	3	3	3
CO4	3	3	3	3	3	1		2	2	2	2	3	3	3

Textbooks / References

1. Aho, A. V., Lam, M. S., Sethi, R., &; Ullman, J. D. (2007). Compilers: Principles, techniques, and tools (2nd ed.). Pearson Education.
2. Cooper, K. D., &; Torczon, L. (2022). Engineering a compiler (3rd ed.). Morgan Kaufmann.
3. Muchnick, S. S. (1997). Advanced compiler design and implementation. Morgan Kaufmann.
4. Louden, K. C., &; Lambert, K. A. (2016). Compiler construction: Principles and practice. Cengage Learning.
5. Nystrom, R. (2021). Crafting interpreters. Genever Benning.

CSL223 Database Management Systems Lab [0-0-2-1]

Course Objectives

- To understand the fundamentals of database systems and data modeling techniques.
- To design and implement relational databases using SQL and advanced database programming constructs.
- To apply normalization and schema design principles for efficient database development.
- To understand query processing, indexing, transaction management, recovery, and database security.
- To explore modern database technologies such as NoSQL, cloud databases, and vector databases for contemporary applications.

Course Outcomes

- CO1: Design conceptual and logical database schemas using ER and Enhanced ER models.
- CO2: Implement relational databases and develop SQL/PL-SQL programs for data management.
- CO3: Apply normalization techniques and evaluate database schema quality.
- CO4: Analyze indexing methods, query execution plans, and optimization strategies.
- CO5: Implement transaction management, concurrency control, security mechanisms, and modern database applications using NoSQL and emerging database technologies.

Syllabus

1. Database Design using ER/EER Models: Design ER and Enhanced ER diagrams for real-world applications.
2. Relational Database Design: Convert ER models into relational schemas and implement constraints.
3. SQL Fundamentals: Practice DDL, DML, DCL, and TCL commands.
4. SQL Queries and Joins: Write simple, nested, and join queries for data retrieval.
5. Advanced SQL: Implement views, set operators, CTEs, and window functions.
6. PL/SQL Programming: Develop programs using control structures, cursors, and exception handling.
7. Stored Procedures and Functions: Create and execute procedures and user-defined functions.
8. Database Triggers: Implement triggers for data validation and auditing.
9. Normalization and Schema Design: Apply functional dependencies and normalization techniques.
10. Indexing and Query Optimization: Create indexes and analyze query execution plans.
11. Transaction Management and Security: Perform transaction control, concurrency control, and privilege management.
12. NoSQL Database Operations: Implement CRUD operations and aggregation using MongoDB.
13. Modern Database Technologies: Explore Cloud Databases, NewSQL Databases, and Vector Databases.
14. Mini Project: Design and implement a complete database application for a real-world problem.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	2	3							2	3	2
CO2	3	3	3	2	3							2	3	3
CO3	3	3	3	2	2							2	3	2
CO4	3	3	2	3	3			1				2	3	2
CO5	3	3	3	3	3	1		1	1	1		3	3	3

Textbooks / References

1. Silberschatz, A., Korth, H. F., & Sudarshan, S., *Database System Concepts*, 7th ed., McGraw-Hill Education, 2019.
2. Elmasri, R., & Navathe, S. B., *Fundamentals of Database Systems*, 7th ed., Pearson Education, 2016.
3. Viescas, J. L., & Hernandez, M. J., *SQL Queries for Mere Mortals: A Hands-On Guide to Data Manipulation in SQL*, 5th ed., Addison-Wesley Professional, 2023.
4. Beaulieu, A., *Learning SQL: Generate, Manipulate, and Retrieve Data*, 3rd ed., O'Reilly Media, 2020.
5. Kleppmann, M., *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*, 1st ed., O'Reilly Media, 2017.
6. Bradshaw, S., Brazil, E., & Chodorow, K., *MongoDB: The Definitive Guide: Powerful and Scalable Data Storage*, 3rd ed., O'Reilly Media, 2019.

7. Thusoo, A., & Sen Sarma, J., *Vector Search: Building Intelligent Search Systems with Embeddings*, 1st ed., O'Reilly Media, 2025.

CSC311 Software Engineering [2-1-0-3]

Course Objectives

- To introduce the fundamental concepts, principles, and practices of software engineering and software development processes.
- To develop the ability to analyze, specify, and manage software requirements using structured methodologies.
- To provide knowledge of software design principles, architectural styles, UML modeling, and object-oriented design techniques.
- To equip students with project management, risk management, and software quality management skills required for successful software project execution.

Course Outcomes

- CO1: Explain software engineering principles, software process models, and software development methodologies for building reliable software systems.
- CO2:Analyze, elicit, document, validate, and manage software requirements, and prepare Software Requirements Specification (SRS) documents.
- CO3: Apply software design principles, architectural patterns, UML modeling, and object-oriented techniques to develop effective software designs.
- CO4: Implement coding best practices and evaluate software quality using appropriate testing strategies, debugging techniques, and software metrics.
- CO5: Apply software project management, risk management, configuration management, and quality assurance techniques to plan, monitor, and deliver software projects successfully.

Syllabus

Introduction to Software Engineering: The evolving role of software, changing nature of software, software myths. A Generic view of process: Software engineering- a layered technology, a process framework, the capability maturity model integration (CMMI), Process models (SDLC): The waterfall model, incremental process models, evolutionary process models, agile methodologies, the unified process (RUP).
Software Requirements: Functional and non-functional requirements, user requirements, system requirements, interface specification, the software requirements specification. Requirements engineering process: Feasibility studies, requirements elicitation and analysis, requirements validation, requirements management.
Design Engineering: Design process and design quality, design concepts- Coupling and cohesion. Creating an architectural design: Software architecture, architectural styles and patterns. Design Methodologies: Function-Oriented Structured Design, UML and Object-Oriented Design.
Coding: Source code control, Incremental Coding Process, Test-driven development (TDD), Pair programming. Testing Strategies: Testing concepts, test strategies for conventional software, black-box and white-box testing, validation testing, system testing, the art of debugging. Product metrics: Software quality, metrics for analysis model, metrics for design model, metrics for source code, metrics for testing, metrics for maintenance.
Software Project Management: Cost estimation- Project scheduling- Staffing-Software configuration management, Quality assurance, Software quality models, Project Monitoring. Risk management: Reactive Vs proactive risk strategies, software risks, RMMM. Quality Management: Quality concepts, software quality assurance, software reviews, formal technical reviews, the ISO 9000 quality standards.

List of Tutorial Activities

1. Library Management System
2. Student Information Management System
3. Online Examination System
4. Hospital Management System
5. Inventory and Stock Management System
6. Event Registration and Management System
7. Online Course Management System
8. E-Commerce Application
9. Recruitment and Placement Management System
10. Hotel Reservation System
11. Vehicle Rental Management System
12. Complaint Management System
13. Digital Banking Application
14. E-Book Management System
15. Smart Campus Management System

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1		2			1				2	2	1
CO2	3	3	2	2	2			1		1		2	3	2
CO3	3	3	3	2	2			1	1	1		2	3	2
CO4	2	3	3	3	3			1	1	1		2	3	3
CO5	2	3	2	2	2			2	3	2	1	3	3	2

Textbooks / References

1. Software Engineering, A practitioner's Approach- Roger S. Pressman, 6th edition, Mc Graw Hill International Edition.
2. Software Engineering- Sommerville, 7th edition, Pearson Education.
3. Software Engineering with Open Source and GenAI, Pankaj Jalote, 2ed
4. The unified modeling language user guide Grady Booch, James Rambaugh, Ivar Jacobson, Pearson Education.
5. Software Engineering, an Engineering approach- James F. Peters, Witold Pedrycz, John Wiley.
6. Software Engineering principles and practice- Waman S Jawadekar, The Mc Graw-Hill Companies.

7. Fundamentals of object-oriented design using UML Meiler page-Jones: Pearson Education.

CAC311 Data Engineering [3-0-0-3]

Course Objectives

- To understand end-to-end data engineering pipelines.
- To learn data acquisition, storage, and processing techniques.
- To design scalable ETL pipelines.
- To apply data pre-processing, analytics, and visualization.
- To implement cloud-native data engineering pipelines and modern data practices

Course Outcomes

- CO1: Explain data engineering concepts and pipeline architecture.
- CO2: Identify data sources and design storage systems.
- CO3: Build ingestion pipelines with pre-processing techniques.
- CO4: Apply data modelling, analytics, and visualization methods.
- CO5: Design and implement cloud-native data engineering pipelines using cloud storage, compute, data integration, and orchestration services.

Syllabus

Introduction to Data Engineering and Data Architecture: Introduction to Data Engineering, Need for Data Engineering, Data Engineering Lifecycle, Roles and Responsibilities of Data Engineers, Overview of Data Engineering Pipelines, Data Architecture, Good Principles, Types of Data Architectures, Data Maturity Models.

Data Types, Sources, Data Storage and Data Management: Data Types, Data Generation, Data Sources: Databases, APIs, Logs, IoT Devices, Streaming Data Sources, Change Data Capture. Data Storage: File, Block and Object Storage, Distributed Storage Systems, NoSQL Databases, Data Warehouses, Data Lakes, Data Lakehouses. Data Management: Data Modelling, Metadata Management, Data Lineage, Data Provenance, Data Catalogs, Data Lifecycle and Retention Policies.

Data Ingestion, Preprocessing and Orchestration: Data Ingestion: Batch Processing, Stream Processing, Micro-batch Processing, Bounded and Unbounded Data, ETL Pipelines, API-based Ingestion, Data Serialization Formats (JSON, Avro, Parquet). Data Pre-processing: Data Quality, Data Cleaning Techniques, Handling Missing Data, Noise Detection and Elimination, Data Integration and Transformation, Feature Selection Techniques. Orchestration: Workflow Scheduling, Pipeline Dependencies, Orchestration Tools (Airflow), Monitoring and Logging, DataOps Principles.

Data Modeling, Transformation, Analytics and Visualization: Data Modelling and Transformation: Conceptual, Logical and Physical Data Models, Data Transformation Techniques, Batch and Stream Processing, Descriptive and Exploratory Data Analysis, Aggregations and Statistical Summaries. Data lineage. Data Visualization: Histogram, Pie Chart, Box Plot, Scatter Plot, Bubble Plot, Word Clouds, Visualization Best Practices.

Cloud-Native Data Engineering with Future Trends: Cloud Computing Concepts, Service Models, Deployment Models, Cloud Storage: Object Storage, Data Lakes, Compute Services, Cloud-Native Data Engineering: Principles and Architecture, Cloud Data Integration: Data Factory Concepts – ELT in the Cloud, Workflow Orchestration, Monitoring and Logging, Future Trends: Serverless Data Pipelines.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1		2							2	2	1
CO2	3	3	2	1	3							2	3	2
CO3	3	3	3	2	3				1			2	3	3
CO4	2	3	3	3	3			1	1	1		2	3	3
CO5	2	3	3	3	3			1	1	1		3	3	3

Textbooks / References

1. Reis, J. and Housley, M., *Fundamentals of Data Engineering*, O'Reilly Media, 2022.
2. Liu, Z., *Data Engineering Fundamentals: Building Scalable Data Solutions with ETL Pipelines and Strategic Data Architecture Design*, 2025.
3. Densmore, J., *Data Pipelines Pocket Reference*, O'Reilly Media, 2021.
4. Akidau, T., Chernyak, S., and Lax, R., *Streaming Systems*, O'Reilly Media, 2018.
5. Kimball, R. and Ross, M., *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling*, John Wiley & Sons, 2013.
6. Cloud Computing: Concepts, Technology & Architecture, Thomas Erl, Ricardo Puttini, and Zaigham Mahmood, Prentice Hall, 2013.

CSC312 Parallel and Distributed Computing [3-0-0-3]

Course Objectives

- To introduce the concept and the basics of parallel and distributed computing.
- To provide knowledge on various available parallel programming models.
- To enable awareness on the utilization of programming models on various scientific applications.

Course Outcomes

- CO1: Understand the concepts, architectures, and programming models used in parallel and distributed computing systems.

- CO2: Apply parallel programming techniques using shared-memory and distributed-memory programming models.
- CO3: Analyze scientific and AI/ML applications for parallelization and performance improvement.
- CO4: Utilize performance analysis and profiling tools for evaluating parallel applications.
- CO5: Explain the applications and performance characteristics of parallel and distributed computing models.

Syllabus

Architectures: Heterogeneous computing architectures, Accelerators (SIMD units, Vectorization, GPUs, TPUs), Goals of parallel systems, introduction to High-Performance Computing (HPC), scalability, parallelism in AI-ML workloads.

Parallel Programming on Shared Memory: OpenMP (C++ languages), Execution Model, Shared and private data, Directives, Barriers, Sections, Run-Time library functions, Scheduling strategies, OpenMP Task Dependencies, Taskloop, Target Offloading.

Parallel Programming on Distributed Memory: MPI, Collective operations, Non-Blocking communications, Collectives, Process topologies, Single sided communications, Persistent Communications.

GPU Computing: Fundamentals of GPU architecture. Data-parallel programming concepts. Introduction to CUDA and OpenCL programming models. Overview of GPU computing for AI applications.

Parallel Algorithms and AI Workloads: Characteristics of parallel algorithms. Data partitioning, task decomposition, synchronization, load balancing, regular and irregular applications, dependency analysis, parallelization strategies. Parallel machine learning concepts, distributed AI workloads, distributed data processing, data-parallel and model-parallel execution, and scalable AI training.

Performance Tools: Concepts, Event-Model execution, Profiling, Tracing, Types of profiling, Profiling tools – Scalasca, EnergyAnalyzer, Tracing tools, Roofline Performance Modeling.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1		2							2	2	1
CO2	3	3	3	2	3				1			2	3	3
CO3	3	3	3	3	3				1	1		2	3	3
CO4	2	3	2	3	3					1		3	3	2
CO5	2	2	2	2	2			1		1		3	2	2

Textbooks / References

1. Michael J. Quinn, *Parallel Programming in C with MPI and OpenMP*, McGraw-Hill, 2004.
2. Ruud van der Pas, Eric Stotzer, and Christian Terboven, *Using OpenMP – The Next Step*, MIT Press, 2017.
3. Norman Matloff, *Parallel Computing for Data Science: With Examples in R, C++, and CUDA*, CRC Press, 2016.
4. Robert Cook, *An Introduction to Parallel Programming with OpenMP, PThreads and MPI*, Kindle Edition, 2011.
5. Peter Pacheco, *An Introduction to Parallel Programming*, Morgan Kaufmann, 2011.
6. *International Journal of Parallel Programming* – Issues and articles devoted to OpenMP, MPI, and CUDA.

CAC313 Advanced Machine Learning [3-0-0-3]

Course Objectives

- To provide a strong foundation in advanced supervised learning, ensemble learning, and probabilistic modeling techniques.
- To develop an understanding of deep neural networks, optimization algorithms, and regularization methods used in modern deep learning systems.
- To introduce advanced unsupervised and representation learning techniques for clustering, dimensionality reduction, anomaly detection, and feature learning.
- To familiarize students with emerging machine learning paradigms including reinforcement learning, semi-supervised learning, self-supervised learning, and federated learning.
- To enable learners to analyze, design, and implement machine learning models for real-world intelligent applications.

Course Outcomes

- CO1: Analyze and apply advanced supervised learning techniques including SVMs, ensemble learning methods, and probabilistic models for classification and prediction tasks.
- CO2: Design and optimize deep neural network models using appropriate activation functions, optimization algorithms, and regularization techniques.
- CO3: Implement advanced unsupervised and representation learning methods for clustering, dimensionality reduction, anomaly detection, and high-dimensional data analysis.
- CO4: Apply reinforcement learning, semi-supervised learning, self-supervised learning, and federated learning concepts to solve modern intelligent system problems.
- CO5: Evaluate machine learning models using suitable performance metrics and select appropriate algorithms for real-world applications.

Syllabus

Advanced Supervised and Ensemble Learning: Overview of Machine Learning and Learning Theory , Basic concept, Probably Approximately Correct (PAC) Learning, Version Space, Computational Complexity of Learning, Introduction to VC Dimension. Margin-Based Models: Support Vector Machines (SVM), Margin maximization, Soft-margin classifiers, Kernel methods, Kernel trick, Linear and non-linear classification techniques. Ensemble Learning: Ensemble learning fundamentals, Bagging techniques, Random Forests, Boosting methods, AdaBoost, Gradient Boosting, XGBoost concepts, Ensemble model evaluation and applications. Probabilistic Models: Fundamentals of probabilistic learning, Bayesian learning and inference, Bayesian Belief Networks, Hidden Markov Models (HMM), Monte Carlo Methods, Probabilistic graphical models and applications.

Deep Neural Networks and Optimization Techniques: Deep Neural Networks: Fundamentals of deep learning, Artificial Neural Networks (ANN), Perceptrons, Feedforward and multilayer neural networks, Activation functions, Loss functions, Backpropagation algorithm, Weight initialization techniques. Optimization Methods: Gradient Descent and its variants, Stochastic Gradient Descent (SGD), Momentum optimization, RMSProp, Adam optimizer, Learning rate scheduling techniques. Regularization Techniques: Overfitting and underfitting problems, L1 and L2 regularization, Dropout techniques, Early stopping, Batch normalization.

Advanced Unsupervised and Representation Learning: Clustering and Dimensionality Reduction: Density-based clustering (DBSCAN), Linear Discriminant Analysis (LDA), t-SNE and manifold learning concepts. Representation Learning: Feature extraction and feature selection, Latent feature representation, Autoencoders, Sparse autoencoders, Variational Autoencoders (VAE) concepts, Representation learning for high-dimensional data. Anomaly and Pattern Detection: Outlier detection techniques, Anomaly detection models, Pattern mining concepts, Applications in fraud detection and healthcare analytics.

Advanced Machine Learning Applications and Emerging Trends: Advanced Learning Paradigms: Reinforcement Learning fundamentals, Markov Decision Process (MDP), Q-Learning concepts, Policy-based learning methods, Exploration and exploitation strategies, Semi-supervised learning techniques, Self-supervised learning concepts, Federated learning architecture and applications.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3					1		2	3	2
CO2	3	3	3	3	3					1		2	3	3
CO3	3	3	3	3	3		1		1	1		2	3	3
CO4	2	3	3	3	3			1	1	1		2	3	3
CO5	3	3	3	3	2			1	1	2		3	3	3

Textbooks / References

1. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
2. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press, 2016.
3. Trevor Hastie, Robert Tibshirani, and Jerome Friedman, The Elements of Statistical Learning, Springer, 2nd Edition.
4. Kevin P. Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012.
5. Richard S. Sutton and Andrew G. Barto, Reinforcement Learning: An Introduction, MIT Press, 2nd Edition.
6. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media.
7. François Chollet, Deep Learning with Python, Manning Publications.
8. Sebastian Raschka and Vahid Mirjalili, Python Machine Learning, Packt Publishing.
9. Charu C. Aggarwal, Neural Networks and Deep Learning, Springer.
10. Ethem Alpaydin, Introduction to Machine Learning, MIT Press.

CAL311 Data Engineering Lab [0-0-2-1]

Course Objectives

- To provide hands-on experience in data engineering and analytics workflows.
- To acquire data from diverse sources, manage and store data using different formats and databases, develop ETL pipelines, automate workflows.
- To perform data pre-processing and exploratory analysis, assess data quality and lineage,
- To implement cloud-native data engineering pipelines and modern data practices.

Course Outcomes

- CO1: Extract, integrate, and manage data from multiple structured and semi- structured sources, including relational databases, APIs, files, and NoSQL DBs.
- CO2: Design, implement, and evaluate data storage solutions using various formats and database technologies, considering storage efficiency and retrieval performance.
- CO3: Develop and automate ETL pipelines using appropriate tools and frameworks, ensuring reliable data transformation, workflow orchestration, and performance monitoring.
- CO4: Apply data pre-processing and perform exploratory data analysis and visualization to derive insights from datasets and communicate findings.
- CO5: Design and implement cloud-native data engineering pipelines using cloud storage, compute, data integration, and orchestration services.

Syllabus

1. Develop a Python program to extract data from a relational database, a public REST API, and CSV files.
2. Store a dataset in CSV, JSON, and Parquet formats and analyze their storage efficiency and retrieval performance.

3. Create a simple NoSQL database collection and implement metadata management for effective data organization.
4. Extract data from multiple sources, perform transformations and load the processed data into a target database. Validate the pipeline output and measure execution performance.
5. Apply pre-processing techniques to prepare the dataset for analytics and machine learning applications.
6. Create an Apache Airflow DAG to automate an ETL process with multiple dependent tasks. Configure scheduling, monitor workflow execution, and analyze logs for successful pipeline management.
7. Compute descriptive statistics and perform exploratory data analysis on a dataset. Create visualizations to identify trends and patterns.
8. Perform data quality assessment and track data lineage within a sample pipeline.
9. Develop a streaming application to ingest real-time data (e.g., sensor, log, or API-generated data) using Apache Kafka or a similar platform. Process the incoming data stream, perform basic transformations and aggregations, and visualize or store the results for real-time analytics.
10. Launch and configure a virtual machine (VM) on a cloud platform. Create a cloud object storage bucket and perform basic operations such as uploading, downloading, organizing, and deleting files.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1		2							2	2	1
CO2	3	3	2	1	3							2	3	2
CO3	3	3	3	2	3					1		2	3	3
CO4	2	3	3	3	3			1	1	1		2	3	3
CO5	2	3	3	3	3		1	2	1	1		3	3	3

Textbooks / References

1. Reis, J. and Housley, M., *Fundamentals of Data Engineering*, O'Reilly Media, 2022.
2. Liu, Z., *Data Engineering Fundamentals: Building Scalable Data Solutions with ETL Pipelines and Strategic Data Architecture Design*, 2025.
3. Densmore, J., *Data Pipelines Pocket Reference*, O'Reilly Media, 2021.
4. Akidau, T., Chernyak, S., and Lax, R., *Streaming Systems*, O'Reilly Media, 2018.
5. Kimball, R. and Ross, M., *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling*, John Wiley & Sons, 2013.

CSL312 Parallel and Distributed Computing Lab [0-0-2-1]

Course Objectives

- To introduce the concept and the basics of parallel and distributed computing.
- To provide knowledge on various available parallel programming models.
- To enable awareness on the utilization of programming models on various scientific applications.

Course Outcomes

- CO1: Gain practical knowledge on various types of HPC applications.
- CO2: Utilize parallel programming languages and frameworks such as OpenMP and MPI.
- CO3: Develop and execute parallel applications using shared-memory and distributed-memory programming models.
- CO4: Apply performance analysis and debugging tools for parallel applications and programs.

Syllabus

1. Study of multi-core and many-core architectures using Linux tools.
2. OpenMP programming basic constructs.
3. OpenMP programming task-related constructs and programs based on the concepts.
4. OpenMP-based applications – Matrix Multiplication, Fibonacci, Merge-Sort, N-Queens Problem.
5. OpenMP-based applications using task-based constructs.
6. MPI programming basics – Point-to-Point communications – Matrix Multiplication.
7. MPI programming – Collective Communications – Parallel Reduction, Broadcast, Scatter-Gather.
8. MPI non-blocking communications – Applications and programs.
9. Hybrid programming – MPI + OpenMP (any use-cases).
10. Performance analysis tools – Scalasca, Vampir, TAU.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2							2	2	1
CO2	3	3	2	2	3					1		2	3	2
CO3	3	3	3	3	3				1	1		2	3	3
CO4	2	3	3	3	3			1	1	2		3	3	3

Textbooks / References

1. Michael J. Quinn, *Parallel Programming in C with MPI and OpenMP*, McGraw-Hill, 2004.
2. Ruud van der Pas, Eric Stotzer and Christian Terboven, *Using OpenMP – The Next Step*, MIT Press, 2017.
3. Norman Matloff, *Parallel Computing for Data Science: With Examples in R, C++, and CUDA*, CRC Press, 2016.
4. Robert Cook, *An Introduction to Parallel Programming with OpenMP, PThreads and MPI*, Kindle Edition, 2011.
5. Peter Pacheco, *An Introduction to Parallel Programming*, Morgan Kaufmann, 2011.

6. *The International Journal of Parallel Programming* – Issues and articles devoted to OpenMP, MPI, and CUDA.

CAL313 Advanced Machine Learning Lab [0-0-2-1]

Course Objectives

- Explore the concepts, terminology, and working principles of advanced machine learning techniques such as linear models, ensembles, neural networksand reinforcement learning.
- Interpret and understand the role of model selection, optimization, and learning strategies in building effective machine learning systems.
- Apply and analyse advanced learning techniques on different types of data, including tabular, image, sequential, and unlabeled datasets.
- Create suitable machine learning workflows by selecting, tuning, and integrating advanced models for real-world problem solving.

Course Outcomes

- CO1: Identify the core concepts, terminology, and techniques used in advanced machine learning models and workflows.
- CO2: Explain the principles and functioning of linear models, ensemble methods, neural networks, reinforcement learning, and unsupervised learning methods.
- CO3: Implement and evaluate advanced machine learning techniques on structured, image-based, sequential, and unlabeled datasets using appropriate optimization and tuning methods.
- CO4: Design and develop an end-to-end advanced machine learning solution by selecting and integrating suitable learning techniques for a given application.

Syllabus

1. Implement and analyze non-linear regression and classification models to study decision boundaries, margin maximization, and model performance on complex datasets.
2. Develop ensemble learning models using techniques such as bagging, random forests, AdaBoost, gradient boosting, and perform comparative performance analysis.
3. Design and train artificial neural networks including perceptrons and multilayer feedforward networks using backpropagation and optimization algorithms.
4. Implement probabilistic models — Naive Bayes classifier, Bayesian Belief Network construction/inference, and Hidden Markov Models for sequential data
5. Apply Monte Carlo methods for probabilistic estimation and sampling-based inference
6. Perform clustering and dimensionality reduction using advanced unsupervised learning techniques such as DBSCAN, LDA, and manifold learning methods.
7. Implement anomaly/outlier detection techniques
8. Investigate the impact of hyperparameter optimization, learning-rate scheduling, regularization, and model refinement strategies on predictive performance.
9. Evaluate models built using appropriate performance metrics, confusion matrix, k-fold cross-validation, bias-variance analysis.
10. Implement a semi-supervised learning technique on a partially labeled dataset and compare against fully supervised baseline.
11. Implement a basic self-supervised learning pretext task to study representation learning.
12. Develop reinforcement learning solutions involving agent – environment interaction, Markov Decision Processes, Q-learning, policy-based learning, and comprehensive model evaluation using appropriate performance metrics.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	1	2					1		3	3	2
CO2	3	3	2	2	2					1		3	3	3
CO3	3	3	3	3	3	1		1	2	2	1	3	3	3
CO4	3	3	3	3	3	2	1	2	3	3	2	3	3	3

Textbooks / References

1. Raschka, S., Liu, Y. (Hayden), and Mirjalili, V. Machine Learning with PyTorch and Scikit-Learn: Develop Machine Learning and Deep Learning Models with Python, 1st Edition, Packt Publishing Ltd., Birmingham, UK, 2022. ISBN: 978-1801819312, 774 pages.
2. Goodfellow, I., Bengio, Y., and Courville, A. Deep Learning, Adaptive Computation and Machine Learning Series, MIT Press, Cambridge, MA, USA, 2016. ISBN: 978-0262035613 (Hardcover), ISBN: 978-0262337373 (eBook), 800 pages.
3. Rojas, R. Neural Networks: A Systematic Introduction, 1st Edition, Springer-Verlag Berlin Heidelberg, Germany, 1996. ISBN: 978-3-540-60505-8 (Softcover), ISBN: 978-3-642-61068-4 (eBook), DOI: 10.1007/978-3-642-61068-4, 502 pages.
4. Krohn, J. Deep Reinforcement Learning and GANs: Advanced Topics in Deep Learning, O'Reilly Media, Sebastopol, CA, USA, 2020. ISBN: 978-1098114831.
5. Sarkar, D., Bali, R., and Ghosh, T. Hands-On Transfer Learning with Python: Implement Advanced Deep Learning and Neural Network Models Using TensorFlow and Keras, Packt Publishing Ltd., Birmingham, UK, 2018. ISBN: 978-1788831307.
6. Foster, D. Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play, 2nd Edition, O'Reilly Media, Sebastopol, CA, USA, 2023. ISBN: 978-1098134181.

CAC321 AI Safety [2-0-0-2]

Prerequisites

1. Fundamentals of Machine Learning
2. Probability, Statistics and Linear Algebra

Course Overview/Description

Artificial Intelligence systems are increasingly deployed in critical domains where reliability, safety, privacy, and ethical decision-making are essential. This course introduces the fundamental principles of AI safety by examining risks associated with data, models, and deployment environments. It covers AI vulnerabilities, adversarial threats, bias, privacy, validation and testing, formal methods, explainability, interpretability, and safety challenges in large language models. The course also introduces responsible AI practices, governance, and human oversight for developing trustworthy AI systems.

Course Objectives

- To introduce the fundamental principles and challenges of AI safety in modern intelligent systems.
- To examine risks arising from data privacy, bias, vulnerabilities, adversarial threats, and large language models.
- To understand AI testing, validation, formal methods, explainability, and interpretability for trustworthy AI systems.
- To familiarise students with ethical principles, governance frameworks, and responsible AI practices.
- To develop the ability to analyse AI systems from a safety and trustworthiness perspective.

Course Outcomes

- CO1: Explain the fundamental concepts and principles of AI safety.
- CO2: Analyse risks arising from data privacy, bias, adversarial threats, vulnerabilities, and large language models.
- CO3: Evaluate AI systems using testing, validation, formal methods, explainability, and interpretability principles.
- CO4: Assess ethical, governance, and responsible AI practices for trustworthy AI deployment.
- CO5: Analyse AI systems from a safety perspective across different application domains.

Syllabus

Foundations of AI safety. AI safety, AI security, trustworthy AI and responsible AI. AI system lifecycle. AI failures and incident case studies. AI Alignment, Hazard, risk, uncertainty, and safety by design. Human oversight. AI governance and regulatory landscape.

Data privacy, data quality, bias, fairness, algorithmic discrimination, distribution shift, and out-of-distribution generalization. AI vulnerabilities and attack surfaces. Introduction to adversarial machine learning. Safety challenges in large language models including hallucinations, prompt injection, jailbreaks, misinformation, and misuse risks.

AI testing methodologies. Dataset quality assessment. AI validation and evaluation frameworks. Robustness evaluation. Uncertainty estimation. AI assurance. Introduction to formal verification and formal methods for AI systems. Runtime monitoring. Human-in-the-loop validation.

Explainability and interpretability for trustworthy AI. Transparent AI systems. Explainability for safety-critical applications. Accountability and responsible AI. Safe deployment practices. Governance frameworks. Emerging challenges and future directions in AI safety.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	1	1	2	2	2	2				2	2	2
CO2	2	3	2	2	2	3	2	2				2	2	3
CO3	2	3	2	3	3	2	2	2		1		2	2	3
CO4	1	2	2	2	2	3	3	3	1	1	1	3	1	2
CO5	2	3	3	2	2	3	3	3	1	1	1	3	2	3

Textbooks / References

1. Russell, S., *Human Compatible: Artificial Intelligence and the Problem of Control*, Viking, 2019.
2. Varshney, K. R., *Trustworthy Machine Learning*, Independently Published, 2022.
3. Molnar, C., *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*, 2022.
4. Barocas, S., Hardt, M., and Narayanan, A., *Fairness and Machine Learning: Limitations and Opportunities*, MIT Press, 2023.
5. National Institute of Standards and Technology (NIST), *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1, 2023.
6. National Institute of Standards and Technology (NIST), *Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations*, NIST AI 100-2, 2024.
7. Amodei, D. et al., "Concrete Problems in AI Safety," arXiv:1606.06565, 2016.
8. Hendrycks, D., *Introduction to AI Safety, Ethics, and Society*, 1st Edition, CRC Press / Taylor & Francis, 2025.

CAC322 Deep Learning [3-0-0-3]

Course Objectives

- To introduce the fundamental concepts, architectures, and mathematical foundations of deep learning.
- To develop an understanding of optimization techniques and learning algorithms for training deep neural networks.
- To study advanced deep learning architectures, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Transformer-based models.

- To explore applications of deep learning in computer vision, natural language processing, generative AI, and intelligent systems.
- To develop the ability to implement and evaluate deep learning models through programming assignments using contemporary deep learning frameworks.

Course Outcomes

- CO1: Explain the concepts, mathematical foundations, and optimization techniques of deep learning.
- CO2: Analyze deep neural network architectures, including feedforward and convolutional neural networks, for computer vision applications.
- CO3: Examine recurrent neural network architectures, including RNNs, LSTMs, and GRUs, for sequential data processing.
- CO4: Interpret the principles and applications of generative deep learning models, including Autoencoders, VAEs, and GANs.
- CO5: Apply contemporary deep learning frameworks such as TensorFlow and PyTorch to explore and evaluate deep learning models.
- CO6: Analyze transformer-based architectures and evaluate emerging trends, ethical considerations, and real-world applications of deep learning and generative AI.

Syllabus

Introduction to Deep Learning: Introduction to deep learning, Deep Feedforward Networks, Forward and backward propagation,

Deep Neural Networks and Sequential Models: Deep Neural Networks, Convolutional Neural Network, Convolution operation and pooling, CNN architectures, LeNet, AlexNet, VGG, ResNet, Transfer learning, Image classification and object detection, Recurrent Neural Networks, Sequence modelling, Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Time-series prediction, NLP applications.
Evaluation metrics for deep models, explainability for deep models.

Graph Neural Networks: Graph Representation, Message passing, Graph Convolutional Networks.

Generative AI, Deep Unsupervised Learning and Recent Trends: Autoencoders (standard, sparse, denoising, contractive, etc.), Variational Autoencoders, Autoencoder and DBM, Multi-task Deep Learning, Multi-view Deep Learning, GANs (Generative Adversarial Networks), Generative Moment Matching Networks.
Transformers and Ethical Issues: Transformers, Transformer architecture, Self-attention and multi-head attention, Vision Transformer (ViT), SWIN, BERT and GPT models, Ethical issues in deep learning.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2					1		3	3	2
CO2	3	3	3	2	3					2		3	3	3
CO3	3	3	3	2	3					2		3	3	3
CO4	3	3	3	3	3			1		2		3	3	3
CO5	2	2	3	3	3				2	2		3	3	3
CO6	3	3	3	3	3	2	2	2	2	3	2	3	3	3

Textbooks / References

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
2. Bishop, C. M., *Pattern Recognition and Machine Learning*, Springer, 2006.
3. Raúl Rojas, *Neural Networks: A Systematic Introduction*, Springer, 1996.
4. Foster, David, *Generative Deep Learning*, O'Reilly Media, Inc., 2022.
5. Rothman, Denis, *Transformers for Natural Language Processing and Computer Vision: Explore Generative AI and Large Language Models with Hugging Face, ChatGPT, GPT-4V, and DALL-E 3*, Packt Publishing Ltd, 2024.

CAC323 Big Data Analytics and Streaming Systems [3-1-0-4]

Course Objectives

- To understand the principles of distributed computing, big data architectures, and scalable data processing frameworks for analyzing large-scale datasets.
- To develop an understanding of batch and stream processing models, real-time analytics, and distributed data processing using contemporary big data technologies.
- To study streaming data analytics techniques, including windowing models, complex event processing, concept drift, and large-scale data analysis.
- To analyze security, privacy, and reliability challenges in big data and streaming systems, including data protection, secure processing, and trustworthy analytics.
- To develop the ability to design and evaluate scalable big data analytics solutions for real-time applications across diverse domains using modern distributed computing platforms.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Explain the principles of distributed computing, big data architectures, and scalable data processing frameworks.
- CO2: Apply batch and stream processing techniques for analyzing and processing large-scale datasets using contemporary big data platforms.

- CO3: Analyze streaming data using windowing models, complex event processing, concept drift analysis, and real-time analytics techniques.
- CO4: Evaluate the security, privacy, and reliability aspects of big data and streaming systems for trustworthy data processing.
- CO5: Evaluate the security, privacy, and reliability aspects of big data and streaming systems for trustworthy data processing.

Syllabus

Introduction to big data analytics. Characteristics and challenges of big data. Batch, interactive, and stream processing paradigms. Distributed computing principles. Distributed file systems and storage architectures. Hadoop ecosystem, HDFS, YARN, and MapReduce programming model. Scalability, fault tolerance, and resource management in distributed analytics.

Apache Spark architecture and distributed data processing. Resilient Distributed Datasets (RDDs), DataFrames, and Spark SQL. Distributed transformations and actions. Performance optimisation techniques. Introduction to Spark MLlib for large-scale analytics. Graph analytics using GraphX. Batch analytics applications and industrial use cases.

Streaming data models and real-time analytics. Characteristics and challenges of streaming data. Windowing techniques including tumbling, sliding, session, and landmark windows. Stream aggregation and continuous query processing. Apache Kafka architecture and distributed messaging. Spark Structured Streaming and introduction to Apache Flink. Complex Event Processing (CEP). Concept drift, incremental learning, and real-time decision support.

Security, privacy, and reliability in big data analytics. Data privacy principles. Encryption techniques for distributed data processing. Authentication and access control. Secure data sharing. Data integrity and provenance. Streaming system vulnerabilities, data poisoning, injection attacks, adversarial threats, and resilient analytics. Trustworthy data analytics and governance considerations.

Large-scale analytics applications and emerging trends. Internet of Things (IoT) analytics. Social media and web-scale analytics. Log analytics and recommendation systems. Time-series and financial analytics. AI-enabled big data analytics. Cloud-native analytics platforms. Lakehouse architecture. Real-time business intelligence. Industrial case studies. Emerging trends in distributed analytics and streaming systems.

List of Tutorial Activities

1. Study and compare the architectures of Hadoop, Apache Spark, and Apache Flink for large-scale data processing.
2. Analyze real-world big data applications from domains such as healthcare, finance, smart cities, e-commerce, and social media.
3. Solve problems related to distributed data partitioning, scalability, fault tolerance, and resource management.
4. Compare batch processing and stream processing models through suitable application scenarios.
5. Analyze different windowing techniques (tumbling, sliding, session, and landmark windows) for streaming data applications.
6. Study the architecture and workflow of Apache Kafka and Spark Structured Streaming using industrial case studies.
7. Evaluate concept drift and incremental learning strategies for evolving data streams.
8. Discuss recent research papers and emerging trends in big data analytics and streaming systems.
9. Analyze security, privacy, and reliability challenges in distributed analytics through case studies involving data breaches, adversarial attacks, and secure data processing.
10. Present a case study on the design of a scalable big data analytics solution for a real-world application using modern distributed computing technologies.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1		2						1	2	3	2
CO2	3	3	3	2	3						2	2	3	3
CO3	3	3	2	3	3						2	2	3	3
CO4	3	3	2	3	3	2		2			3	2	3	3
CO5	3	3	2	3	3	2		2			3	3	3	3

Textbooks / References

1. Akidau, T., Chernyak, S., and Lax, R., Streaming Systems: The What, Where, When, and How of Large-Scale Data Processing, O'Reilly Media, 2018.
2. Damji, J. S., Wenig, B., Das, T., and Lee, D., Learning Spark: Lightning-Fast Data Analytics, 2nd Edition, O'Reilly Media, 2020.
3. White, T., Hadoop: The Definitive Guide, 4th Edition, O'Reilly Media, 2015.
4. Kleppmann, M., Designing Data-Intensive Applications, O'Reilly Media, 2017.
5. Leskovec, J., Rajaraman, A., and Ullman, J. D., Mining of Massive Datasets, 3rd Edition, Cambridge University Press, 2020.
6. Shapira, G., Palino, T., Sivaram, R., and Petty, K., Kafka: The Definitive Guide: Real-Time Data and Stream Processing at Scale, 2nd Edition, O'Reilly Media, 2021.
7. Baesens, B., Analytics in a Big Data World: The Essential Guide to Data Science and its Applications, Wiley, 2014.

CAC324 Number Theory and Cryptography [3-1-0-4]

Course Objectives

- To expose the students to the fundamental algorithms for integer arithmetic, greatest common divisor calculation, modular arithmetic, and other number-theoretic computations.
- To understand fundamental algorithms from symmetric-key and public-key cryptography.
- To explore the applications of cryptography including symmetric and public-key cryptosystems.
- To investigate the methods of attacks and implement proper cryptosystem.

Course Outcomes

- CO1: Solve problems in elementary number theory.
- CO2: Apply elementary number theory to cryptography.
- CO3: To understand the number-theoretic foundations of modern cryptography and the principles behind their security.
- CO4: To understand fundamental algorithms for symmetric key and public-key cryptography.
- CO5: Develop a deeper conceptual understanding of the theoretical basis of number theory and cryptography.
- CO6: To implement and analyze cryptographic and number-theoretic algorithms.

Syllabus

Divisibility, Greatest Common Divisors, Euclidean Algorithm, Least Common Multiple, Solving Linear Diophantine Equations, Group, Ring. Prime Numbers, Unique Prime Factorization, Test of Primality by Trial Division, The concept of congruences, Congruence Classes, Applications of Congruences: Check digits, Solving (single) linear congruence, Solving system of linear congruences, the Chinese Remainder Theorem. Fermat's Little Theorem, The general case: Euler's theorem.

The multiplicative order, Primitive Roots (mod n), The modulus n which does not have primitive roots, The Existence Theorems, Applications: The use of primitive roots. Euler's Criterion, The Legendre Symbol and its properties, Examples of computing the Legendre symbol, Jacobi Symbol, Quadratic Residues and Primitive Roots.

Cryptography Introduction: Kerckhoff's principle, Shift cipher, Affine cipher, Vigenere Cipher, Hill cipher, Playfair cipher.

Symmetric Key Cryptography: Block cipher, Data encryption standard (DES), 3-DES, Finite field, Advanced encryption standard (AES).

Asymmetric/ Public Key cryptography: RSA cryptosystem, Discrete logarithm-based crypto-systems, Elgamal Cryptosystem, Elliptic curve cryptosystem, Signature schemes, Digital signature standard, RSA Signature scheme.

Integrity, Authentication, and Key Management: Message and message digest, Random oracle model, Message authentication code, Cryptographic hash function and its properties, Iterated hash function, SHA-1, SHA-512, Digital signature, RSA digital signature, Elgamal digital signature, Symmetric key distribution, Kerberos, Diffie-Hellman key agreement, Public key distribution.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2							2	3	2
CO2	3	3	3	2	3			1		1		2	3	3
CO3	3	3	3	3	3			2		1		3	3	3
CO4	3	3	3	2	3			2		1		3	3	3
CO5	3	3	2	2	2			2		2		3	3	2
CO6	3	3	3	3	3	2	2	3	2	2	1	3	3	3

Textbooks / References

1. James S. Kraft, Lawrence C. Washington An Introduction to Number Theory with Cryptography,Chapman and Hall/CRC.
2. Neal Koblitz, A Course in Number Theory and Cryptography, Springer.
3. Behrouz A. Forouzan, Debdeep Mukhopadhyay, Cryptography and Network Security, 3rd Edition, Mc Graw Hill Education, 2016.
4. William Stallings, Cryptography and Network security: Principles and Practice, 7/e, Pearson Education Asia, 2017.
5. I. Niven, H.S. Zuckerman, H.L. Montgomery, An Introduction to theory of numbers, Wiley.
6. L. C. Washington, Elliptic curves: number theory and cryptography, Chapman & Hall/CRC.
7. J. Pipher, J. Hoffstein and J. H. Silverman , An Introduction to Mathematical Cryptography, Springer-Verlag.
8. G.A. Jones and J.M. Jones, Elementary Number Theory, Springer-Verlag.

CAL325 Agentic AI Lab [0-0-2-1]

Course Objectives

- To introduce the concepts, architectures, and development approaches of Agentic AI systems using Large Language Models (LLMs), APIs, and modern AI frameworks.
- To provide hands-on experience in designing and implementing AI agents with capabilities such as reasoning, planning, memory management, tool integration, and autonomous task execution.
- To enable students to develop intelligent applications using Retrieval-Augmented Generation (RAG), embeddings, vector databases, multimodal models, and multi-agent collaboration techniques.
- To develop the ability to build, evaluate, and deploy responsible Agentic AI solutions by considering reliability, safety, ethical practices, and real-world application requirements.

Course Outcomes

- CO1: Develop LLM-powered applications using prompt engineering and API-based interfaces.
- CO2: Design AI agents capable of reasoning, planning, memory management, and tool usage.
- CO3: Implement Retrieval-Augmented Generation (RAG) and integrate external knowledge sources into AI applications.
- CO4: Develop single-agent and multi-agent systems for solving real-world tasks using modern Agentic AI frameworks.

Syllabus

- Exploring LLM APIs (OpenAI/Gemini/Ollama) and basic prompt engineering.
- Developing an interactive chatbot using an LLM with effective prompting techniques.
- Implementing Retrieval-Augmented Generation (RAG) using embeddings and a vector database for document question answering.
- Designing an AI agent with function/tool calling capabilities.
- Building an agent with conversational memory and context management.
- Developing an autonomous task-oriented agent capable of planning and executing multi-step workflows.
- Implementing a multimodal AI application using Vision-Language Models.
- Designing and implementing a multi-agent system for collaborative problem solving using an Agentic AI framework.
- Developing an end-to-end Agentic AI application integrating LLMs, RAG, memory, and external tools.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3			1		1		3	3	3
CO2	3	3	3	2	3	1		2	2	2		3	3	3
CO3	3	3	3	3	3			2		2		3	3	3
CO4	3	3	3	3	3	2	2	3	3	3	2	3	3	3

Textbooks / References

1. Generative AI with Python, Bonny P. McClain, Manning Publications, 2025.
2. AI Engineering, Chip Huyen, O'Reilly Media, 2025.
3. Hands-On Large Language Models, Jay Alammar and Maarten Grootendorst, O'Reilly Media, 2025.
4. Documentation of LangChain, LangGraph, Ollama, etc

CSL321 FOSS Project and Innovation Lab [0-1-2-2]

Course Objectives

- To understand the principles of open hardware and the role of Free and Open Source Software (FOSS) tools in embedded system and product development.
- To develop embedded and Internet of Things (IoT) applications using open hardware platforms, sensors, actuators, and communication interfaces.
- To apply digital fabrication techniques using open-source CAD and PCB design tools for rapid prototyping.
- To design, implement, test, and demonstrate an innovative hardware-software system addressing a practical engineering problem.
- To develop technical documentation and collaboratively execute engineering projects using open engineering practices.

Course Outcomes

- CO1: Design and develop embedded applications using open hardware platforms and peripheral interfaces.
- CO2: Integrate sensors, actuators, communication protocols, and edge computing concepts to build Internet of Things (IoT)-based systems.
- CO3: Utilize open-source tools for 3D modelling, PCB design, and rapid prototyping to realize functional hardware systems.
- CO4: Develop, test, document, and demonstrate an innovative hardware prototype through collaborative project-based learning.

Syllabus

Open Hardware Platforms and Embedded Systems: Introduction to open hardware and the Free and Open Source Software (FOSS) ecosystem, Overview of embedded systems, Arduino architecture and programming, ESP32 architecture and development, Raspberry Pi as an edge computing platform, GPIO programming, Sensor and actuator interfacing, Embedded communication interfaces including UART, SPI, and I^2C, Hardware debugging and testing.

Internet of Things and Edge Computing: Internet of Things (IoT) architecture and design principles, Device connectivity and communication, MQTT and HTTP protocols, Node-RED for IoT application development, Data acquisition and visualization, Edge computing fundamentals, Local data processing and automation, Integration of hardware and software components, Secure communication and deployment considerations.

Digital Fabrication and Rapid Prototyping: Digital fabrication workflow, Parametric 3D modelling using FreeCAD and OpenSCAD, PCB schematic capture and layout using KiCad, PCB design verification and manufacturing workflow, 3D printing process and rapid prototyping, Prototype assembly, Integration and testing, Design documentation and engineering drawings.

Innovation Project: Team-based development of an open hardware solution for a real-world application including problem identification and requirement analysis, System architecture and component selection, Prototype design and implementation, Integration of embedded hardware, Internet of Things (IoT), and fabricated components, Testing, validation, and performance evaluation, Technical documentation, Demonstration and project showcase. Only Free and Open Source Software (FOSS) tools shall be used for design, development, prototyping, and documentation.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	3							2	3	2
CO2	3	3	2	2	3	1						2	3	3
CO3	2	2	3	3	3							2	3	3
CO4	3	3	3	3	3	1	1	2	2	2	2	3	3	3

Textbooks / References

1. Michael Margolis, *Arduino Cookbook*, 3rd Edition, O'Reilly Media, 2020.
2. Dogan Ibrahim, *ESP32 Development Using the Arduino IDE*, Elektor, 2023.
3. Joan Horvath and Rich Cameron, *Mastering 3D Printing*, 2nd Edition, Apress, 2020.
4. Peter Waher, *Learning Internet of Things*, Packt Publishing, 2015.
5. Official Arduino Documentation.
6. Node-RED Documentation.

CAC411 Machine Learning Operations (MLOps) [3-0-0-3]

Course Objectives

- To introduce DevOps principles and their role in machine learning system development.
- To understand the complete MLOps lifecycle, including data, model, and pipeline management.
- To design, deploy, monitor, and maintain machine learning systems in production environments.
- To explore scalable, automated, and edge-enabled machine learning deployment practices.

Course Outcomes

- CO1: Explain DevOps and MLOps principles and their application in machine learning workflows.
- CO2: Develop reproducible and automated machine learning pipelines using versioning and lifecycle management techniques.
- CO3: Apply data versioning and dataset management techniques to ensure reproducibility, traceability, and effective lifecycle management of machine learning projects.
- CO4: Deploy, monitor, and maintain machine learning models in production environments.
- CO5: Apply MLOps concepts to real-time, edge, and large-scale machine learning applications.

Syllabus

DevOps and Infrastructure Foundations: Software Development Life Cycle (SDLC), Agile development practices, DevOps principles, DevOps culture and practices, Version Control using Git, Continuous Integration (CI), Continuous Delivery and Deployment (CD), Infrastructure as Code, Automated testing and monitoring.

MLOps Concepts and Lifecycle: Introduction to MLOps, Evolution of MLOps, MLOps versus DevOps, Data-centric AI, Data quality and management, Machine Learning Development Lifecycle, Experiment tracking, Model versioning, Reproducibility in ML systems, Feature stores, Model registry, MLOps maturity levels, ML artifacts and metadata management.

ML Pipelines, Data Version Control and Automation: Data Version Control (DVC), Dataset tracking and versioning, Reproducible ML pipelines, Data ingestion techniques, Batch and streaming data processing, Data validation, Feature engineering pipelines, Hyperparameter tuning, Experiment management, Testing of ML components, Model packaging, Continuous Integration and Continuous Deployment for machine learning, Automated model training and evaluation pipelines, automated testing for ML: code linting, unit tests, data integration tests.

Model Deployment, Monitoring and Feedback: Model serving concepts, REST APIs for ML applications, Containerized model deployment, Model deployment strategies, Blue-Green deployment, Canary deployment, A/B testing, Model monitoring and observability, Data drift and concept drift, Performance monitoring, Logging and alerting, Feedback loops, Continuous retraining, Human-in-the-Loop (HITL) systems, Workflow orchestration for ML pipelines, Cloud-native MlOps architectures, overview of LLMOps (Large Language Model Operations).

Streaming Systems: Real-time machine learning systems, Stream processing concepts, Event-driven architectures, Online and batch inference, Edge AI fundamentals, Case studies in MLOps for IoT and intelligent edge systems, recommendation systems, anomaly detection, predictive maintenance.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2						1	2	2	3
CO2	3	3	2	2	3						2	2	3	3
CO3	3	3	2	2	3						2	2	3	3
CO4	3	3	3	2	3						2	2	3	3
CO5	3	3	3	3	3				1	1	2	3	3	3

Textbooks / References

1. Treveil, M. et al., *Introducing MLOps*, O'Reilly Media, 2020.
2. Huyen, C., *Designing Machine Learning Systems*, O'Reilly Media, 2022.
3. Gift, N. and Deza, A., *Practical MLOps*, O'Reilly Media, 2021.
4. Burkov, A., *Machine Learning Engineering*, True Positive Inc., 2020.
5. Kleppmann, M., *Designing Data-Intensive Applications*, O'Reilly Media, 2017.
6. Google Cloud, *Practitioner's Guide to MLOps*, 2021.

CAC412 Generative Artificial Intelligence Techniques [3-0-0-3]

Course Description

This course introduces the foundations, architectures, and applications of Generative AI, with emphasis on Large Language Models, prompt engineering, Retrieval-Augmented Generation (RAG), and multimodal AI. Students explore modern generative models, ethical considerations, and real-world use cases, enabling them to design, evaluate, and responsibly deploy AI-powered intelligent systems.

Course Objectives

- Understand the fundamental concepts, evolution, architectures, and applications of Generative Artificial Intelligence.
- Explain the principles of deep learning, transformer architectures, foundation models, and Large Language Models (LLMs) that underpin modern generative AI systems.
- Analyze prompt engineering techniques, Retrieval-Augmented Generation (RAG), multimodal AI, and agentic AI for developing intelligent and autonomous AI applications.
- Develop a strong theoretical foundation to critically assess emerging trends and advancements in generative AI and related intelligent systems.

Course Outcomes

Upon completion of the course, learners will be able to:

- CO1: Explain the fundamental concepts, architectures, and applications of generative artificial intelligence, including discriminative and generative models, foundation models, and the generative AI ecosystem.
- CO2: Describe the principles of deep learning, transformer architectures, autoencoders, diffusion models, and Large Language Models (LLMs) that underpin modern generative AI systems.
- CO3: Analyze Large Language Models, prompt engineering techniques, and Retrieval-Augmented Generation (RAG) to understand knowledge-enhanced AI systems and their applications.
- CO4: Explain the concepts, architectures, and applications of multimodal AI and agentic AI, including vision-language models, multimodal learning, intelligent agents, planning, reasoning, memory, and autonomous decision-making.
- CO5: Evaluate the capabilities, limitations, opportunities, challenges, and emerging trends of generative AI for solving real-world problems in a responsible and informed manner.

Syllabus

Introduction to Generative AI, brief recap of self-attention and transformer architecture as the backbone for LLMs, Review of Generative Models- GAN, Conditional GAN (cGAN), brief recap of Autoencoders and VAE latent-space representation as foundation for Latent Diffusion Models.

Diffusion Models: Motivation, Forward and reverse diffusion process, Denoising Diffusion Probabilistic Models, Latent Diffusion Models, Conditioning.

Language Models and LLM Architecture, GPT: Pre-training and Fine-tuning strategies, Inference, Emergent Capabilities of LLMs, Knowledge Limitations of LLMs, Hallucinations in LLMs, Semantic Search, Retrieval-Augmented Generation, RAG Architecture, RAG Pipeline, Applications of RAG, Benchmarking & Evaluating Models Prompt Engineering, Zero-shot Prompting, One-shot Prompting, Few-shot Prompting, Chain-of-Thought Prompting, Structured Prompting, ReACT, Prompt Optimisation, Prompt Evaluation, Vector Databases and application development frameworks like LangChain

Introduction to Multimodal AI, Unimodal and Multimodal Learning, Foundation Models for Multimodal AI, Vision-Language Models (VLMs), Cross-modal Representation Learning, Multimodal Embeddings, Cross-modal Attention, Multimodal Fusion, Image Captioning, Visual Question Answering, Text-to-Image Generation, Image-to-Text Generation, LLaVA, BLIP, CLIP

Ethical considerations and challenges: Ethical Considerations in Generative AI Understanding the ethical implications of generative models, Deepfakes, Addressing bias and fairness in generative AI systems Ensuring responsible use and deployment of generative models, Generative AI Use Cases

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2							2	3	3
CO2	3	3	2	2	2							2	3	3
CO3	3	3	3	2	3							2	3	3
CO4	3	3	3	2	3	1						2	3	3
CO5	3	3	2	2	2	2	2	3			1	3	3	3

Textbooks / References

1. Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). Deep learning (Vol. 1, No. 2, pp. 1-800). Cambridge: MIT press.
2. Foster, David. Generative deep learning. " O'Reilly Media, Inc.", 2022.
3. Tunstall, Lewis, Leandro Von Werra, and Thomas Wolf. Natural language processing with transformers. " O'Reilly Media, Inc.", 2022.
4. Raschka, Sebastian. Build a large language model (from scratch). Simon and Schuster, 2024.
5. Huyen, C., 2024. AI engineering: Building applications with foundation models." O'Reilly Media, Inc.".
6. Rothman, Denis. Transformers for Natural Language Processing and Computer Vision: Explore Generative AI and Large Language Models with Hugging Face, ChatGPT, GPT-4V, and DALL-E 3. Packt Publishing Ltd, 2024.

CAL411 Machine Learning Operations Lab [0-0-2-1]

Course Objectives

- Provide hands-on experience in managing the machine learning lifecycle using MLOps practices and tools.
- Enable students to design, automate, and deploy machine learning workflows using containerization and CI/CD techniques.
- Familiarize students with experiment tracking, dataset versioning, and reproducible machine learning practices.
- Develop skills in building scalable and maintainable ML pipelines using orchestration frameworks and automation tools.
- Expose students to industry-standard tools and practices for model deployment, monitoring, and lifecycle management.

Course Outcomes

- CO1: Apply experiment tracking and dataset versioning techniques to ensure reproducibility in machine learning workflows.
- CO2: Implement automated model training, evaluation, and validation pipelines using CI/CD practices and orchestration tools.
- CO3: Develop and deploy machine learning models as scalable inference services using containerization technologies.
- CO4: Configure and manage MLOps infrastructure for model lifecycle management, automation, and workflow orchestration.
- CO5: Design end-to-end MLOps pipelines integrating data management, experiment tracking, deployment, and performance verification.

Syllabus

- Perform experiment tracking and management for machine learning workflows.
- Implement automated model training, evaluation, and metric reporting using continuous integration techniques.
- Develop and deploy machine learning models as inference services using containerization technologies.
- Implement automated model deployment and update workflows using continuous deployment pipelines.
- Configure and manage self-hosted automation and orchestration environments.
- Design and deploy machine learning workflow pipelines for model training and deployment.
- Develop automated validation pipelines for model inference and performance verification.
- Perform dataset versioning and management for reproducible machine learning experiments.
- Integrate data versioning with local and cloud-based storage systems.
- Establish end-to-end experiment tracking and model lifecycle management infrastructure.
- Explore workflow orchestration frameworks for machine learning pipeline automation.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	3							2	3	3
CO2	3	3	3	2	3						1	2	3	3
CO3	3	2	3	2	3						1	2	3	3
CO4	3	3	3	3	3						2	2	3	3
CO5	3	3	3	3	3				1	1	2	3	3	3

CAP411 B.Tech Project Phase I [0-0-6-3]

Course Objectives

- To enable students to identify and formulate a relevant engineering problem through a comprehensive literature survey and problem analysis.
- To develop the ability to design, implement, and validate innovative hardware and/or software solutions using modern engineering tools and techniques.
- To cultivate skills in experimentation, data analysis, project management, teamwork, technical documentation, and professional communication.

Course Outcomes

- CO1: Identify, analyze, and formulate a complex engineering problem by applying knowledge of mathematics, science, and engineering fundamentals.
- CO2: Design, develop, and implement an appropriate hardware/software system or process to address the identified problem using modern engineering tools and methodologies.
- CO3: Conduct experiments, analyze and interpret data, evaluate system performance, and draw valid conclusions based on the obtained results.
- CO4: Prepare technical reports and research documentation, and effectively communicate project outcomes through oral presentations while demonstrating professional and ethical practices

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2				1	1		2	3	2
CO2	3	3	3	2	3			1	2	2	2	2	3	3
CO3	3	3	2	3	3			1	2	2	2	2	3	3
CO4	2	2	2	2	2	1	1	3	2	3	2	3	2	2

CAP421 B.Tech Project Phase II [0-0-18-9]

Course Objectives

- To enable students to identify and formulate a relevant engineering problem through a comprehensive literature survey and problem analysis.

- To develop the ability to design, implement, and validate innovative hardware and/or software solutions using modern engineering tools and techniques.
- To cultivate skills in experimentation, data analysis, project management, teamwork, technical documentation, and professional communication.

Course Outcomes

- CO1: Identify, analyze, and formulate a complex engineering problem by applying knowledge of mathematics, science, and engineering fundamentals.
- CO2: Design, develop, and implement an appropriate hardware/software system or process to address the identified problem using modern engineering tools and methodologies.
- CO3: Conduct experiments, analyze and interpret data, evaluate system performance, and draw valid conclusions based on the obtained results.
- CO4: Prepare technical reports and research documentation, and effectively communicate project outcomes through oral presentations while demonstrating professional and ethical practices

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2				1	1		2	3	2
CO2	3	3	3	3	3			1	2	2	2	2	3	3
CO3	3	3	3	3	3		1	1	2	2	2	2	3	3
CO4	2	2	2	2	2	2	1	3	3	3	3	3	2	2

CAE302 Pattern Recognition [3-0-0-3]

Course Objectives

- To introduce the mathematical foundations of pattern recognition systems.
- To develop understanding of statistical and structural pattern analysis techniques.
- To study classification and clustering methodologies for data interpretation.
- To explore feature selection and dimensionality reduction approaches.
- To familiarize students with learning paradigms used in intelligent recognition systems.
- To enable students to analyze and evaluate recognition models for practical applications.

Course Outcomes

- CO1: Explain the concepts and components of pattern recognition systems.
- CO2: Apply statistical methods for pattern classification and decision making.
- CO3: Analyze feature extraction and dimensionality reduction techniques.
- CO4: Evaluate contemporary techniques and tools used in pattern recognition systems.
- CO5: Design and implement basic pattern recognition models for engineering applications.

Syllabus

Introduction to Pattern Recognition Systems: Introduction to pattern recognition systems, Pattern representation and description, Feature vectors and feature spaces, Types of learning: supervised, unsupervised, reinforcement, Decision-making processes, Applications of pattern recognition, Training and test sets, Feature selection, Clustering vs. Classification, Applications, Linear Algebra, Vector spaces, Probability theory, Estimation techniques.

Feature Analysis and Dimensionality Reduction: Feature generation and transformation, Feature selection methods, Principal Component Analysis (PCA), Fisher Linear Discriminant, Dimensionality reduction techniques, Curse of dimensionality.

Parameter Estimation and Clustering Methods: Maximum-Likelihood estimation: Gaussian case, Maximum Posteriori estimation, Bayesian estimation: Gaussian case, Unsupervised learning and clustering, Criterion functions for clustering, Algorithms for clustering: K-Means, Hierarchical and other methods, Cluster validation, Gaussian mixture models, Expectation-Maximization method for parameter estimation, Maximum entropy estimation, Mixtures of Gaussians (MoG).

Sequential Pattern Recognition and Non-Parametric Methods: Sequential Pattern Recognition, Hidden Markov Models (HMMs): Discrete HMMs, Continuous HMMs, Nonparametric techniques for density estimation, Parzen-window method, K-Nearest Neighbour method, Non Parametric methods, Density estimation, Parzen Windows, Probabilistic Neural Networks (PNNs).

Applications of Pattern Recognition: Recommender systems, Pattern recognition in speech and text analysis, Biometric recognition systems, Intelligent diagnostic systems, Machine vision systems.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							2	3	2
CO2	3	3	2	2	3							2	3	3
CO3	3	3	2	3	3							2	3	3
CO4	3	3	2	3	3					1		2	3	3
CO5	3	3	3	3	3	1		1	1	1		3	3	3

Textbooks / References

1. Bishop, Christopher M., *Pattern Recognition and Machine Learning*, Springer Science+Business Media, LLC, 2006.
2. Duda, Richard O. and Peter E. Hart, *Pattern Classification*, John Wiley & Sons, 2006.

3. Bishop, Christopher M., *Neural Networks for Pattern Recognition*, Oxford University Press, 1995.
4. Fu, King-Sun, *Syntactic Pattern Recognition*, Applications of Pattern Recognition, CRC Press, 2019.

CAE303 Artificial Intelligence: Knowledge Representation and Reasoning [3-0-0-3]

Course Objectives

- To understand the principles and techniques of knowledge representation and reasoning in Artificial Intelligence.
- To explore structured knowledge representation methods, including semantic networks, frames, ontologies, and knowledge graphs.
- To analyze advanced reasoning paradigms such as non-monotonic, probabilistic, evidential, temporal, and spatial reasoning for intelligent decision-making.
- To examine the role of knowledge representation in explainable AI and emerging neuro-symbolic intelligent systems.

Course Outcomes

- CO1: Explain the principles, characteristics, and formalisms of knowledge representation and their role in intelligent systems.
- CO2: Design and model domain knowledge using semantic networks, frames, ontologies, and knowledge graph representations.
- CO3: Apply non-monotonic, probabilistic, and evidential reasoning techniques to address reasoning under incomplete and uncertain information.
- CO4: Analyze temporal, spatial, and commonsense reasoning approaches and evaluate their applicability in solving real-world AI problems.
- CO5: Assess the significance of knowledge representation in explainable AI and neuro-symbolic AI for building transparent and intelligent systems.

Syllabus

Foundations of Knowledge Representation: Introduction to Knowledge Representation and Reasoning, Role of Knowledge in Artificial Intelligence, Knowledge-Based Systems, Knowledge Engineering Process, Types of Knowledge, Characteristics of a Good Knowledge Representation Scheme, Knowledge Representation Schemes and Formalisms, Issues and Challenges in Knowledge Representation.

Structured Knowledge Representation and Knowledge Graphs: Semantic Networks, Frames, Ontologies, Introduction to Knowledge Graphs, Knowledge Graph Representation, Real-world Knowledge Graphs, Knowledge Graph Applications.

Non-Monotonic Reasoning: Limitations of Classical Reasoning, Non-Monotonic Reasoning, Default Logic, Circumscription, Defeasible Reasoning, Closed World Assumption vs Open World Assumption, Belief Revision and AGM Framework.

Probabilistic and Evidential Reasoning: Limitations of Logical Reasoning under Uncertainty, Probability Theory for Reasoning, Bayesian Networks, Inference in Bayesian Networks, Markov Logic Networks, Dempster-Shafer Theory.

Temporal, Spatial and Commonsense Reasoning: Introduction to Temporal Reasoning, Situation Calculus, Event Calculus, Allen's Interval Algebra, The Frame Problem, Spatial Reasoning, Commonsense Reasoning, Explainable AI and Knowledge Representation, Neuro-Symbolic AI.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							2	3	2
CO2	3	3	3	2	3							2	3	3
CO3	3	3	2	3	3			1				2	3	3
CO4	3	3	2	3	3	1		1		1		2	3	3
CO5	3	3	3	3	3	2	1	2	1	1		3	3	3

Textbooks / References

1. Ronald J. Brachman and Hector J. Levesque, *Knowledge Representation and Reasoning*, Morgan Kaufmann Publishers, 1st Edition, 2004.
2. Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach*, 4th Edition, Pearson Education, 2021.
3. Grigoris Antoniou and Frank van Harmelen, *A Semantic Web Primer*, 3rd Edition, MIT Press, 2023.
4. Pascal Hitzler, Markus Krötzsch and Sebastian Rudolph, *Foundations of Semantic Web Technologies*, Chapman & Hall/CRC, 2009.
5. Aidan Hogan et al., *Knowledge Graphs*, Springer, 2022.
6. Elaine Rich, Kevin Knight and Shivashankar B. Nair, *Artificial Intelligence*, 3rd Edition, Tata McGraw-Hill Education, 2010.
7. Deepak Khemani, *A First Course in Artificial Intelligence*, McGraw Hill Education (India), 2013.
8. John F. Sowa, *Knowledge Representation: Logical, Philosophical, and Computational Foundations*, Brooks/Cole, 2000.

CAE304 Computer Vision [3-0-0-3]

Course Objectives

- To understand the principles of image formation, image processing, and visual perception systems.
- To analyze feature extraction, segmentation, and stereo vision techniques for visual data interpretation.
- To apply computer vision algorithms for object detection, recognition, and scene understanding.

- To develop computer vision solutions for real-world applications using modern machine learning approaches.

Course Outcomes

- CO1: Explain the fundamentals of image formation, image processing, and visual representation techniques.
- CO2: Apply feature extraction, image matching, segmentation, and stereo vision methods to analyze visual data.
- CO3: Analyze object detection, recognition, and classification techniques for solving computer vision problems.
- CO4: Design and evaluate computer vision systems for real-world applications such as surveillance, OCR, and face recognition.

Syllabus

Image Formation and Processing: Introduction to Computer Vision, Brief History, Geometric primitives and transformations, Photometric image formation, Digital camera, Point operators, Linear and nonlinear filtering, Fourier transforms, Pyramids and wavelets, Geometric transformation, Global optimizations.

Early Vision: Texture representation, Local and pooled texture representation, Texture synthesis, Image denoising, Shape from texture, Epipolar geometry, Binocular reconstruction, Human stereopsis, Local and global methods for binocular fusion, Structure from motion, Internally calibrated perspective cameras, Uncalibrated weak-perspective cameras, Uncalibrated perspective cameras.

Mid-Level Vision: Feature extraction and feature representation, Histogram of Oriented Gradients (HOG), Scale Invariant Feature Transform (SIFT), Background subtraction techniques, Image matching, Principal Component Analysis (PCA), Grouping and Gestalt principles, Image segmentation by clustering pixels, Hough transform, Fitting lines and curves, Robust fitting, RANSAC, Deformable contours, Interactive segmentation.

High-Level Vision and Applications: Object, scene, and activity categorization, Object detection, Supervised classification algorithms, Probabilistic models for sequence data, Visual attributes, Active learning, Face Recognition, Facial Expression Recognition, Optical Character Recognition (OCR), Automated Video Surveillance.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							2	3	2
CO2	3	3	2	2	3							2	3	3
CO3	3	3	2	3	3			1				2	3	3
CO4	3	3	3	3	3	2	1	2	1	1		3	3	3

Textbooks / References

1. Richard Szeliski, *Computer Vision: Algorithms and Applications*, Springer, 2021.
2. Valliappa Lakshmanan, Martin Görner, and Ryan Gillard, *Practical Machine Learning for Computer Vision*, O'Reilly Media, 2021.
3. Robert Laganiere, *OpenCV Computer Vision Application Programming Cookbook*, 2nd Edition, Packt Publishing, 2014.
4. David A. Forsyth and Jean Ponce, *Computer Vision: A Modern Approach*, Prentice Hall, 2015.
5. Md Atiqur Rahman Ahad, *Computer Vision and Action Recognition: A Guide for Image Processing and Computer Vision Community for Action Understanding*, Springer, 2014.
6. James J. DiCarlo, Davide Zoccolan, and Nicole C. Rust, "How Does the Brain Solve Visual Object Recognition?", *Neuron*, vol. 73, no. 3, pp. 415–434, 2012.

CSE302 Cloud Computing [3-0-0-3]

Course Objectives

- To gain knowledge on virtualization techniques.
- To frame VM clusters.
- To learn OS-level virtualization tools, including dockers.
- To understand the working methodology of existing clouds, such as, Amazon, Opennebula, GCE, and so forth.
- To learn how to program clouds using new programming models.

Prerequisite

- Computer Networks
- Parallel and Distributed Computing
- Fundaments of Programming

Course Outcomes

Upon successful completion of the course, the students will be able to:

- CO1: The students will learn the base technologies for cloud.
- CO2: Apply appropriate cloud services for their applications.
- CO3: Design cloud services using golang or nodejs.
- CO4: Learn how to program public clouds such as AWS or GCE.

Syllabus

Introduction: Base Technologies-Review: Introduction, Grid Computing, Cluster, P2P computing, and so forth, System Models for Distributed and Cloud computing.

Virtualization: Virtualization concepts, levels of Virtual ization, VM server consolidation, VirtualBox, VMWare Vsphere- Datacenter Automation, Performance impacts of virtualization, Docker containers, orchestration platforms– Kubernetes, Docker Swarm.

Cloud Infrastrcuture: Design Challenges of Clouds, Public cloud platforms– GCE, AWS, Azure, Resource Management in Clouds, Private cloud environments- openstack and opennebula, security aspects of clouds, Storage aspects of clouds, introduction to programming models.

Advanced Topics: DevOps, CI/CD Pipelines, Cloud Automation Tools–Terraform, Jenkins, GitHub Actions, SDN, HPC-in-the-cloud, IoT cloud, Serverless cloud.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							2	2	3
CO2	3	3	3	2	3			1				2	2	3
CO3	3	3	3	2	3			1	1	1		2	2	3
CO4	3	3	3	3	3	1		1	1	1		3	2	3

Textbooks / References

1. Kai Hwang, Geoffrey C. Fox, Jack K. Dongarra, Distributed and Cloud Computing: From parallel processing to Internet of Things, Morgen Kauffmann 2013.
2. Rajkumar Buyya, James Broberg, Andrzej Goscinski, Cloud Computing: Principles and Paradigms, 2011 John Wiley & Sons, Inc.
3. William Stallings, Foundations of Modern Networking: SDN, NFV, QoE, IoT, and Cloud, Pearson publishers, 2016
4. Jonathan Baier, Getting Started with Kubernetes: 2nd Edition, Packt publishers, 2015.
5. Vaishnavi P. Cloud Computing: An Efficient Load Balancing and Resource Provisioning in the Cloud Environmen, Scholars press, 2023.
6. Kevin Hoffman and Dan Nemeth, Cloud Native Go: Building Web Applications and Microservices for the Cloud with Go and React (Developer's Library), Pearson publishers, 2016.
7. Bob Familiar, Microservices, IoT, and Azure: Leveraging DevOps and Microservice Architecture to deliver SaaS Solutions, Apress publishers, 2015.
8. Dirk Slama, Frank Puhlmann, Jim Morrish, and Rishi M. Bhatnagar, Enterprise IoT: Strategies and Best practices for Connected products and services, O'Rielly publishers, 2015.
9. Oluyemi James Odeyinka, AWS Cloud Automation: In depth guide to automation using Terraform infrastructure as code solutions, Bpb publishers, 2024.

CSE303 Computer Systems for Machine Learning [3-0-0-3]

Course Objectives

- To understand the architecture, workflows, and performance requirements of modern machine learning systems.
- To analyze hardware platforms, storage systems, and distributed computing techniques that support scalable machine learning workloads.
- To explore advanced machine learning infrastructures, including edge AI, federated learning, model optimization, and large-scale AI systems.

Course Outcomes

- CO1: Explain the components, workflows, and performance characteristics of machine learning systems and data processing pipelines.
- CO2: Analyze and evaluate hardware architectures, parallel computing platforms, and distributed machine learning frameworks for scalable model training and inference.
- CO3: Apply machine learning system concepts to design efficient and scalable solutions using modern infrastructures such as edge AI, federated learning, and large language model systems.

Syllabus

Foundations of Machine Learning Systems : Introduction to Machine Learning Systems – Components of ML Systems – ML Training and Inference Workflows – Data Pipelines for Machine Learning – Batch and Stream Processing – Feature Engineering and Feature Stores – Performance Metrics for ML Systems – Challenges in Large-Scale Machine Learning. Hardware Systems for Machine Learning: Computer Architecture for ML – CPU and Parallel Processing – GPU Architecture and Programming Models – Tensor Processing Units (TPUs) and AI Accelerators – Memory Hierarchy and Storage Systems – Communication Bottlenecks – Performance Evaluation of ML Workloads – Energy-Efficient Computing for AI.

Distributed and Scalable Machine Learning: Distributed Computing Fundamentals – Data Parallelism – Model Parallelism – Distributed Training Architectures – Parameter Servers – Communication and Synchronization Techniques – Resource Management – Fault Tolerance in ML Systems – Scalable Deep Learning Frameworks.

Modern ML Systems and Applications: Machine Learning Frameworks and Execution Models – Hardware Acceleration for Deep Learning – Efficient Inference Systems – Edge AI and TinyML – Federated Learning Systems – Large Language Model (LLM) Infrastructure – Model Compression and Quantization – Case Studies in Industrial ML Systems.

Textbooks / References

1. Huyen, C. (2022). Designing machine learning systems: An iterative process for production-ready applications. O'Reilly Media.
2. Burkov, A. (2020). Machine learning engineering. True Positive Inc.
3. Hennessy, J. L., & Patterson, D. A. (2019). Computer architecture: A quantitative approach (6th ed.). Morgan Kaufmann.
4. Panda, D. K., & Balaji, P. (Eds.). (2018). High performance computing: Modern systems and practices. Morgan Kaufmann.

5. Warden, P., & Situnayake, D. (2019). TinyML: Machine learning with TensorFlow Lite on Arduino and ultra-low-power microcontrollers. O'Reilly Media.
6. Yang, Q., Fan, L., & Yu, H. (Eds.). (2020). Federated learning. Springer.
7. Geron, A. (2022). Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow: Concepts, tools, and techniques to build intelligent systems (3rd ed.). O'Reilly Media.

CSE304 Blockchain Technology and Applications [3-0-0-3]

Course Objectives

- To explain and analyze the fundamental concepts, architecture, and working principles of blockchain technology.
- To examine and compare the evolution of blockchain across different generations, highlighting their features, advancements, and applications.
- To apply blockchain technologies and platforms to design and develop solutions for real-world problems in various domains.

Course Outcomes

- CO1: Explain the fundamental principles of blockchain technology and its impact on industries.
- CO2:Compare different blockchain architectures and understand their key characteristics.
- CO3:Analyze the role of consensus mechanisms in ensuring trust and security in blockchain networks.
- CO4: Understand the concept of smart contracts and their significance in blockchain applications.
- CO5: Evaluate real-world blockchain applications and assess challenges related to scalability, security, and adoption.

Syllabus

Introduction to Blockchain- History and evolution of blockchain, Key Characteristics (Decentralization, Transparency, Immutability). Base technologies- Docker, peer-to-peer networks, Docker setup for blockchain nodes.
Cryptographic foundations for Blockchain- Basics of Cryptography in Blockchain, Cryptographic hash functions (SHA-256, Merkle Trees), Digital signatures and Public-Key Cryptography, Role of Hashing and Encryption in Blockchain
Blockchain Architecture and Components- Blockchain Structure (Blocks, Transactions, Nodes, Ledger), Types of Blockchain Architectures: Public, Private, Consortium, Hybrid
Consensus Mechanisms- Proof of Work (PoW), Proof of Stake (PoS), Proof of Burn (PoB).
Cryptocurrency Ecosystems- Bitcoin Fundamentals- UTXOs, wallets, transactions. Bitcoin scripting and transaction lifecycle. Ethereum fundamentals- Ethereum Virtual Machine (EVM), gas, accounts. Transactions and smart contract execution.
Smart Contracts- Writing smart contracts with Solidity. Security best practices
Enterprise Blockchain Solutions: Hyperledger Fabric architecture, Channels, Chaincode, Comparing Public and Enterprise Blockchains, Industry Use Cases.
Blockchain for Society and Industry – Blockchain applications, E-governance, smart cities, and identity management.
Scalability and Cloud Integration- Blockchain on cloud platforms (AWS QLDB or Azure Blockchain or any other).

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2	1		1				2	2	3
CO2	3	3	2	2	3	1		1				2	2	3
CO3	3	3	2	3	3	2		2				2	2	3
CO4	3	3	3	2	3	2		2	1	1		2	2	3
CO5	3	3	3	3	3	3	2	2	1	1		3	2	3

Textbooks / References

1. Douglas R Stinson, "Cryptography Theory and Practice",CRC Press, 2018.
2. William Stallings,' Cryptography and Network Security: Principles and Practice", Pearson, 2017
3. Antonopoulos, A. M. (2017). Mastering Bitcoin: Programming the Open Blockchain. O'Reilly.
4. Narayanan, A., Bonneau, J., Felten, E., Miller, A., & Goldfeder, S. (2016). Bitcoin and Cryptocurrency Technologies. Princeton University Press.

CAE305 Data Handling, Analysis and Visualization [3-0-0-3]

Course Objectives

- To expose to visual representation methods and techniques that increase the understanding of complex data.
- To quickly examine large amounts of data, expose trends efficiently, exchange ideas with key players, and influence decisions.
- To design effective visualizations, dashboards, and interactive analytics for communicating data-driven insights.

Course Outcomes

- CO1: Understanding of good design practices for visualization, tools for visualization of data from a variety of fields.
- CO2: Analyze and interpret data-driven results to support decision-making while adhering to principles of effective and responsible data visualization.
- CO3: Develop effective visualizations, dashboards, and interactive analytical reports using appropriate visualization tools and techniques.

Syllabus

Data Definitions and Analysis Techniques: Elements, Variables, and Data categorization, Levels of Measurement, Missing Data Handling, Data Cleaning and Preprocessing, Data management and indexing, Introduction to statistical learning and R-Programming.

Exploratory Data Analysis: Univariate, Bivariate, Multivariate analysis, Summary statistics, correlation matrices, EDA with R.

Descriptive Statistics: Measures of central tendency, Measures of location of dispersions, Measures of Shape, Practice and analysis with R. **Basic Analysis Techniques:** Statistical hypothesis generation and testing, Chi-Square test, t-Test, Analysis of variance, Correlation analysis, Maximum likelihood test, Practice and analysis with R. Data analysis techniques: Regression analysis, Classification techniques, Clustering, Association rules analysis, Practice and analysis with R.

Data Visualization: Basic Plotting: Line plot - Bar plot - Pie Chart - Scatter Plot - Histogram - Stacked Bar Charts - Sub Plots - Matplotlib, Searborn, Plotly - Seaborn Styles, Applied Visualizations: Box plot - Density Plot - Area Chart - Heat map - Tree map - Graph Networks-Violin Plot-Geospatial Visualization.

Interactive Visualizations and Animations: Dynamic charts - Dynamic maps - Animation types - 2D, 3D, Motion Animation - Animation Principles - Altair Package - Statistical Visualizations.

Principles of Information Visualization: Visual Perception and Cognition - Gestalt's Principles - Tufte's Principles - Applications of Principles.

Storytelling and Dashboard Design: Data storytelling frameworks, Executive dashboard design, KPI selection and visualization, Interactive dashboards.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	3					1		2	2	3
CO2	3	3	2	2	3	1		1		2		2	2	3
CO3	3	3	3	2	3	1		1	1	3		3	2	3

Textbooks / References

1. Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye, "Probability and Statistics for Engineers and Scientists", 9th Edition, Prentice Hall, 2011.
2. G James, D. Witten, T Hastie, and R. Tibshirani, "An Introduction to Statistical Learning: With Applications in R", Springer, 2013.
3. Hastie, T., Tibshirani, R., Friedman, J.H. and Friedman, J.H., "The Elements of Statistical Learning: Data Mining, Inference, and Prediction", Springer, 2nd Edition, 2017.
4. A. Rajaraman and J. Ullman, "Mining of Massive Datasets", Cambridge University Press, 2012.
5. E. Tufte, "The Visual Display of Quantitative Information", 2nd Edition, Graphics Press, 2001.
6. Cole Nussbaumer Knaflic, "Storytelling with Data: A Data Visualization Guide for Business Professionals", 2nd Edition, John Wiley & Sons, 2023.
7. Stephen Few, "Information Dashboard Design: Displaying Data for At-a-Glance Monitoring", 2nd Edition, Analytics Press, 2013.
8. Steve Wexler, Jeffrey Shaffer and Andy Cotgreave, "The Big Book of Dashboards: Visualizing Your Data Using Real-World Business Scenarios", John Wiley & Sons, 2017.

CSE309 Augmented & Virtual Reality Systems [3-0-0-3]

Course Objectives

- To understand the fundamental concepts, technologies, and evolution of Virtual Reality (VR) and Augmented Reality (AR).
- To explain the components, architectures, and development platforms used in immersive systems.
- To apply VR/AR tools and techniques to create interactive virtual and augmented environments.
- To analyze the applications, challenges, and emerging trends of VR/AR technologies in real-world domains.

Course Outcomes

- CO1: Describe the concepts, history, components, and architectures of Virtual Reality and Augmented Reality systems.
- CO2: Utilize VR/AR development tools, modeling techniques, and interaction methods to create basic immersive applications.
- CO3: Analyze the functionality and performance of VR/AR systems, including input/output devices, rendering techniques, and user interactions.
- CO4: Evaluate emerging VR/AR technologies and design simple solutions for real-world applications considering usability and human factors.

Syllabus

Virtual Reality: Introduction to Virtual Reality, The Three Is of Virtual Reality, Early VR, First Commercial VR, VR at the Turn of the Millennium, VR in the 21st Century, Components of Classical and Modern VR Systems. Input Devices: Three-Dimensional Position Trackers, Navigation and Manipulation Interfaces, Gesture Interfaces, Neural Interfaces. Output Devices: Graphics Display Characteristics, Display Technologies, Personal Graphics Displays, Large-Volume Displays, Micro-LED Walls and Holographic Displays, Three-Dimensional Sound Displays, Haptic Displays, Olfactory Displays.

VR Architecture, Modeling and Programming: Rendering pipeline, Graphics Benchmark, Distributed VR Architectures. Geometric Modeling, Kinematics Modeling, Physical Modeling, Model Management. Scene Graphs and Toolkits, Unity 3D Game Engine.

Augmented Reality: Introduction to Augmented Reality, Relationship between AR and other technologies, Concepts related to AR, How Augmented Reality works, Ingredients of an AR system. Major Components and Content: Sensors, Projectors, Displays, Software Components, Visual Content Creation, Audio Content Creation, Content Creation for Other Senses.

Interaction, Applications and Advancements: Interaction in the real world, Manipulation, Navigation, Communication and Multiperson AR Applications. Applications: Magic Books, Magic Mirrors, Magic Windows and Doors, Magic Lens, Navigation Assistance. Advanced Topics: Augmented Reality and Artificial Intelligence, Computer Vision in AR/VR, Internet of Things (IoT) Integrated AR/VR, Mixed Reality, Future Trends and Innovations in AR/VR/MR, Human Factor Issues, User Performance, Sensorial Conflict Aspects of VR/AR/MR.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							2	2	3
CO2	3	3	3	2	3	1			1	1		2	2	3
CO3	3	3	2	3	3	1			1	1		2	2	3
CO4	3	3	3	3	3	2	1	1	1	1		3	2	3

Textbooks / References

1. Burdea, G. C. and Coffet, P., *Virtual Reality Technology*, 2nd Edition, Wiley-IEEE Press, 2003.
2. Craig, A. B., *Understanding Augmented Reality: Concepts and Applications*, Morgan Kaufmann, 2013.
3. Craig, A., Sherman, W., and Will, J., *Developing Virtual Reality Applications: Foundations of Effective Design*, Morgan Kaufmann, 2009.
4. Geroimenko, V., *Augmented Reality and Artificial Intelligence*, Springer Nature Switzerland, 2024.
5. Shinde, G. R., Dhotre, P. S., Mahalle, P. N., and Dey, N., *Internet of Things Integrated Augmented Reality*, Springer Nature Singapore, 2020.
6. Maurya, R. K., *Computer Graphics with Virtual Reality System*, 3rd Edition, Wiley, 2018.
7. Sherman, W. R. and Craig, A. B., *Understanding Virtual Reality: Interface, Application, and Design*, 2nd Edition, Morgan Kaufmann, Elsevier, 2019.
8. Burdea, G. C. and Coiffet, P., *Virtual Reality Technology*, 2nd Edition, Wiley, 2017.
9. Hale, K. S. and Stanney, K. M., *Handbook of Virtual Environments*, 2nd Edition, CRC Press, 2015.

CSE311 Internet of Things & Edge Intelligence [3-0-0-3]

Course Objectives

- To understand the principles and architectures of edge-enabled IoT systems.
- To study edge computing paradigms, platforms, and service orchestration frameworks.
- To analyze distributed intelligence, edge analytics, context-aware computing, and adaptive service mechanisms.
- To understand Edge AI concepts, TinyML fundamentals, and intelligent processing in resource-constrained environments.
- To examine trust, security, privacy, resilience, and sustainability considerations in edge-enabled systems.
- To evaluate applications, emerging trends, and future directions in edge intelligence.

Course Outcomes

- CO1: Explain the architecture and operational principles of edge-enabled IoT systems.
- CO2: Analyze edge computing paradigms and distributed intelligence frameworks for IoT applications.
- CO3: Evaluate distributed intelligence, edge analytics, context-aware systems, and adaptive service mechanisms for IoT applications.
- CO4: Explain Edge AI concepts, TinyML fundamentals, and intelligent processing approaches for resource-constrained environments.
- CO5: Analyze trust, security, privacy, resilience, and sustainability requirements of intelligent edge systems.
- CO6: Design and evaluate edge-enabled IoT solutions for industrial and societal applications while assessing emerging trends in edge intelligence.

Syllabus

IoT and Edge Computing Foundations: IoT ecosystem and architecture review, Limitations of cloud-centric IoT systems, Edge computing concepts and characteristics, Fog computing, Cloudlets, Mobile Edge Computing (MEC), Edge-cloud continuum, Edge intelligence concepts, Architectural layers of edge-enabled IoT systems, Benefits and challenges of edge-enabled intelligence.

Edge Platforms and Service Management: Edge computing platforms and deployment environments, Resource-constrained edge environments, Edge service lifecycle, Workload placement concepts, Service orchestration principles, Containerization and microservices at the edge, Edge-cloud collaboration, Edge storage and data management, Quality of Service (QoS) considerations, Scalability and performance optimization, Service continuity in distributed edge systems.

Distributed Intelligence and Edge Analytics: Distributed sensing environments, Sensor fusion and multi-source data integration, Edge data processing models, Stream analytics and event-driven architectures, Data filtering and aggregation at the edge, Context-aware computing, Context modelling and reasoning, Situational awareness systems, Adaptive service delivery mechanisms, Collaborative edge intelligence, Distributed decision-making and knowledge sharing among edge nodes, Human-in-the-loop intelligence, Introduction to Edge AI concepts, TinyML fundamentals, Lightweight inference, Federated learning.

Trustworthy and Resilient Edge Systems: Security and privacy challenges in edge-enabled IoT systems, Trust management frameworks, Identity and access management concepts, Data privacy and local processing strategies, Reliability and resilience in distributed edge systems, Fault tolerance principles, Explainability and transparency in edge AI decision-making, Accountability considerations, Ethical and responsible edge intelligence, Sustainable and energy-efficient edge computing practices.

Emerging Edge Intelligence Systems: Industrial IoT (IIoT) and smart manufacturing, Industry 4.0 and Industry 5.0 perspectives, Edge-enabled digital twins, Autonomous monitoring systems, Collaborative edge intelligence, AI-native edge ecosystems, Intelligent edge-cloud continuum, Edge intelligence for cyber-physical systems, 6G-enabled edge services, Next-generation intelligent infrastructure, Future trends and challenges in edge intelligence.

Textbooks / References

1. Shajulin Benedict, *Edge Intelligence: Deep Learning-Enabled Edge Computing*, IOP Publishing, 2024.
2. Perry Lea, *IoT and Edge Computing for Architects: Implementing Edge and IoT Systems from Sensors to Clouds with Communication Systems, Analytics, and Security*, 2nd Edition, Packt Publishing, 2020.
3. Rute C. Sofia and John Soldatos (Eds.), *Shaping the Future of IoT with Edge Intelligence: How Edge Computing Enables the Next Generation of IoT Applications*, River Publishers, 2024.
4. Rajkumar Buyya and Satish Narayana Srirama (Eds.), *Fog and Edge Computing: Principles and Paradigms*, Wiley, 2019.
5. Sudip Misra, Anandarup Mukherjee, and Arijit Roy, *Introduction to IoT*, Cambridge University Press, 2021.
6. Pethuru Raj and Anupama Raman, *The Internet of Things: Enabling Technologies, Platforms and Use Cases*, CRC Press, 2017.

CSE312 Computational Complexity [3-0-0-3]

Course Objectives

- To understand major complexity classes and methods used to determine the difficulty of problems.
- To explore the role of randomness, counting, and hierarchy concepts in computation.
- To gain exposure to advanced topics and important open questions in computational complexity theory.

Course Outcomes

- CO1: Identify and classify computational problems into major complexity classes such as P, NP, coNP, NL, and PSPACE.
- CO2: Use reductions to determine the computational difficulty of problems and recognize NP-complete and PSPACE-complete problems.
- CO3: Describe how randomness and counting are used in computational complexity.
- CO4: Understand important open problems, including the P versus NP question.

Syllabus

Foundations: Models of computation, Algorithms and complexity, Time and space measures, Deterministic and nondeterministic computation.

Complexity Classes: P, NP, coNP, Verifiers and certificates, Reductions, NP-completeness, Cook-Levin theorem, Standard NP-complete problems.

Time Complexity: Time hierarchy theorem, Polynomial hierarchy, Other complexity classes.

Space Complexity: L, NL, PSPACE, Savitch's theorem, NL-completeness, NL = coNL, PSPACE-completeness, Space hierarchy theorem.

Randomization and Counting: Randomized algorithms, Primality testing, BPP, RP, coRP, #P and #P-completeness.

Advanced Topics: Communication complexity, Interactive proofs, PCP theorem (overview), Open problems including P vs NP.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	2	1							2	3	2
CO2	3	3	2	3	1							2	3	2
CO3	3	3	1	3	1							2	3	2
CO4	3	3	1	2								3	3	2

Textbooks / References

1. Sanjeev Arora and Boaz Barak, *Computational Complexity: A Modern Approach*, Cambridge University Press, 2009.
2. Oded Goldreich, *Computational Complexity: A Conceptual Perspective*, Cambridge University Press, 2008.
3. Christos H. Papadimitriou, *Computational Complexity*, Pearson, 1993.

CSE313 Online Algorithms [3-0-0-3]

Course Objectives

- To understand the online computation model and its distinction from offline computation.

- To learn competitive analysis as the primary tool for evaluating online algorithms.
- To design and analyze deterministic and randomized online algorithms.
- To study classical online optimization problems and their lower bounds.
- To explore applications of online algorithms in caching, scheduling, networking, cloud computing, and resource allocation.

Course Outcomes

- CO1: Explain the online computation model, adversarial inputs, and competitive analysis.
- CO2: Analyze deterministic and randomized online algorithms and derive competitive ratios.
- CO3: Design online algorithms for caching, searching, matching, and scheduling problems.
- CO4: Apply primal-dual and probabilistic techniques in online optimization problems.
- CO5: Evaluate online algorithms for real-world applications such as cloud resource management, load balancing, and network routing.

Syllabus

Introduction to Online Computation: Introduction to online computation, Online versus offline algorithms, Adversarial models, Competitive analysis, Deterministic and randomized online algorithms, Lower bounds.

Fundamental Online Problems: Ski-rental problem, Search problems, Secretary problems, Data management and packing problems, List accessing algorithms, Move-to-front heuristic, Online bin-packing algorithms.

Paging and Caching Algorithms: LRU, FIFO, Randomized paging, Competitive analysis, Paging with predictions.

Metric and Matching Problems: Metrical task systems, K-server problem, Online knapsack, Online bipartite matching algorithms.

Advanced Online Algorithms: Online set cover, Load balancing, Online scheduling.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	2	1							2	3	2
CO2	3	3	2	3	2							2	3	2
CO3	3	3	3	2	2	1						2	3	3
CO4	3	3	3	3	2							2	3	3
CO5	3	3	3	3	3	2	1			1		3	3	3

Textbooks / References

1. *Online Algorithms*, Cambridge University Press, 2023.
2. Allan Borodin and Ran El-Yaniv, *Online Computation and Competitive Analysis*, Cambridge University Press, 1998.
3. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein, *Introduction to Algorithms*, 4th Edition, MIT Press, 2022.

CSE315 Network Optimization [3-0-0-3]

Course Objectives

- To provide fundamental knowledge of graphs, network flow models, and their real-world applications.
- To introduce classical optimization problems such as shortest path, maximum flow, and minimum cost flow with standard algorithms.
- To develop understanding of advanced solution techniques including simplex, dual, and primal-dual methods for network optimization.
- To expose students to nonlinear and modern extensions of network optimization including convex, integer, and multicommodity flow problems.

Course Outcomes

- CO1: Formulate network optimization problems using graph-based models and flow representations.
- CO2: Apply standard algorithms and shortest path techniques to solve network problems.
- CO3: Analyze and solve min-cost flow and related optimization problems using simplex, dual ascent, and primal-dual approaches.
- CO4: Evaluate and apply advanced network optimization methods for nonlinear, convex, and integer-constrained network problems.

Syllabus

Introduction: Graphs and Flows, Network Flow Models, Examples, Algorithms - An Overview. Shortest Path Problems: Problem Formulation and Applications, A Generic Shortest Path Algorithm, Label Setting Methods, Label Correcting Methods, Single Origin and Single Destination Methods, Auction Algorithms.

Network Flow: Price-Based Augmenting Path Algorithms, The Min-Cost Flow Problem, Transformations and Equivalences, Duality.

Simplex Methods for Min-Cost Flow: Main Ideas in Simplex Methods, The Basic Simplex Algorithm, Extension to Problems with Upper and Lower Bounds. Dual Ascent Methods for Min-Cost Flow: Dual Ascent, The Primal-Dual (Sequential Shortest Path) Method, The Relaxation Method, Auction Algorithms for Min-Cost Flow.

Nonlinear Network Optimization: Convex and Separable Problems, Problems with Side Constraints, Multicommodity Flow Problems, Integer Constraints, Networks with Gains, Optimality Conditions, Duality, Algorithms and Approximations.

Convex Separable Network Problems: Convex Functions of a Single Variable, Optimality Conditions, Duality, Dual Function Differentiability, Algorithms for Differentiable Dual Problems, Auction Algorithms. Network Problems with Integer Constraints: Introduction to modern extensions in network optimization.

Textbooks / References

1. Dimitri P. Bertsekas, *Network Optimization: Continuous and Discrete Models*, Athena Scientific, 1st Edition, 1998.
2. R. K. Ahuja, T. L. Magnanti, and J. B. Orlin, *Network Flows: Theory, Algorithms and Applications*, Prentice Hall, 1993.
3. Elias Munapo, Santosh Kumar, Philimon Nyamugure, and Trust Tawanda, *Network Optimization: An Introduction to the Network Reconstruction Approach*, River Publishers, 2025.
4. David P. Williamson, *Network Flow Algorithms*, Cambridge University Press, 2019.

CSE316 Introduction to Quantum Machine Learning [3-0-0-3]

Course Objectives

- To establish a solid theoretical foundation in quantum computing principles specifically tailored for computer science workflows.
- To conceptualize the paradigm shift from classical artificial intelligence to quantum-enhanced machine learning frameworks.
- To master quantum data representation techniques, including state preparation and data encoding mechanisms.
- To analyze, design, and implement foundational quantum machine learning algorithms using modern quantum programming frameworks.

Course Outcomes

By the end of this course, students will be able to:

- CO1: Analyze quantum circuits and evaluate basic quantum algorithms using qubits, gates, and measurement principles.
- CO2: Differentiate between CC, CQ, QC, and QQ AI variants and identify problem domains where quantum speedups are mathematically viable.
- CO3: Implement optimal quantum encoding strategies (amplitude, angle, basis) to translate classical data into quantum states.
- CO4: Develop and simulate variational quantum algorithms and quantum neural networks for classification and optimization tasks.

Syllabus

Foundations of Quantum Computing: Complex vector spaces, Bra-Ket notation, inner products, Hilbert spaces, and Hermitian matrices. Mathematical representation of a qubit, superposition, and the Bloch Sphere visual representation. Single-qubit gates (X, Y, Z, H, Phase gates) and multi-qubit gates (CNOT, CZ). Reversibility of quantum circuits. EPR pairs, Bell states, quantum entanglement as a computational resource, Born's rule, and projective measurements. High-level overview of Deutsch-Jozsa and Grover's Search algorithm to demonstrate classical vs. quantum complexity bounds.

The AI-Quantum Intersection & Taxonomy: Quick review of classical empirical risk minimization, optimization landscapes, and computational bottlenecks in deep learning. The Four Quadrants of Quantum AI: Detailed classification of standard machine learning vs. quantum approaches: CC: Classical Data, Classical Algorithm (Standard ML), QC: Quantum Data, Classical Algorithm (Quantum characterization), CQ: Classical Data, Quantum Algorithm (Quantum-enhanced ML), QQ: Quantum Data, Quantum Algorithm (Pure quantum processing), Core Differences (ML vs. QML): State space scaling (2n hilbert space advantages), processing primitives (unitary transformations vs. non-linear activations), and the nature of quantum optimization landscapes.

Quantum Data Pipeline: Representation & Encoding: The No-Cloning Theorem, Core limitations in handling data inside quantum states and its implications for data pipelines. Quantum State Preparation, Data Encoding Techniques - Basis Encoding, Amplitude Encoding, Angle / Feature Encoding. Quantum Kernels & Feature Maps - Transforming data into high-dimensional Hilbert spaces using quantum circuits; introduction to explicit vs. implicit quantum feature spaces.

Deep Dive into Quantum Machine Learning Algorithms: Variational Quantum Circuits (VQCs) - The foundational backbone of NISQ-era QML; Parametrized Quantum Circuits (PQCs), cost function evaluation, and hybrid quantum-classical optimization loops. The Barren Plateau Problem - gradient vanishing phenomena in quantum optimization landscapes and mitigation strategies. Quantum Neural Networks (QNNs) - Designing quantum classifiers, parameter shift rules for computing quantum gradients, and quantum convolutional layers (QCNNs). Quantum Support Vector Machines (QSVM) - Quantum kernel estimation methods and running kernel-based classification on hybrid architectures. Unsupervised QML & Quantum Clustering- Introduction to Quantum K-Means, Quantum Principal Component Analysis (QPCA), and Quantum Generative Adversarial Networks (QGANs).

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	2	2							2	3	2
CO2	3	3	2	2	2							3	3	3
CO3	3	3	3	2	3							3	3	3
CO4	3	3	3	3	3	1		1	1			3	3	3

Textbooks / References

1. Jesse Van Griensven Thé, Roydon Andrew Fraser, and Jose Rosas-Bustos. Quantum Computing and Quantum Machine Learning for Engineers and Developers. Springer, 2025.
2. Nielsen, Michael A., and Isaac L. Chuang. Quantum computation and quantum information. Cambridge university press, 2010.
3. Schuld, Maria, and Francesco Petruccione. Supervised learning with quantum computers. Vol. 17. Berlin: Springer, 2018..
4. Combarro, Elias F., Alberto Di Meglio Samuel González-Castillo, and Alberto Di Meglio. A practical guide to quantum machine learning and quantum optimization. Packt Publishing, 2023.
5. Melnikov, Alexey, et al. "Quantum machine learning: from physics to software engineering." Advances in Physics: X 8.1 (2023): 2165452.

CAE402 Dependable and Secure AI [3-0-0-3]

Course Objectives

- To understand the concepts of dependable AI, including reliability, robustness, safety, and security.
- To analyze vulnerabilities, adversarial threats, and defense mechanisms in machine learning systems.
- To apply privacy-preserving, fair, and accountable AI techniques to mitigate risks and biases.
- To design and evaluate secure AI deployment and governance strategies for responsible AI systems.

Course Outcomes

- CO1: Explain the principles of dependable, secure, and trustworthy AI systems and evaluate their reliability and robustness.
- CO2: Analyze and mitigate adversarial attacks, security threats, privacy risks, and failures in AI models.
- CO3: Apply fairness, privacy-preserving, and responsible AI techniques to develop ethical AI solutions.
- CO4: Design secure AI pipelines and recommend governance, monitoring, and deployment practices for trustworthy AI applications.

Syllabus

Introduction to Dependable AI: Resilience, robustness, safety, and security, Reliable Neural Networks, Fault Models, Assessing Fault Tolerance, Redundancy, Reliability during the Learning Phase, Methodology for Fault Tolerance, Visualization Levels for Neural Networks, Fault Locations, Fault Manifestations, Fault Coverage, AI threat landscape and attack taxonomy (NIST AI 100-2), Security in Machine Learning (ML) context using the CIA triad, Reliability and safety in AI systems, Real-world failure case studies.

Adversarial Attacks and Defenses: Evasion attacks including FGSM, PGD, and C&W attacks, Physical-world adversarial examples, Poisoning and backdoor attacks including BadNets and clean-label poisoning, Model extraction attacks, Membership inference and privacy leakage, Defense mechanisms including adversarial training, randomized smoothing, certified robustness, IBM ART toolkit.

Privacy-Preserving and Responsible AI: Differential privacy, Formal definition, Laplace and Gaussian mechanisms, DP-SGD, Federated learning protocols, Privacy benefits, Byzantine robustness, Secure aggregation, Fairness in AI, Sources of bias, Metrics including demographic parity, equalized odds, and calibration, Algorithmic accountability, LLM safety including jailbreaks, prompt injection, red-teaming, and systemic AI risks.

Secure AI Systems, MLOps, Deployment and Governance: Securing the ML pipeline including data collection, training, and serving, Model registry and artifact signing, Homomorphic encryption basics, Secure Multi-Party Computation (SMPC) for ML (overview), AI model watermarking, Concept drift monitoring in production, MLOps security, Algorithmic impact assessments, Responsible AI governance and disclosure.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	2	2	2	1	2				2	3	3
CO2	3	3	2	3	3	2	1	2				2	3	3
CO3	3	3	3	2	3	3	2	3	1	1		2	3	3
CO4	3	3	3	3	3	3	2	3	1	1		3	3	3

Textbooks / References

1. K. R. Varshney, *Trustworthy Machine Learning*, Independently Published, 2022.
2. Solon Barocas, Moritz Hardt, and Arvind Narayanan, *Fairness and Machine Learning*, MIT Press.
3. Christoph Molnar, *Interpretable Machine Learning*, 2nd Edition.
4. Christopher Bishop and Hugh Bishop, *Deep Learning: Foundations and Concepts*, Springer, 2024.
5. Kevin Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022.
6. Cynthia Dwork, "Differential Privacy," *ICALP*, 2006.
7. Martin Abadi et al., "Deep Learning with Differential Privacy," *ACM CCS*, 2016.

CAE403 Explainable AI & Interpretable ML [3-0-0-3]

Course Objectives

- To define and explain the importance of explainability and interpretability in artificial intelligence systems.
- To introduce and critically analyze techniques for enhancing model transparency, including feature importance, rule-based models, and surrogate models.

- To cover post hoc explanation methods (LIME, SHAP, counterfactuals) and attention/concept-based interpretability for deep neural networks.
- To provide hands-on experience in applying XAI techniques to real-world datasets and problems across diverse domains.
- To examine the ethical, legal, and societal implications of AI transparency, including fairness, the right to explanation, and the right to be forgotten.

Course Outcomes

- CO1: Effectively apply interpretability techniques including feature importance analysis, rule-based models, and model-agnostic methods to enhance the transparency of machine learning models.
- CO2: Evaluate and critically compare different interpretability methods with respect to fidelity, stability, and computational cost.
- CO3: Develop a critical understanding of the trade-offs between model complexity and interpretability across diverse scenarios and application domains.
- CO4: Demonstrate practical skills in applying explainability techniques to real-world datasets and challenges using industry-standard tools and frameworks.
- CO5: Gain awareness of the ethical implications related to AI transparency, fairness, accountability, and interpretability in sociotechnical systems.

Syllabus

Introduction to Explainable AI: Science of Interpretable Machine Learning, Motivation, Challenges, and Mythos of Model Interpretability, Human Factors in Explainability, Interpreting Interpretability.

Post hoc Explanations: Explaining the Predictions of Any Classifier, Pitfalls, Challenges, and Evaluation of Feature Attributions, LIME and SHAP, OpenXAI, The Disagreement Problem in Explainable Machine Learning, Counterfactual Explanations (or) Algorithmic Recourse, Learning Model-Agnostic Counterfactual Explanations for Tabular Data.

Attention and Concept Based Explanations: Quantifying Interpretability of Deep Visual Representations, Interpretability Beyond Feature Attribution, Data Attribution and Interactive Explanation, Equitable Valuation of Data, Explainable Active Learning (XAL), Theory of Explainability and Interpreting Generative Models.

Explainability for Fair Machine Learning: Connections with Robustness, Privacy, Fairness, and Unlearning, Right to Explanation and the Right to be Forgotten, Fairness via Explanation Quality, Mechanistic Interpretability and Compiled Transformers, Understanding and Reasoning in Large Language Models.

Advanced Topics and Applications: Hands-on Application of Explainability Techniques to Real-World Datasets, Case Studies in Healthcare, Finance, and Legal Domains, Ethical Implications of AI Transparency and Accountability, Emerging Trends: XAI for Generative AI and Foundation Models, Integrating XAI into End-to-End Machine Learning Pipelines.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3	1		1				2	3	3
CO2	3	3	2	3	3	1		2				2	3	3
CO3	3	3	3	2	3	2	1	2	1			2	3	3
CO4	3	3	3	3	3	2	1	2	1	1		3	3	3
CO5	2	3	2	2	2	3	2	3	1	1		3	2	3

Textbooks / References

1. Michael Munn and David Pitman, *Explainable AI for Practitioners*, O'Reilly Media, Inc., 2022.
2. Leonida Gianfagna and Antonio Di Cecco, *Explainable AI with Python*, Springer, 2021.
3. Christoph Molnar, *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*, Leanpub, 2023.
4. Denis Rothman, *Hands-On Explainable AI (XAI) with Python*, Packt Publishing, 2020.

CAE404 Reinforcement Learning [3-0-0-3]

Course Objectives

- To understand the Reinforcement Learning (RL) framework, Markov Decision Process (MDP) formulation, Bellman equations, and exploration-exploitation trade-off.
- To apply classical RL algorithms including Dynamic Programming (DP), Monte Carlo (MC), Temporal Difference (TD), Q-Learning, and SARSA to sequential decision problems.
- To analyze and compare RL algorithms for sample efficiency, stability, and applicability.
- To apply advanced paradigms including Reinforcement Learning from Human Feedback (RLHF), Multi-Agent Reinforcement Learning (MARL), and Model-Based RL to real-world Artificial Intelligence (AI) problems.

Course Outcomes

- CO1: Explain the RL framework, MDP formulation, Bellman equations, and value functions.
- CO2: Implement classical RL algorithms including DP, MC, TD, Q-Learning, and SARSA on benchmark environments.
- CO3: Critically compare RL algorithms for efficiency, convergence, stability, and problem suitability.
- CO4: Apply advanced RL paradigms including RLHF, MARL, and Model-Based RL to real-world domains and reflect on ethical implications.

Syllabus

Introduction to RL: RL framework, Agent, Environment, Reward, Return, Policy, Value function, Model, Exploration versus exploitation, RL versus supervised learning, Survey of applications including AlphaGo, ChatGPT, and robotics.

MDPs and Dynamic Programming: Markov property, MDP formulation (S, A, P, R, γ), Bellman Expectation and Optimality Equations, Policy Evaluation, Policy Iteration, Value Iteration, Generalized Policy Iteration, Limitations of Dynamic Programming.

Model-Free Prediction and Control: Monte Carlo prediction and control, Temporal Difference learning methods, TD(0), SARSA, Q-Learning, Expected SARSA, Double Q-Learning, Eligibility traces TD(λ), n-step TD methods, Stochastic approximation and convergence of TD and Q-Learning.

Deep Reinforcement Learning: Function approximation, Deep Q-Network (DQN), Experience replay, Target networks, Double DQN, Dueling DQN, Rainbow, Prioritized Replay DDQN, Distributional DQN, Noisy DQN.

Policy Gradient and Actor-Critic Methods: REINFORCE algorithm and stochastic policy search, Vanilla Policy Gradient (VPG), Asynchronous Advantage Actor-Critic (A3C), Deep Deterministic Policy Gradient (DDPG), Twin-Delayed DDPG (TD3), Soft Actor-Critic (SAC), Proximal Policy Optimization (PPO).

Advanced Topics and Applications: Model-Based RL, Offline RL (CQL), Hierarchical RL, Multi-Agent RL, Safe RL, RLHF pipeline, Reward model, PPO fine-tuning, InstructGPT.

Textbooks / References

1. Richard S. Sutton and Andrew G. Barto, *Reinforcement Learning: An Introduction*, 2nd Edition, MIT Press, 2018.
2. Maxim Lapan, *Deep Reinforcement Learning Hands-On*, 3rd Edition, Packt Publishing, 2024.
3. Stefano V. Albrecht, Filippos Christianos, and Lukas Schäfer, *Multi-Agent Reinforcement Learning*, MIT Press, 2024.
4. Nikhil Sanghi, *Deep Reinforcement Learning with Python: RLHF for Chatbots and Large Language Models*, 2nd Edition, Apress, 2024.
5. Laura Graesser and Wah Loon Keng, *Foundations of Deep Reinforcement Learning*, Addison-Wesley, 2020.

CAE405 Federated Learning and Privacy Preserving AI [3-0-0-3]

Course Objectives

- To understand the fundamentals of distributed and federated machine learning.
- To analyze privacy risks and security challenges in AI systems.
- To apply federated learning algorithms for decentralized model training.
- To implement privacy-preserving techniques such as differential privacy and secure aggregation.
- To evaluate the trade-offs between privacy, security, accuracy, communication efficiency, and scalability.
- To design privacy-aware AI solutions for real-world applications.

Course Outcomes

Upon successful completion of the course, the students will be able to:

- CO1: Explain the principles, architectures, and challenges of federated learning systems.
- CO2: Implement federated optimization algorithms for distributed machine learning tasks.
- CO3: Analyze privacy threats and security vulnerabilities in collaborative AI systems.
- CO4: Apply privacy-preserving techniques including differential privacy, secure aggregation, and cryptographic methods.
- CO5: Evaluate federated learning frameworks and compare their performance under different deployment scenarios.
- CO6: Design and develop privacy-preserving AI solutions for practical applications in healthcare, finance, edge computing, and IoT.

Syllabus

Introduction to Federated Learning and Privacy: Preserving AI-Motivation for distributed AI-Data privacy challenges in machine learning-Centralized vs distributed learning paradigms-Introduction to federated learning-Federated AI ecosystem and applications-Edge computing and federated learning-Privacy regulations and ethical considerations.

Fundamentals of Federated Learning: Federated learning architectures-Cross-device federated learning-Cross-silo federated-learning-Federated optimization -Federated Averaging (FedAvg)-Communication-efficient learning-Client selection strategies-Handling non-IID data-Personalization in federated learning-Model aggregation techniques-Scalability challenges.

Privacy and Security in Federated Learning: Privacy threats in machine learning-Membership inference attacks-Model inversion attacks-Data reconstruction attacks-Poisoning attacks-Data poisoning-Model poisoning-Backdoor attacks-Adversarial machine learning-Security requirements in federated systems-Trust models and threat modelling.

Privacy-Preserving Techniques: Differential Privacy (Local DP-Global DP)-Privacy budget and noise mechanisms-Secure Multi-Party Computation (SMPC)-Homomorphic Encryption-Secure Aggregation protocols-Trusted Execution Environments (TEE)-Privacy-utility trade-offs-Combining

FL with differential privacy.

Advanced Federated Learning Methods: Personalized Federated Learning-Federated Transfer Learning-Federated Meta-Learning-Federated Reinforcement Learning-Hierarchical Federated Learning-Split Learning-Decentralized and Peer-to-Peer Learning-Resource-aware federated learning-Federated foundation models and generative AI.

Tools, Frameworks, and Applications: Federated learning frameworks-TensorFlow Federated-Flower-FedML-PySyft-Experimental setup and benchmarking-Performance evaluation metrics-Federated learning different applications-Emerging trends and research challenges-Case studies and recent developments.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	2	2	1		1				2	3	3
CO2	3	3	3	2	3	1		1				2	3	3
CO3	3	3	2	3	3	2	1	2				2	3	3
CO4	3	3	3	2	3	2	1	2	1			2	3	3
CO5	3	3	2	3	3	2	1	2	1	1		3	3	3
CO6	3	3	3	3	3	3	2	3	1	1		3	3	3

Textbooks / References

1. Q. Yang, Y. Liu, Y. Cheng, Y. Kang, T. Chen, and H. Yu, Federated Learning. San Rafael, CA, USA: Morgan & Claypool Publishers, 2019.
2. J. Li, P. Li, Z. Liu, X. Chen, and T. Li, Privacy-Preserving Machine Learning. Singapore: Springer Nature Singapore, 2022. doi: 10.1007/978-981-16-9139-3.
3. Y. Jin, H. Zhu, J. Xu, and Y. Chen, Federated Learning: Fundamentals and Advances. Singapore: Springer Nature Singapore, 2023. doi: 10.1007/978-981-19-7083-2.
4. G. D. Samaraweera, D. Zhuang, and J. M. Chang, Privacy-Preserving Machine Learning. Shelter Island, NY, USA: Manning Publications, 2023.
5. S. R. Aravilli, Privacy-Preserving Machine Learning: A Use-Case-Driven Approach to Building and Protecting ML Pipelines from Privacy and Security Threats. Birmingham, U.K.: Packt Publishing, 2024

CAE406 Graph Neural Network and Geometric Deep Learning [3-0-0-3]

Course Objectives

- Unify diverse deep learning architectures under a single geometric framework by applying group theory, representation theory, and Felix Klein's Erlangen Program.
- Formulate advanced relational architectures.
- Design specialized neural network operators that preserve structural and spatial symmetries across sets, point clouds, continuous manifolds, and 3D geometric spaces.

Course Outcomes

- CO1: To Construct custom Message Passing Neural Networks (MPNNs) including GCNs, GraphSAGE, and GATs
- CO2: To Generalize localized deep learning operators across continuous manifolds, point clouds, and discrete meshes by integrating principles of topology and differential geometry
- CO3: To Design invariant and equivariant network layers (such as Deep Sets, PointNet, and Tensor Field Networks) to process coordinate-free set structures and 3D spatial orientations.
- CO4:To Formulate complex geometric pipelines using Heterogeneous Graphs, Knowledge Graphs, and SE(3)-Transformers to solve domain-specific relational and directional tasks.

Syllabus

Groups and Symmetries: Foundations of group theory, representation theory, and invariant/equivariant functions.The Erlangen Program: Using Felix Klein's geometric blueprint to unify deep learning architectures (e.g., CNNs, RNNs, GNNs).Topology & Differential Geometry: Manifolds, meshes, and continuous spaces.
Graph Basics: Adjacency matrices, degree matrices,Classical Embedding Methods: Node2Vec, DeepWalk, and spectral clustering.
Message Passing Neural Networks (MPNN):General framework for neighborhood aggregation, transformation,Graph Convolutional Networks (GCN), GraphSAGE, Graph Attention Networks.
Heterogeneous Graphs & Knowledge Graphs
(E(n))-equivariant GNNs, Tensor Field Networks, and SE(3)-Transformers, Sets and Point Clouds: Deep Sets and PointNet, Transformers as GNNs

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3							2	3	3
CO2	3	3	2	3	3							2	3	3
CO3	3	3	3	2	3	1						2	3	3
CO4	3	3	3	3	3	1			1			3	3	3

Textbooks / References

1. Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges, Authors: Michael M. Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković
2. Graph Neural Networks: Foundations, Frontiers, and Applications, Authors: Lingfei Wu, Peng Cui, Jian Pei, and Liang Zhao
3. Graph Representation Learning, Authors: William L. Hamilton

CAE407 Medical Image Analysis [3-0-0-3]

Course Objectives

- To understand medical imaging modalities and image formation principles
- To apply image processing techniques for medical images
- To perform segmentation, registration, and feature extraction

Course Outcomes

- CO1: Explain the principles of medical imaging modalities, image acquisition and fundamental image processing techniques used in medical image analysis.
- CO2: Apply preprocessing techniques such as image enhancement, denoising, and contrast improvement to improve the quality of medical images
- CO3: Analyze medical images using region selection, segmentation, deformable models, and mathematical morphology techniques to identify structures of clinical interest.
- CO4: Implement image registration and feature extraction methods to derive meaningful representations from medical images.
- CO5: Evaluate medical image classification approaches by employing similarity measures, feature engineering techniques, and pattern classification methods in disease diagnosis applications.

Syllabus

Introductory Concepts: Medical Imaging Modalities, Image acquisition and artifacts, Image processing techniques:- Point processing and Neighbourhood processing, Image Interpolation, Processing of colour images and conversions, DICOM, Histopathology images.

Preprocessing of Medical Images: Measures of Image Quality, Adaptive Histogram Equalization, CLAHE, Image Denoising, Speckle reduction techniques, Case study:- X-ray, Ultrasound.

Region selection and Analysis: Classical Methods: -Region-Based methods, Edge-Based methods, Watershed: -Marker controlled watershed, Parametric Deformable Models: - Active Contour Models (Snakes), Geometric Deformable Models: - Level Set Methods and Level Set Evolution, Mathematical Morphology, Case study: MRI.

Image Registration & Feature Engineering: Image Registration: -Rigid and Non-Rigid, Feature extraction techniques: - Intensity-based, Texture features (GLCM), Shape features, Shape Descriptors, Chain Codes, Similarity Measures, Pattern Classification, Case study: - Classification of Tumours.

Lab Practice:

- Implementation of basic medical image processing operations and DICOM image handling.
- Image enhancement and preprocessing using histogram equalization, CLAHE, and denoising techniques.
- Segmentation using region-based, edge-based, watershed, and active contour methods.
- Extraction of Shape, Intensity and texture features from segmented Region of Interest.
- Case studies on X-ray, ultrasound, MRI, and tumour classification datasets.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2	1						2	2	3
CO2	3	3	2	2	3	1						2	3	3
CO3	3	3	2	3	3	2						2	3	3
CO4	3	3	3	3	3	2			1			2	3	3
CO5	3	3	3	3	3	3	1	1	1	1		3	3	3

Textbooks / References

1. Rangayyan, Rangaraj M. Biomedical image analysis. CRC press, 2004.
2. Jiri Jan "Medical Image Processing, Reconstruction and Analysis: Concepts and Methods", Vol. 2. CRC Press, 2019.
3. John L. Semmlow, and Benjamin Griffel. Biosignal and medical image processing. CRC press, 2014.
4. Kayvan Najarian and Robert Splinter, "Biomedical Signal and Image Processing", Second Edition, CRC Press, 2005.
5. Atam Dhawan,"Medical image analysis" 2nd ed. John Wiley & Sons, 2011.
6. Jerry L. Prince and Jonathan Links, "Medical Imaging Signals and Systems", First Edition, Prentice Hall, 2005.

CSC411 Information Retrieval Semantic Search [3-0-0-3]

Course Objectives

- Explain the transition from traditional, keyword-based information retrieval to modern dense vector spaces and semantic search.
- Deconstruct multi-stage search pipelines, including embedding generation, hybrid indexing, and neural reranking architectures.
- Formulate production-grade Retrieval-Augmented Generation (RAG) applications by integrating vector databases and Large Language Models (LLMs).
- Design autonomous, tool-augmented search agents capable of executing multi-step reasoning loops to solve complex queries.

Course Outcomes

- CO1: Implement lexical rankers (BM25) and neural text embedding pipelines to capture semantic meaning in unstructured data
- CO2: Evaluate chunking strategies, vector database indexing methodologies (HNSW/IVF), and hybrid score fusion techniques for optimal retrieval recall.
- CO3: Build an integrated, hallucination-resistant RAG pipeline that securely links enterprise document sets to generative AI contexts.

- CO4: Critique the performance of intelligent retrieval systems using standardized evaluation metrics like faithfulness and context relevance

Syllabus

Introduction, IR pipeline, Lexical search, Inverted indexing and sparse retrieval, TF-IDF, Statistical ranking, BM25 algorithm, Limitations of lexical search.

Dense Representations: Bi-Encoders vs. Cross-Encoders. Generating text embeddings via neural models (BERT, Sentence-Transformers).

Vector Indexing & Approximate Nearest Neighbor (ANN) Architecture and operations of modern Vector Database, Hybrid search and re-ranking.

Retrieval-Augmented Generation (RAG) pipeline, Query Rewriting/Expansion, SelfRAG.

From Passive Retrieval to Agentic Search, Reasoning and Acting (ReAct Framework), Tool-Augmented Search, Multi-step/Routing Agents.

Multi-Modal IR, Evaluation of Search & RAG Systems: Precision, Recall, mAP, RAGTraid, RAGAs.

Lab Practice:

- Baseline Keyword Search
- Probabilistic Ranking with BM25
- Vector Spaces & Dense Embedding Generation
- Data Preprocessing for AI Search (chunking strategies)
- Vector Databases Part 1: Indexing & ANN Storage
- Two-Stage Hybrid Retrieval & Cross-Encoder Re-ranking
- Building a Foundations RAG System
- Autonomous Search Agents (ReAct Framework)

Textbooks / References

1. Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze, Introduction to Information Retrieval, Cambridge University Press
2. Stanford CS276: Information Retrieval and Web Search, Platform: Stanford Online / YouTube.

CSE402 Embedded AI and TinyML [3-0-0-3]

Course Objectives

- Fundamentals of ML and embedded devices
- Data collection from sensors; dataset creation and preparation
- Training and deploying tiny ML models
- Programming with TensorFlow Lite for Microcontrollers

Course Outcomes

- CO1: Optimizing models for resource-constrained devices
- CO2: Conceiving and designing your own TinyML application
- CO3: Awareness of cutting-edge TinyML R&D

Syllabus

Embedded AI Definition, Advantages, Challenges, and Applications

Fundamentals of TinyML: Introductions, Applications, challenges, ML Paradigm, Building block of Deep learning, Exploring machine learning scenario, Building a computer vision model and Responsible AI

Applications of TinyML: AI lifecycle and ML workflow, ML on mobile and edge devices, Keyword Spotting (KWS), Visual Wake Words (VWW), Anomaly Detection, Data Engineering, Responsible AI Development

Deploying TinyML: C++ for SW/HS Setup, Sensors INterfacing, Overview of Embedded Hardware and Software, TensorFlow Lite Micro, hands-on deployment of KWS, VWW and gesture recognition on Arduino/ Raspberry Pi

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3							2	3	3
CO2	3	3	3	2	3	1			1	1		2	3	3
CO3	2	2	1	2	3	1	1	1				3	2	3

Textbooks / References

1. Pete Warden, Daniel Situnayake, TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers, O'Reilly Media, 2020.
2. Daniel Situnayake, Jenny Plunkett, AI at the Edge, O'Reilly Media, 2023.
3. Gian Marco Iodice, TinyML Cookbook, Second Edition, Packt Publishing, 2023

CSE403 Robotics and Autonomous Systems [3-0-0-3]

Course Objectives

- To introduce the basic concepts of robotics and its specifications.
- To introduce the different sensors and actuators used in robots.
- To introduce the various end effector configuration and their design considerations for a particular application.
- To impart knowledge of the basic kinematic model of robotic manipulators.

Course Outcomes

- CO1: Describe the anatomy and specifications of Robots.
- CO2: Summarize different sensors and actuators for robots.
- CO3: Choose a robotic configuration and grippers for a particular application.
- CO4: Obtain basic kinematic model of robotic manipulators.

Syllabus

Definition and origin of robotics, Robot Anatomy, Robot specifications, Robot characteristics, Accuracy, Precision, and Repeatability, Classification of robots, Anatomy of a robotic manipulator, Links, Joints, Actuators, Sensors, End effector, Controller.

Sensors in robotics: Sensor classification, Internal sensors, Position sensors, Velocity sensors, Acceleration sensors, Force sensors, External sensors, Range, Proximity, Touch, Force-torque sensing.

Actuators for robots: Actuators classification, Electric, Hydraulic, Pneumatic actuators, Their advantages and disadvantages, End effectors classification, Mechanical grippers, Magnetic grippers, Vacuum grippers, Adhesive grippers, Selection and design considerations of grippers in robot.

Computer Vision and Image Processing for Robotics: Object Detection and Recognition, Motion Detection and Tracking, Introduction to Deep Learning for Vision, Vision Sensors and Cameras in Robotics, Applications, Traffic Sign Recognition, Obstacle Detection, Autonomous Driving Vision Systems.

Robot Kinematics and Localization Techniques: Homogenous coordinates and transfer representations, Kinematics chains, Introduction to forward kinematics and inverse kinematics, Localization Techniques, Mapping Concepts, Simultaneous Localization and Mapping (SLAM), Path Planning Algorithms, Dijkstra Algorithm, A* Algorithm, Obstacle Detection and Avoidance.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							1	2	2
CO2	3	2	2	2	3	1						1	2	3
CO3	3	3	3	2	3	2			1	1		2	3	3
CO4	3	3	2	3	3	2			1			2	3	3

Textbooks / References

1. Mikell P. Groover et al., *Industrial Robots Technology, Programming and Applications*, McGraw Hill, New York, 2008.
2. Saeed B. Niku, *Introduction to Robotics*, Pearson, 2011.
3. S. K. Saha, *Introduction to Robotics*, McGraw Hill Education.
4. John J. Craig, *Introduction to Robotics*, Pearson, 2009.
5. R. K. Mittal and I. J. Nagrath, *Robotics and Control*, Tata McGraw Hill, New Delhi, 2003.
6. Spong and Vidyasagar, *Robot Dynamics and Control*, McGraw Hill.
7. Fu K. S., Gonzalez R. C., and Lee C. S. G., *Robotics: Control, Sensing, Vision and Intelligence*, McGraw Hill, 1987.
8. Gonzalez and Woods, *Digital Image Processing*.
9. Corke, P., *Robotics, Vision and Control: Fundamental Algorithms in MATLAB*, 2nd Edition, Springer, Cham, 2017.

CSE404 Quantum Computing & Post-Quantum Cryptography [3-0-0-3]

Course Objectives

- To introduce the fundamental principles of quantum mechanics relevant to computing and cryptography.
- To explain and analyze the working of quantum algorithms that impact classical cryptographic systems.
- To examine the threat posed by quantum computing to existing public-key cryptosystems such as RSA, ECC, etc.
- To explore and evaluate post-quantum cryptographic algorithms designed to resist quantum attacks.
- To apply post-quantum cryptography in practical security solutions and understand standardization efforts.

Course Outcomes

- CO1: Explain the basic concepts of quantum computing, including qubits, superposition, entanglement, and quantum gates.
- CO2: Analyze and compare major quantum algorithms such as Shor's and Grover's algorithms and their impact on cryptography.
- CO3: Assess the vulnerabilities of classical cryptographic systems in a post-quantum era.
- CO4: Understand and differentiate between various families of post-quantum cryptographic algorithms, such as lattice-based, hash-based, code-based, multivariate, etc.
- CO5: Evaluate real-world readiness, performance, and adoption challenges of post-quantum cryptography in existing infrastructure.

Syllabus

Introduction to Quantum Computing: Basics of quantum mechanics, motivation-from classical bits to qubits, superposition, entanglement, measurement, Quantum gates and circuits, Comparison with classical computing.

Mathematical Foundations: Complex vector spaces, Dirac notation, bra-ket, tensor products, unitary matrices, Probability amplitudes, Review of linear algebra and group theory for quantum applications.

Quantum Algorithms and their Cryptographic Impact: Quantum Fourier Transform, Shor's algorithm and its threat to RSA, ECC, Diffie-Hellman, Grover's algorithm and its effect on symmetric key lengths, Overview of Deutsch-Jozsa, Bernstein-Vazirani, and Simon's algorithm.

Quantum Cryptography: Quantum Key Distribution (QKD): BB84 protocol, E91 protocol, Principles of no-cloning theorem, Limitations and practical challenges of QKD.

Post-Quantum Cryptography (PQC) Fundamentals: Motivation-why quantum computers break classical public-key crypto, NIST Post-Quantum Cryptography Standardization process, Security models for PQC.

Post-Quantum Algorithms: Lattice-based cryptography, Kyber, Dilithium, Hash-based signatures, XMSS, SPHINCS+, Code-based cryptography, McEliece, BIKE, Multivariate cryptography, Rainbow, GeMSS, Isogeny-based cryptography, SIKE, Comparison of key sizes, signature sizes, performance, and security assumptions.

Hybrid Approaches and Migration Strategies: Hybrid classical and post-quantum protocols, Cryptographic agility, Challenges in migrating TLS and Blockchain, Post-quantum Signatures, PQC in secure messaging, Cloud storage, IoT devices.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							2	3	2
CO2	3	3	2	2	2	1						2	3	3
CO3	3	3	2	3	2	2	1					2	3	3
CO4	3	2	3	2	3	1	1					2	3	3
CO5	2	3	3	3	3	2	2	1	1	1		3	2	3

Textbooks / References

1. Rieffel, E. G., Polak, W. H., Quantum Computing: A Gentle Introduction. MIT Press, 2011.
2. Kaye, P., Laflamme, R., Mosca, M., An introduction to quantum computing, 1st ed., Oxford University Press, 2007.
3. Nielsen, M. A., Chuang, I. L., Quantum Computation and Quantum Information: 10th Anniversary Edition. Cambridge University Press, 2010.
4. Benenti, G., Casati, G., Rossini, D., Strini, G., Principles of quantum computation and information: A comprehensive textbook, 2nd ed., World Scientific, 2019.
5. Bernstein, D. J., Lange, T. (2017). Post-Quantum Cryptography. Springer.
6. Yan, S. Y., Quantum Computing for Computer Scientists. Cambridge University Press, 2022.

CSE405 Malware Analysis & Reverse Engineering [3-0-0-3]

Course Objectives

- To learn to architect secure, multi-OS isolated analysis labs to contain sophisticated threats during execution.
- To gain a thorough understanding of x86, x64, and ARM architectures and how they translate to machine-level instructions.
- To familiarize the methodologies for reverse engineering proprietary and malicious software across Windows, Linux, and Mobile platforms.
- To identify and bypass complex obfuscation, anti-debugging, and anti-VM triggers used by high-level threat actors.
- To explore the application of machine learning and deep learning for automated malware classification and large-scale detection.
- To learn earn to document Indicators of Compromise (IoCs) and TTPs to support enterprise incident response teams

Course Outcomes

- CO1: Set up analysis environemts with multi-AV scanners and sandboxes.
- CO2: Cconduct static triage and dynamic monitoring to map malware behavior.
- CO3: Understand the assembly coding and analyze code for x86, x64, and ARM systems to identify malicious logic, regardless of the target platform.
- CO4: Decompile and analyze bytecode for various applications.
- CO5: Understand and evaluate ML and DL-based malware detectors using extracted behavioral features.
- CO6: Use knowledge of malware development to create custom detection scripts or rules for proactive hunting.

Syllabus

Introduction, Types of Malware, Malware Functionalities and Persistence,Code Injection and and Hooking, Malware Obfuscation Techniques, Malware Classification Methods, Setting up the environment.

Basic analysis, Assembly language and Disassembly,Debugging malicious binaries, Sandboxes and Multi AV Scanners.

Reverse Engineering Foundations, comparing x86, x64,ARM, windows & linux fundamentals, Applied Reversing, Beyond the Documentation, Deciphering File Formats, Auditing Program Binaries,Reversing malware,Java ,.NET,Mobile and linux malware analysis.

The role of AI/ML in malware analysis.understanding ml based malware detectors,bulding and evaluating malware dtectors, Application of deep leaning in malware detection and analysis, Introduction to malware development.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	3	1		1				2	3	3
CO2	3	3	2	3	3	1		1				2	3	3
CO3	3	3	2	3	3	1		1				2	3	3
CO4	3	3	3	3	3	1		1				2	3	3
CO5	3	3	2	3	3	2	1	1				3	2	3
CO6	3	3	3	3	3	2	2	2	1	1		3	3	3

Textbooks / References

1. Monnappa, K. A. (2018). Learning Malware Analysis: Explore the concepts, tools, and techniques to analyze and investigate Windows malware. Packt Publishing.
2. Sikorski, M., & Honig, A. (2012). Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software. No Starch Press.
3. Ligh, M., Adair, S., Hartstein, B., & Richard, M. (2010). Malware Analyst's Cookbook and DVD: Tools and Techniques for Fighting Malicious Code. Wiley Publishing. ISBN: 978-0-470-61303-
4. Dang, B., Gazet, A., Bachaalany, E., & Josse, S. (2014). Practical reverse engineering: x86, x64, ARM, windows kernel, reversing tools, and obfuscation. John Wiley & Sons

5. Eilam, E. (2005). Reversing: Secrets of reverse engineering. John Wiley & Sons.
6. Andriesse, D. (2018). Practical Binary Analysis: Build Your Own Linux Tools for Binary Instrumentation, Analysis, and Disassembly. No Starch Press.
7. Han, Q., Mandujano, S., Porst, S., Subrahmanian, V. S., & Tetali, S. D. (2023). The Android Malware Handbook: Detection and Analysis by Human and Machine. No Starch Press.
8. Joshua Saxe and Hillary Sanders. 2018. Malware Data Science: Attack Detection and Attribution. No Starch Press, USA.

CAE408 Applied Predictive Analytics [3-0-0-3]

Course Objectives

- To extend student's knowledge in the area of Data Science with emphasis on Predictions utilizing associated statistical methods and software tools.
- Students shall better understand the use in its commercial application to a domain such as Telecommunication/ Insurance and should be able to independently work on a dataset using statistical tools.

Course Outcomes

- CO1: Ability to apply specific statistical and regression analysis methods applicable to predictive analytics to identify new trends and patterns, uncover relationships, create forecasts, predict likelihoods, and test predictive hypotheses.
- CO2: Ability to develop and use various quantitative and classification predictive models based on various regression and decision tree methods.
- CO3: Develop familiarity with popular tools and software used in industry for predictive analytics.
- CO4: Learn how to apply the models to interpret and report on results for a management audience for a domain such as Telecommunications / Insurance

Syllabus

Overview: Supervised, Unsupervised and Semi-supervised modelling / learning.

Overview: Regression model building framework: Problem definition, Data pre-processing; Model building; Diagnostics and validation

Introduction to Statistical Inference: Null Hypothesis (H_0) and Alternative Hypothesis (H_1), Significance Level (α), p-value and Decision Making, Type I and Type II Errors, Statistical Power, Parametric vs. Non-Parametric Tests, Assumptions of Statistical Testing

Simple Linear Regression: Coefficient of determination,Significance tests, Residual analysis, Confidence and Prediction intervals.

Multiple Linear Regression: Multiple coefficient of determination, Interpretation of regression coefficients, Categorical variables, Heteroscedasticity, Multi-collinearity, outliers, Auto regression and transformation of variables, Regression model building.

Logistic and Multinomial Regression: Logistic function, Estimation of probability using logistic regression, Deviance, Wald Test, Hosmer Lemshow Test, Classification table.

Model Comparison Tests: Paired t-test , Wilcoxon Signed-Rank Test , McNemar's Test, Friedman Test, Nemenyi Post-hoc Test

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3							2	3	3
CO2	3	3	3	2	3					1		2	3	3
CO3	2	2	2	2	3				1	1		2	2	3
CO4	2	3	2	3	3	1			2	2		2	2	3

Textbooks / References

1. James, Witten, Hastie and Tibshirani "An Introduction to Statistical Learning: with Applications in R" by, Springer, 1st. Edition, 2013.
2. Alberto Cordoba, "Understanding the Predictive Analytics Lifecycle", Wiley, 2014.
3. Eric Siegel, Thomas H. Davenport, "Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die", Wiley, 2013.
4. James R Evans, "Business Analytics – Methods, Models and Decisions", Pearson 2013.
5. R. N. Prasad, Seema Acharya, "Fundamentals of Business Analytics", Wiley, 2015.

CAE409 Time Series Analytics [3-0-0-3]

Course Objectives

- To develop the ability to preprocess, model, and forecast time series data using classical statistical methods and modern machine learning techniques
- To develop the capability to build data-driven forecasting models and evaluate their effectiveness in solving real-world problems.

Course Outcomes

- CO1: Understand the fundamental concepts of time series data, stationarity, trend, seasonality, and autocorrelation.
- CO2: Analyze and preprocess time series datasets by identifying temporal patterns and performing data preparation operations.
- CO3: Apply classical statistical and autoregressive methods for time series forecasting.
- CO4:Perform feature engineering and exploratory analysis for forecasting applications.
- CO5:Develop machine learning and deep learning-based forecasting models and evaluate their performance.

Syllabus

Introduction - Time series, time series analysis, and time series forecasting. Stationary Models and the Autocorrelation Function, Estimation and Elimination of Trend and Seasonal Components.

Overview of Time Series Forecasting, Flavors of Machine Learning for Time Series Forecasting, Supervised Learning for Time Series Forecasting. Time Series Forecasting Template-Business Understanding and Performance Metrics, Data Ingestion, Data Pre-processing and Feature Engineering, An Overview of Demand Forecasting Modeling Techniques.

Time Series Data Preparation, Common Data Preparation Operations for Time Series. Time stamps vs. Periods, Time Series Exploration and Understanding, Time Series Feature Engineering. Introduction to Autoregressive and Automated Methods for Time Series Forecasting. Introduction to deep learning models for Time Series Forecasting. RNN, LSTM, GRU, Temporal Convolutional Networks, and Temporal Fusion Transformers.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							2	2	3
CO2	3	3	2	2	3							2	3	3
CO3	3	3	2	3	3							2	3	3
CO4	2	3	2	3	3				1	1		2	2	3
CO5	3	3	3	3	3	1			1	1		3	3	3

Textbooks / References

1. Francesca Lazzeri (2020), Machine Learning for Time Series Forecasting with Python, John Wiley & Sons, Inc.
2. Peter J. Brockwell, Richard A. Davis (2016), Introduction to Time Series and Forecasting, Springer, 3rd Edition.
3. Chris Chatfield (2003), The Analysis of Time Series–An Introduction, Chapman and Hall / CRC, 6th Edition.
4. George E.P. Box, Gwilym M. Jenkins, Gregory C. REINSEL, Greta M. Ljung (2015), Time Series Analysis-Forecasting and Control, Pearson, 5th Edition.
5. S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," arXiv preprint arXiv:1803.01271, 2018.
6. Bryan Lim, Sercan Ö. Arık, Nicolas Loeff, Tomas Pfister, Temporal Fusion Transformers for interpretable multi-horizon time series forecasting, International Journal of Forecasting, Volume 37, Issue 4, 2021,Pages 1748-1764, ISSN 0169-2070, https://doi.org/10.1016/j.ijforecast.2021.03.012

CAE410 Scientific Machine Learning [3-0-0-3]

Course Objectives

- To introduce the algorithmic and computational foundations of Scientific Machine Learning.
- To understand the role of scientific computing and differential equations in machine learning-based modelling.
- To study automatic differentiation and optimization techniques used in scientific machine learning.
- To explore Physics-Informed Neural Networks and Physics-Informed Gaussian Processes for solving scientific and engineering problems.

Course Outcomes

- CO1: Understand the algorithmic and computational foundations of Scientific Machine Learning.
- CO2: Apply basic scientific computing techniques for modelling systems governed by differential equations.
- CO3: Understand automatic differentiation and its role in optimization and neural network training.
- CO4: Analyse Physics-Informed Neural Networks and Physics-Informed Gaussian Processes for solving scientific and engineering problems.

Syllabus

Introduction to Scientific Machine Learning, Scientific laws, mathematical models, and governing equations, Review of Machine Learning, Least-squares parametric regression using linear and neural network models, Non-parametric regression using Gaussian Processes, Applications of Scientific Machine Learning in science and engineering.

Basic notions of Ordinary Differential Equations (ODEs) and Partial Differential Equations (PDEs), Initial and boundary value problems, Numerical solutions using classical discretization schemes, Introduction to scientific computing for differential equation-based modelling.

Introduction to automatic differentiation, Forward-mode and reverse-mode automatic differentiation, Computational graphs and gradient computation, Backpropagation and optimization in neural network training.

Physics-Informed Neural Networks (PINNs), PINN architecture and training methodology, Initialization, gradient-based optimization, adaptive learning rates, learning rate scheduling, and second-order optimization methods, Self-Adaptive PINNs, Forward and inverse problem modelling using PINNs.

Physics-Informed Gaussian Processes (PIGPs), Introduction to Reproducing Kernel Hilbert Spaces (RKHS), Solving differential equations using Gaussian Processes, Gaussian Processes as optimal recovery in RKHS, Comparison of PINNs and PIGPs for scientific modelling.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							2	3	2
CO2	3	3	2	3	3							2	3	3
CO3	3	2	2	3	3							2	3	3
CO4	3	3	3	3	3	1			1			3	3	3

Textbooks / References

1. Baker et al., *Workshop Report on Basic Research Needs for Scientific Machine Learning: Core Technologies for Artificial Intelligence*, 2019.
2. Raissi, M., Perdikaris, P., and Karniadakis, G. E., *Physics-Informed Neural Networks: A Deep Learning Framework for Solving Forward and Inverse Problems Involving Nonlinear Partial Differential Equations*, Journal of Computational Physics, 2019.
3. Yifan et al., *Solving and Learning Nonlinear PDEs with Gaussian Processes*, Journal of Computational Physics, 2021.
4. Iserles, A., *A First Course in the Numerical Analysis of Differential Equations*, Cambridge University Press, 2009.
5. Boyd, J. P., *Chebyshev and Fourier Spectral Methods*, Dover Publications, 2013.
6. Griewank, A. and Walther, A., *Evaluating Derivatives: Principles and Techniques of Algorithmic Differentiation*, SIAM, 2008.

CAE411 Streaming Data Analytics [3-0-0-3]

Course Objectives

- To introduce the principles, challenges, and computational models for processing and analyzing continuous data streams in real-time environments.
- To develop the ability to design and implement stream analytics solutions using machine learning, complex event processing, and distributed stream processing techniques.
- To provide practical experience with modern stream processing platforms and tools for extracting actionable insights from high-velocity data.

Course Outcomes

- CO1: Understand the fundamental concepts, characteristics, and challenges of streaming data and real-time analytics.
- CO2: Apply data stream processing techniques, including sampling, sliding windows, and synopsis construction, for stream analytics.
- CO3: Develop machine learning models for streaming data using classification, clustering, and frequent pattern mining techniques.
- CO4: Design and implement complex event processing solutions for real-time event detection and analysis.
- CO5:Build and evaluate scalable stream analytics applications using modern platforms such as Apache Kafka for real-world data streams.

Syllabus

Introduction to Data Streams: Characteristics of the data streams, Challenges in mining data streams Requirements and principles for real time processing, Concept drift Incremental learning Basic Streaming Methods, Counting the Number of Occurrence of the Elements in a Stream, Counting the Number of Distinct Values in a Stream, Bounds of Random Variables, Poisson Processes, Maintaining Simple Statistics from Data Streams, Sliding Windows, Tumbling windows, Data Synopsis.

Machine Learning from Data Streams: Decision Tree (Very Fast Decision Tree Algorithm, The Base Algorithm), Clustering (Partitioning, Hierarchical), Frequent Pattern Mining (Mining Frequent Itemset from Data Streams - Landmark window), Concept Drift Detection.

Complex Event Processing: Introduction to Complex Event Processing, Features of CEP, Need for CEP, CEP Architectural Layers, Scaling CEP, Events, Timing and Causality, Event Patterns, Rules and Constraints, STRAW-EPL, Event Hierarchies.

Advanced concepts on Stream Analytics: Synopsis construction in data streams- sampling methods- wavelets sketches - histograms. Join processing in data streams, indexing and querying data streams, dimensionality reduction, forecasting on streams, event-time processing, watermarking, and distributed mining of data streams.

Apache Kafka with case study: Kafka Architecture, Installation, Configuration, Performance Tuning, Kafka Client APIs and Stream APIs, Kafka Connect API, and Kafka Integration with Big Data ecosystems. Case Study on Twitter Streaming, Stock Market Prediction.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							2	2	3
CO2	3	3	2	2	3							2	3	3
CO3	3	3	3	3	3							2	3	3
CO4	2	3	3	3	3	1			1	1		2	3	3
CO5	2	3	3	3	3	1			1	1		3	2	3

Textbooks / References

1. Charu C. Aggarwal, “Data Streams: Models and Algorithms”, Kluwer Academic Publishers, 2007.
2. Joao Gama, “Knowledge Discovery from Data Streams”, CRC Press, 2010.
3. Fischer, Josh, and Ning Wang. Grokking Streaming Systems: Real-time Event Processing. Simon and Schuster, 2022.
4. Bifet, Albert, Ricard Gavalda, Geoffrey Holmes, and Bernhard Pfahringer. Machine learning for data streams: with practical examples in MOA. MIT Press, 2023.
5. Byron Ellis, “Real Time Analytics: Techniques to Analyze and Visualize Streaming Data”, John Wiley &

Sons, 2014.
6. Shilpi Saxena, Saurabh Gupta, "Practical Real-Time Data Processing and Analytics", Packt Publishing, 2017.
7. Neha Narkhede, Gwen Shapira, Todd Palino, "Kafka: The Definitive Guide: Real-Time Data and Stream Processing at Scale", O'Reilly Media, 2017.

CAE412 Bioinformatics & Computational Biology [3-0-0-3]

Course Objectives

- To provide fundamental knowledge of molecular biology concepts relevant to bioinformatics and computational biology.
- To introduce biological databases and computational tools for storage, retrieval, and analysis of biological data.
- To develop skills in sequence alignment, genome analysis, and phylogenetic studies using bioinformatics techniques.
- To familiarize students with computational approaches in proteomics, structural biology, and systems biology.
- To enable students to apply data analytics and machine learning concepts for solving biological research problems.

Course Outcomes

- CO1: Explain the fundamental concepts of molecular biology, genomics, proteomics, and computational biology.
- CO2: Retrieve, manage, and analyze biological information using major bioinformatics databases and tools.
- CO3: Apply sequence alignment and phylogenetic methods to study evolutionary relationships among organisms.
- CO4: Analyze protein structures, gene expression data, and biological networks using computational approaches.
- CO5: Design computational solutions and employ data-driven techniques for biological data analysis and research applications.

Syllabus

Introduction to Bioinformatics and Biological Databases: Introduction to Bioinformatics and Computational Biology- Scope, applications, and challenges, central Dogma of Molecular Biology ,DNA, RNA, and Protein Structure- Genomics, Transcriptomics, and Proteomics Biological Databases: Primary, Secondary, and Composite Databases , Nucleotide Databases (GenBank, EMBL, DDBJ), Protein Databases (UniProt, PDB), Literature Databases and Data Retrieval Techniques

Sequence Analysis and Comparative Genomics: Biological Sequence Representation, Pairwise Sequence Alignment, Global and Local Alignment Needleman-Wunsch Algorithm, Smith-Waterman Algorithm, Scoring Matrices (PAM and BLOSUM), Multiple Sequence Alignment Comparative Genomics, Sequence Similarity Search using BLAST and FASTA.

Phylogenetics and Genome Analysis: Molecular Evolution Concepts, Phylogenetic Tree Construction -Distance-Based Methods, Character-Based Methods, Genome Organization and Annotation Gene Prediction Techniques, Comparative Genome Analysis, Next-Generation Sequencing (NGS) Overview, Applications in Precision Medicine

Proteomics and Structural Bioinformatics: Protein Structure Levels, Protein Structure Prediction- Homology Modeling, Protein Folding Concepts, Structural Databases, Protein Function Prediction Molecular Docking Basics, Protein-Protein Interactions, Introduction to Drug Discovery and Computational Pharmacology

Systems Biology and Computational Approaches: Systems Biology Concepts, Biological Networks and Pathway Analysis, Gene Expression Analysis, Microarray and RNA-Seq Data Analysis, Biological Data Mining, Machine Learning Applications in Bioinformatics, Artificial Intelligence in Computational Biology, Personalized Medicine, Emerging Trends in Bioinformatics and Computational Biology.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2	1						2	2	3
CO2	3	3	2	2	3	1						2	3	3
CO3	3	3	2	3	3	1						2	3	3
CO4	3	3	3	3	3	2			1			2	3	3
CO5	3	3	3	3	3	2	1		1	1		3	3	3

Textbooks / References

1. Bioinformatics: Sequence and Genome Analysis – David W. Mount, Cold Spring Harbor Laboratory Press.
2. Introduction to Bioinformatics – Arthur M. Lesk, Oxford University Press.
3. Bioinformatics and Functional Genomics – Jonathan Pevsner, Wiley-Blackwell.
4. Fundamentals of Bioinformatics and Computational Biology, PHI Learning.
5. Bioinformatics: A Practical Guide to the Analysis of Genes and Proteins, Wiley.
6. Introduction to Computational Biology, Chapman & Hall/CRC. Computational Biology: A Hypertextbook, CRC Press.

CAE413 Social Network Analysis & Graph Mining [3-0-0-3]

Course Objectives

- Understand the components of the network
- Model and visualize the network.
- Learn various graph mining techniques for extracting knowledge from networked data.
- Apply social network analysis and graph mining methods to real-world problems.

Course Outcomes

- CO1: Illustrate the basic concepts of network science.
- CO2:Develop social network applications.
- CO3:Develop graph mining methods.
- CO4:Analyse and solve real-world problems using social network analysis and graph mining approaches.

Syllabus

Introduction: Introduction to network analysis, Fundamental concepts in network analysis, Social network data, Handling real-world network datasets.

Network Measures and Social Network Analysis: Degree Distribution, Characteristic Path Length and Diameter, Network Efficiency, Centrality Measures, Clustering Coefficient and Transitivity, Modularity and Assortativity, Hubs and Authorities, Network Centralization and Density, Reciprocity, Ego Networks and Measures for Ego Networks, Dyadic and Triadic Networks, Cliques, Groups and Cohesive Subgroups, Core-Periphery Structure, Rich Club Phenomenon, Motifs, Influence and Information Diffusion, Search in Networks, Signed Networks and Structural Balance, Strength of Weak Ties, Homophily, Link Analysis, Random Walks, PageRank Centrality.

Network Models: Random Networks: Erdős–Rényi Model. Small-World Networks: Milgram's Experiment, Six Degrees of Separation, Watts–Strogatz Model. Scale-Free Networks: Power Law Behaviour, Preferential Attachment, Barabási–Albert Model. Network Evolution Models: Dynamical Models, Growing Models. Nodal Attribute Models: Exponential Random Graph Models, Hybrid Models of Network Formation.

Community networks: Community structure - modularity, overlapping communities -detecting communities in social networks – discovering communities: methodology, applications - community measurement - evaluating communities – Applications.

Graph Mining: Motivation for Graph Mining, Applications of Graph Mining, Mining Frequent Subgraphs, Transaction-Based Graph Mining, BFS/Apriori Approaches (FSG and Related Algorithms), DFS-Based Approaches (gSpan and Related Algorithms), Mining Significant Graph Patterns, Branch-and-Bound Approaches.

Semantic Web: Modelling and aggregating social network data – developing social semantic application – evaluation of web-based social network extraction – Data Mining – Text Mining in social network – Tools –case study.

Interpretable graph intelligence: Responsible Computing - Introduction, Robustness and Explainability, Fairness, Bias and Privacy, Best Practices.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							2	2	3
CO2	3	3	3	2	3	1			1	1		2	3	3
CO3	3	3	3	3	3	1						2	3	3
CO4	3	3	3	3	3	2	1	1	1	1		3	3	3

Textbooks / References

1. Tanmoy Chakraborty, Social Network Analysis, Wiley India Pvt Ltd, 2021.
2. Nilanjan Dey, Samarjeet Borah, Rosalina Babo, Amira S. Ashour, Social Network Analytics, Computational Research Methods and Techniques, 2018.
3. Stanley Wasserman, Katherine Faust, Social network analysis: Methods and applications, Cambridge university press, 2012.
4. Albert-László Barabási. Network Science. Cambridge University Press, 2016.
5. J. Han and M. Kamber, Data mining–Concepts and Techniques, 2ndEdition, Morgan kaufmanPublishers, 2006
6. Bing Liu, Web Data Mining: Exploring Hyperlinks, Contents, and Usage Data, Springer publishing, 2009
7. Jure Leskovec, Anand Rajaraman, Jeff Ullman. Mining of Massive Datasets. Book 2nd edition. Cambridge University Press
8. Deepayan Chakrabarti and Christos Faloutsos. Graph Mining: Laws, Tools, and Case Studies. Synthesis Lectures on Data Mining and Knowledge Discovery, Morgan & Claypool Publishers,2012
9. Samek, Wojciech, Grégoire Montavon, Andrea Vedaldi, Lars Kai Hansen, and KlausRobert Müller, eds. Explainable AI: interpreting, explaining and visualizing deep learning. Vol. 11700. Springer Nature, 2019.
10. Solon Barocas, Moritz Hardt, and Arvind Narayanan, Fairness and Machine Learning: Limitations and Opportunities, (Available Online).
11. Kearns, Michael, and Aaron Roth, The ethical algorithm: The science of socially aware algorithm design. Oxford University Press, 2019.

CSE406 Multimedia Security & Forensics [3-0-0-3]

Course Objectives

- Understand the fundamentals of multimedia systems and Generative AI technologies relevant to digital media applications.
- Examine multimedia security mechanisms and information hiding techniques used for protecting multimedia content.
- Understand forensic principles and techniques for analyzing the authenticity and integrity of multimedia content.
- Explore security and privacy challenges associated with AI-generated multimedia and synthetic media.
- Understand emerging approaches for content authentication, provenance, and trust in modern multimedia ecosystems.

Course Outcomes

- CO1: Explain the fundamental concepts of multimedia systems, multimedia content representation, and Generative AI technologies.

- CO2: Apply multimedia security mechanisms and information hiding techniques for protecting digital multimedia content.
- CO3: Analyze multimedia content using forensic techniques to identify manipulation and establish content authenticity.
- CO4: Evaluate security and privacy challenges associated with AI-generated multimedia and synthetic media.
- CO5: Assess approaches for multimedia content authentication, provenance, traceability, and trustworthy multimedia ecosystems.

Syllabus

Multimedia and Generative AI Foundations: Introduction to Multimedia, Multimedia Data Types and Characteristics, Multimedia Representation, Storage, Transmission and Compression, Multimedia Applications and Ecosystem, Fundamentals of Generative AI, Generative Approaches for Multimedia Content Creation, Multimodal AI Systems and Applications.

Multimedia Security and Information Hiding: Security Requirements in Multimedia Systems, Confidentiality, Integrity, Authentication and Availability, Multimedia Encryption and Content Protection, Digital Watermarking and Fingerprinting, Information Hiding Techniques and Analysis, Digital Rights Management, Privacy and Security Challenges in Multimedia Systems.

Multimedia Forensics and Content Authentication: Principles of Multimedia Forensics, Metadata and File Structure Analysis, Forensic Analysis of Multimedia Content, Multimedia Tampering Detection, Source Attribution and Device Identification, Multimedia Authentication, Forensic Tools and Case Studies.

Security of Generative AI and Synthetic Media: Generative Techniques for Multimedia Synthesis, AI-generated Multimedia Content, Synthetic Media and Associated Threats, Detection and Verification of AI-generated Multimedia Content, Security and Privacy Challenges in Generative AI Systems.

Trustworthy Multimedia and AI Ecosystems: Authentication and Provenance of AI-generated Content, Content Credentials and Traceability, Ethical and Legal Considerations, Emerging Standards and Future Trends in Multimedia Security and Forensics.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							2	2	3
CO2	3	3	3	2	3	1		1				2	3	3
CO3	3	3	2	3	3	2		1		1		2	3	3
CO4	2	3	2	3	3	3	2	3	1	1		3	2	3
CO5	2	3	3	2	3	3	2	3	1	1	1	3	2	3

Textbooks / References

1. Frank Y. Shih, Multimedia Security: Watermarking, Steganography, and Forensics, CRC Press.
2. Anthony T. S. Ho and Shujun Li (Eds.), Multimedia Forensics, Springer.
3. Borko Furht and Darko Kirovski (Eds.), Multimedia Security Handbook, CRC Press.
4. Husrev Taha Sencar and Nasir Memon, Digital Image Forensics, Springer.
5. Eoghan Casey, Handbook of Digital Forensics and Investigation, Academic Press.
6. William Stallings and Dejan Milojicic, Artificial Intelligence for Cybersecurity, Pearson.
7. Kush R. Varshney, Trustworthy Machine Learning, Cambridge University Press.

CAE414 Responsible and Inclusive Computing [3-0-0-3]

Course Objectives

- To identify and analyze the short-term, generative, and long-term existential risks posed by advanced AI and foundation models.
- To equip students with the mathematical and engineering tools required to audit, probe, explain, and defend deep learning systems against adversarial vectors.
- To formalize fairness metrics, explore privacy-preserving technologies, and study techniques for building digitally accessible architectures for low-resource environments.
- To understand the global and localized socio-legal landscape regulating AI deployment and its macro-economic impact on society.

Course Outcomes

- CO1: Formally evaluate AI systems for both immediate alignment issues (toxicity, bias) and long-term systemic vulnerabilities.
- CO2: Implement defense mechanisms against adversarial and poisoning attacks, and use diagnostic tools to edit or probe neural representations.
- CO3: Apply frameworks like SHAP, LIME, and formal mathematical fairness metrics to audit algorithmic outcomes for disparate impact.
- CO4: Build accessible systems incorporating participatory design, catering to low-resource languages, and respecting compute efficiency guidelines.
- CO5: Architect deployment workflows that actively comply with major geopolitical frameworks, specifically the Indian DPDP Act and the EU AI Act.

Syllabus

Introduction to AI Capabilities: Overview of modern state-of-the-art architectures and the emergence of complex behaviors, Core Principles of RAI, Imminent Risks from AI Models, Generative AI Vector Risks: Unique alignment and safety risks specific to massive Language and Vision generative frameworks, Long-Term & Alignment

Risks: Misuse potential, misgeneralization, the problem of Rogue AGI, Instrumental Convergence, and Power-Seeking behaviors.

Adversarial Vector Analysis: Deep dive into Adversarial Attacks across Vision, NLP, and reinforcement learning systems, ML Poisoning & Supply Chain Attacks: Data poisoning, backdoor vulnerabilities, Trojans, and their broader implications for current/future enterprise AI safety. Explainability & Transparency Techniques: Foundational evaluation tools including LIME, SHAP, and GradCAM, Alignment Engineering & XAI Frontiers: Exploring the imminent and long-term potential of Mechanistic Interpretability, Representation Engineering, model editing, and direct probing, Algorithmic Fairness, Privacy-Preserving AI.

Inclusive & Sustainable Development: Green computing, addressing the carbon footprint of massive compute models, and mitigating the centralization of hardware resources. Societal Inclusivity & Algorithmic Exclusion, Designing for low-resource environments, accessibility compliance, vernacular language computing, and local digital public goods.

Operationalizing Ethics: Participatory design paradigms, ethical approvals, and establishing informed consent frameworks in data collection. System Audits: Establishing formal audit mechanisms and compliance structures within technology development pipelines, The Indian Regulatory Context: Comprehensive breakdown of the Digital Personal Data Protection (DPDP) Act and country-specific policy frameworks.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	2	1	3				3	2	3
CO2	3	3	3	3	3	2		2				2	3	3
CO3	3	3	2	3	3	3	2	3		1		3	3	3
CO4	2	2	3	2	2	3	3	3	2	2		3	2	3
CO5	2	3	3	2	2	3	2	3	1	2	2	3	2	3

Textbooks / References

1. Introduction to AI Safety, Ethics, and Society (2025) – Dan Hendrycks (CRC Press)
2. The Alignment Problem: Machine Learning and Human Values (2020) – Brian Christian (W. W. Norton & Company)
3. Interpretable Machine Learning: A Guide for Making Black Box Models Explainable (2nd Edition, 2022) – Christoph Molnar
4. Adversarial Robustness Toolbox (ART) Documentation & Literature – Linux Foundation AI & Data
5. Design Justice: Community-Led Practices to Build the Worlds We Need (2020) – Sasha Costanza-Chock (MIT Press)
6. Data Feminism (2020) – Catherine D'Ignazio and Lauren F. Klein (MIT Press)
7. The Digital Personal Data Protection (DPDP) Act, 2023 – Ministry of Electronics and Information Technology (MeitY), Government of India
8. The EU Artificial Intelligence Act (Regulation EU 2024/1689) – European Parliament Official Texts

CAE415 Computational Neuroscience [3-0-0-3]

Course Objectives

- Understand biological neural systems.
- Develop mathematical neuron models.
- Analyze neural coding mechanisms.
- Study learning and memory models.
- Apply computational methods to brain signals.
- Explore brain-inspired AI applications.

Course Outcomes

- CO1: Explain biological foundations of neural computation.
- CO2: Model neuronal behavior computationally.
- CO3: Analyze learning and plasticity mechanisms.
- CO4: Process and interpret neural signals.
- CO5: Develop brain-inspired computational models.
- CO6: Evaluate BCI and neuromorphic systems.

Syllabus

Foundations of Neuroscience: Structure of the nervous system, Neuron anatomy and physiology, Synaptic transmission, Brain organization, Neural communication mechanisms

Mathematical Foundations: Differential equations, Linear algebra for neural systems, Probability and stochastic processes, Information theory, Dynamical systems

Computational Models of Neurons: Integrate-and-Fire model, Leaky Integrate-and-Fire model, Hodgkin-Huxley model, FitzHugh-Nagumo model, Spike generation mechanisms

Neural Coding and Information Processing: Rate coding, Temporal coding, Population coding, Information representation, Sensory processing models

Learning and Synaptic Plasticity: Hebbian learning, Synaptic adaptation, Spike-Timing Dependent Plasticity (STDP), Reinforcement learning in biological systems, Memory formation

Neural Networks and Brain Dynamics: Recurrent neural networks, Associative memory models, Attractor networks, Oscillatory neural systems, Large-scale brain networks.

Neural Signal Processing: EEG signal acquisition, Signal preprocessing, Feature extraction, Brain connectivity analysis, Neuroimaging fundamentals

Brain-Inspired Computing and Applications: Spiking Neural Network, Neuromorphic computing, Brain-Computer Interfaces, Cognitive computing, Healthcare and assistive technologies

Practical Components / Assignments:

1. Simulation of neuron models
2. Spike train generation and analysis
3. Neural coding experiments

4. Hebbian learning implementation
5. STDP simulation
6. EEG signal processing
7. Brain connectivity analysis
8. Spiking neural network implementation
9. Neuromorphic computing case study
10. Mini-project using Python/MATLAB

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	2							2	3	2
CO2	3	3	2	2	3							2	3	3
CO3	3	3	2	3	3	1						2	3	3
CO4	3	3	2	3	3	2	1					2	3	3
CO5	3	3	3	2	3	2	1	1				3	3	3
CO6	2	3	3	2	3	3	2	1	1	1		3	2	3

Textbooks / References

1. P. Dayan and L. F. Abbott, Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems, 1st ed. Cambridge, MA, USA: MIT Press, 2001.
2. J. M. Bower and D. Beeman, Eds., Computational Neuroscience: A Comprehensive Approach, Revised ed. New York, NY, USA: Springer, 1998.
3. W. Gerstner, W. M. Kistler, R. Naud, and L. Paninski, Neuronal Dynamics: From Single Neurons to Networks and Models of Cognition, 1st ed. Cambridge, U.K.: Cambridge University Press, 2014.
4. E. R. Kandel, J. H. Schwartz, T. M. Jessell, S. A. Siegelbaum, and A. J. Hudspeth, Principles of Neural Science, 5th ed. New York, NY, USA: McGraw-Hill Education, 2013.
5. W. Gerstner and W. M. Kistler, Spiking Neuron Models: Single Neurons, Populations, Plasticity, 1st ed. Cambridge, U.K.: Cambridge University Press, 2002.
6. E. M. Izhikevich, Dynamical Systems in Neuroscience: The Geometry of Excitability and Bursting, 1st ed. Cambridge, MA, USA: MIT Press, 2007.
7. C. Koch, Biophysics of Computation: Information Processing in Single Neurons, 1st ed. New York, NY, USA: Oxford University Press, 1999.

CSE408 Approximation Algorithms [3-0-0-3]

Course Objectives

- To understand the need for approximation algorithms in solving NP-hard optimization problems.
- To learn fundamental approximation techniques and performance analysis.
- To apply LP-based, randomized, and primal-dual approaches to design approximation algorithms.
- To analyze the quality and limitations of approximation solutions.
- To understand the hardness of approximation and approximation-preserving reductions.

Course Outcomes

- CO1: Design approximation algorithms for NP-hard optimization problems.
- CO2: Analyze approximation ratios and performance guarantees of algorithms.
- CO3: Apply LP rounding, randomized rounding, and primal-dual techniques to solve optimization problems.
- CO4: Evaluate the computational complexity and approximability of optimization problems.
- CO5: Explain hardness of approximation results and approximation-preserving reductions.

Syllabus

Optimization Problems and Approximation Algorithms: Approximation ratio, Polynomial-Time Approximation Scheme (PTAS), Fully Polynomial-Time Approximation Scheme (FPTAS), NP-hard optimization problems.

Classical Approximation Algorithms: Vertex Cover, Set Cover, Traveling Salesman Problem, Knapsack Problem, Scheduling Problems.

LP-Based Approximation Algorithms: Linear Programming Relaxation, Deterministic Rounding, Vertex Cover, Set Cover, Facility Location (overview).

Randomized Techniques: Randomized Rounding, MAX-SAT, Facility Location or Set Cover.

Primal-Dual Algorithms: Vertex Cover, Set Cover, Facility Location, Shortest Path Variants.

Hardness of Approximation: Approximation-preserving reductions, PCP theorem (overview), APX-hardness, Inapproximability results for Set Cover and TSP.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	2							2	3	2
CO2	3	3	2	3	2							2	3	2
CO3	3	3	3	2	3							2	3	2
CO4	3	3	2	3	2							3	3	2
CO5	3	3	1	3	2							3	3	2

Textbooks / References

1. David P. Williamson and David B. Shmoys, *The Design of Approximation Algorithms*, Cambridge University Press, 2011.
2. Vijay V. Vazirani, *Approximation Algorithms*, Springer, 2013.

CSE409 Coding Theory [3-0-0-3]

Course Objectives

- Understand the principles of information theory and source coding that govern efficient information representation and transmission.
- Develop proficiency in applying coding techniques for data compression, error detection, and error correction in communication systems.
- Construct encoder and decoder systems for linear block codes to enhance error detection and correction capabilities.
- Analyze encoding and decoding of the cyclic codes to improve the reliability of the system.
- Analyze encoding and decoding of the convolution and turbo codes to improve system reliability.

Course Outcomes

Students who successfully complete this course will be able to:

- CO1: Explain the concepts of information theory, entropy, mutual information, and channel capacity in digital communication systems.
- CO2: Apply source coding techniques to improve data compression and transmission efficiency.
- CO3: Analyze the encoding and decoding processes of linear block, cyclic, BCH, and Reed-Solomon codes for error control.
- CO4: Construct encoder and decoder structures for block codes used in reliable digital communication.
- CO5: Evaluate the performance of convolutional and turbo coding techniques for error correction in noisy communication channels.

Syllabus

Information Theory: Amount of information, Entropy, Joint entropy, Conditional entropy, Relative entropy, Mutual information, Relationship between entropy and mutual information.

Channel capacity: Binary Symmetric Channel (BSC), Binary Erasure Channel (BEC), Gaussian channel, Bandwidth-SNR trade-off, Shannon's channel coding theorem.

Source Coding: Classification of codes, Kraft inequality, Coding efficiency, Shannon-Fano coding, Huffman coding, Shannon-Fano-Elias coding, Arithmetic coding, Lempel-Ziv coding, Run-length encoding, JPEG standard for lossy compression.

Block Codes: Introduction to error control codes, error detection and correction, automatic repeat request, forward error correction codes: systematic linear block codes encoding, syndrome decoding, Hamming code, perfect codes, construction of low-density parity check codes, Tanner graph, decoding of LDPC codes, belief propagation algorithm introduction.

Cyclic Codes: Introduction, cyclic codes generation, Matrix description of cyclic codes, Burst error correction, Fire codes, CRC codes, Golay codes, Circuit implementation of cyclic codes, Linear feedback shift registers.

BCH and RS Codes: Minimal Polynomials, Generator polynomials in terms of Minimal polynomials, Examples of BCH codes, Decoding of BCH codes, Berlekamp-Massey algorithm overview, Reed-Solomon (RS) codes, Implementation of RS encoders and decoders.

Convolutional and Turbo Codes: Encoding of Convolutional codes, state table, trellis structure, trellis termination, puncturing, decoding using Viterbi algorithm, Iterative design of Turbo codes, Decoding of Turbo codes: Iterative map method, The BCJR algorithm.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	2	2							2	3	2
CO2	3	3	2	2	3							2	3	2
CO3	3	3	3	3	3							2	3	3
CO4	3	2	3	2	3							2	3	3
CO5	3	3	2	3	3							3	3	3

Textbooks / References

1. Thomas M.Cover, Joy A Thomas, Elements of Information Theory, 2nd Edition, Wiley, 2015.
2. Bose, Ranjan. Information theory, coding and cryptography, 2nd Edition, Tata McGraw-Hill Education, 2008.
3. Lin, Shu, and Daniel J. Costello, Error control coding, 2nd Edition, Prentice hall, 2001.
4. SimonHaykin, Communication Systems, 4th edition, Wiley Publications, 2001.
5. R. P. Singh, S. D. Sapre, Communication Systems, 2nd edition, Tata McGraw-Hill Education, 2008.
6. K. Deergha Rao, Channel coding Techniques for wireless communications, 2nd edition,Springer, 2019.
7. Biswas, Nripendra N. Logic design theory, Prentice-Hall, Inc., 1993.

CSE410 Combinatorial Optimization [3-0-0-3]

Course Objectives

- To introduce basic concepts of combinatorial optimization, including exact and approximation algorithms, preprocessing, and complexity analysis.
- To study network flow problems such as maximum flow, minimum cut, and minimum cost flow with applications.
- To understand matching theory, including alternating paths, maximum matching, and minimum weight perfect matching.
- To introduce polyhedral combinatorics and matroid theory with linear and integer programming approaches.

Course Outcomes

- CO1: Apply fundamental principles of combinatorial optimization to analyze the efficiency and correctness of exact and approximation algorithms.
- CO2: Solve network flow problems such as maximum flow, minimum cut, and multicommodity flow using appropriate algorithmic techniques.

- CO3: Implement and analyze algorithms for matching problems, including maximum matching and weighted matching in graphs.
- CO4: Formulate and solve combinatorial optimization problems using polyhedral theory, linear programming, integer programming, and matroid-based approaches.

Syllabus

Introduction to Combinatorial Optimization: Exact and Approximation Algorithms, Preprocessing Techniques, Complexity and Running Time Analysis.

Flow Problems: Maximum flow problems, Network flow, Maximum flow, Applications of Maximum Flow and Minimum Cut, Minimum cuts in undirected graphs, Multicommodity flows, Minimum cost flow problems.

Matching Theory: Matchings and alternating paths, Maximum matching, Minimum weight perfect matchings.

Polyhedral Combinatorics: A unifying approach to combinatorial optimization, Basic polyhedral theory, Linear Programming, Overview of duality, Algorithms for Linear Programming, Equivalence of optimization and separation, Integer Programming, Totally Unimodular Matrices (TUM), Total Dual Integrality (TDI), Cutting plane theory, Branch and bound, Branch and cut algorithms, Applications of linear and integer programming theory.

Matroids: Matroids and the Greedy Algorithm, Matroid properties, Axioms, Constructions, Matroid Intersection, Applications of Matroid Intersection, Weighted Matroid Intersection.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2							2	3	2
CO2	3	3	3	3	3							2	3	3
CO3	3	3	3	2	3							2	3	3
CO4	3	3	3	3	3							3	3	3

Textbooks / References

1. William J. Cook, William H. Cunningham, William R. Pulleyblank, and Alexander Schrijver, *Combinatorial Optimization*, Wiley-Interscience Publication, 1997.
2. Ding-Zhu Du, Panos M. Pardalos, Xiaodong Hu, and Weili Wu, *Introduction to Combinatorial Optimization*, Springer, 2022.
3. Bernhard Korte and Jens Vygen, *Combinatorial Optimization: Theory and Algorithms*, 5th Edition, Springer, 2012.

CSEXXX AI for Climate Change and Sustainable Environment [3-0-0-3]

Course Objectives

- To understand climate and environmental data and their computational challenges.
- To apply machine learning techniques for climate and environmental prediction.
- To develop AI solutions for Earth observation and environmental monitoring.
- To explore foundation models and emerging AI approaches for Earth systems.
- To understand explainability, sustainability, and governance aspects of climate AI.

Course Outcomes

- CO1: Explain the characteristics of climate and environmental datasets and perform appropriate preprocessing.
- CO2: Apply machine learning methods for climate forecasting and environmental prediction.
- CO3: Design AI solutions for Earth observation, environmental monitoring, and disaster assessment.
- CO4: Analyse foundation models and emerging AI techniques for Earth system applications.
- CO5: Evaluate climate AI systems from explainability, sustainability, and governance perspectives.

Syllabus

Climate and environmental data as a computational challenge. Earth observation and climate datasets (ERA5, CMIP6, NASA EarthData). Scientific data formats including NetCDF, HDF5, and GeoTIFF. Data preprocessing, feature engineering, and geospatial data analytics. AI-enabled monitoring for sustainability applications.

Spatio-temporal data analysis. Machine learning techniques for climate forecasting and environmental prediction. Physics-informed machine learning. Anomaly detection, extreme event analysis, and uncertainty estimation for environmental systems.

Earth observation systems and satellite imagery. Geospatial data processing. Land use and land cover classification. Semantic segmentation and change detection. Applications in disaster monitoring, environmental assessment, and climate resilience.

Vision Transformers and foundation models for Earth observation. Prithvi, ClimaX, GraphCast, and related climate AI systems. Large Language Models for environmental intelligence. Multimodal learning and digital twins for sustainability applications.

Explainable AI techniques for climate and environmental systems. Model interpretability and uncertainty estimation. Green AI principles and energy-efficient machine learning. Responsible AI, governance, and policy considerations for climate applications.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3	1	2					2	3	3
CO2	3	3	3	2	3	2	2					2	3	3
CO3	3	3	3	3	3	2	3	1				2	3	3
CO4	3	3	2	3	3	2	2	1				3	3	3
CO5	2	3	2	3	3	3	3	3	1	1	1	3	2	3

Textbooks / References

1. Goodfellow, I., Bengio, Y., and Courville, A., *Deep Learning*, MIT Press, 2016.
2. Bishop, C. M., *Pattern Recognition and Machine Learning*, Springer, 2006.
3. Brunton, S. L. and Kutz, J. N., *Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control*, Cambridge University Press, 2022.
4. Reichstein, M., Camps-Valls, G., Stevens, B., Jung, M., Denzler, J., Carvalhais, N., and Prabhat, *Deep Learning and Process Understanding for Data-Driven Earth System Science*, Nature, 566, 195–204, 2019.
5. Rolnick, D. et al., *Tackling Climate Change with Machine Learning*, ACM Computing Surveys, 55(2), 2022.
6. Nguyen, T. et al., *ClimaX: A Foundation Model for Weather and Climate*, Proceedings of the International Conference on Machine Learning (ICML), 2023.
7. Lam, R. et al., *Learning Skillful Medium-Range Global Weather Forecasting*, Science, 382(6677), 2023.
8. Lundberg, S. M. and Lee, S.-I., *A Unified Approach to Interpreting Model Predictions*, Advances in Neural Information Processing Systems (NeurIPS), 2017.
9. Schwartz, R. et al., *Green AI*, Communications of the ACM, 63(12), 54–63, 2020.

Design Thinking and Innovation [2-1-0-3]

Course Objectives

- To introduce students to the principles, processes, and applications of design thinking and innovation.
- To develop skills in user-centered problem identification, empathy building, and user research methods.
- To foster creativity and innovation through brainstorming, ideation, prototyping, and testing activities.
- To equip students with practical skills in interaction design, usability evaluation, and design communication.
- To encourage interdisciplinary collaboration, ethical thinking, and socially responsible innovation through project-based learning.

Course Outcomes

Upon completion of the course, learners will be able to:

- CO1: Explain the concepts, stages, and significance of the design thinking process in solving real-world problems.
- CO2: Apply user research methods, empathy techniques, and problem-definition strategies to understand user needs effectively.
- CO3: Generate innovative ideas using brainstorming and ideation techniques and develop prototypes for testing and evaluation.
- CO4: Design user-centered solutions using interaction design principles, usability evaluation methods, and iterative improvements.
- CO5: Collaborate effectively in interdisciplinary teams and communicate design solutions through presentations, reports, and project demonstrations.

Syllabus

Introduction to Design Thinking - Introduction to design - Characteristics of good and bad design - Importance of design thinking in engineering and problem solving - Human-centered design approach - The five stages of design thinking: Empathize, Define, Ideate, Prototype, and Test - Design thinking versus traditional problem-solving approaches - Individual and team-based problem solving.

User Empathy and Problem Identification - Understanding user needs and stakeholder analysis - Empathy development techniques - User research methods including interviews, observations, surveys, and focus groups - Problem definition and reframing techniques - User-centered design principles - Ergonomics and accessibility considerations.

Ideation and Concept Development - Brainstorming methods and innovation heuristics - Creativity enhancement techniques - Affinity diagrams, empathy maps, mind maps, and customer journey maps - Concept generation and evaluation - Scenario building and storytelling techniques - Collaborative ideation activities.

Prototyping and Usability Testing - Low-fidelity and high-fidelity prototyping methods - Prototype development tools and techniques - Wireframes - Usability testing methods - Gathering and analyzing user feedback - Iterative design improvement - Heuristic evaluation techniques - Basics of data analysis using SPSS for design evaluation.

Interaction Design and User Experience - Fundamentals of interaction design - History and evolution of interaction design - User Interface (UI) and User Experience (UX) principles - Information architecture - Personas and task flows - Laws and principles of interaction design - Qualitative and quantitative research methods - Usability evaluation techniques.

Innovation, Ethics, and Social Impact - Principles and processes of successful innovation - Modes of design innovation - Social entrepreneurship and innovation for social good - Ethics in design and responsible innovation - Inclusive and sustainable design practices - Industry case studies on innovation and design impact - Familiarization with modern design tools - Design-based mini project development and presentation.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	1			1		1		2		2	1	
CO2	2	3	2	2	1	2	1	2	1	2		2	1	1
CO3	2	3	3	2	2	2	2	2	2	2	1	2	2	2
CO4	2	3	3	3	2	2	2	2	2	2	1	2	2	2
CO5	1	2	2	2	1	2	2	2	3	3	2	2	2	2

Textbooks / References

1. Kumar, K., & Kurni, M. (Eds.), Design Thinking: A Forefront Insight, CRC Press, 2022.
2. Vijay Kumar, 101 Design Methods: A Structured Approach for Driving Innovation in Your Organization, John Wiley & Sons, 2013.
3. Dan Saffer, Designing for Interaction: Creating Innovative Applications and Devices, New Riders Publishing, 2010.
4. Lewrick, M., Link, P.,& Leifer, L., The Design Thinking Toolbox: A Guide to Mastering the Most Popular and Valuable Innovation Methods, Wiley, 2020.
5. Metts, M. J., & Welfle, A., Writing is Designing: Words and the User Experience, Rosenfeld Media, 2020.
6. Uebernickel, F., Jiang, L., Brenner, W., Pukall, B., Naef, T., & Schindlholzer, B., Design Thinking: The Handbook, World Scientific, 2020.
7. Gallagher, A., & Thordarson, K., Design Thinking in Play: An Action Guide for Educators, ASCD, 2020.
8. Regine Gilbert, Inclusive Design for a Digital World: Designing with Accessibility in Mind, Apress, 2019.
9. Michael G. Luchs, Design Thinking: New Product Development Essentials from the PDMA, Wiley, 2015.
10. D'Source: Digital Online Environment for Design Learning – https://www.dsource.in

CSO001 Human Computer Interaction [3-0-0-3]

Course Objectives

- To understand the theoretical foundations and principles of Human-Computer Interaction and usability engineering.
- To develop skills in designing, prototyping, and evaluating user-centred interactive systems.
- To apply interaction design models, usability principles, and evaluation techniques to enhance user experience.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Analyse human factors, cognitive processes, and usability requirements in interactive system design.
- CO2: Design user interfaces using established HCI principles, models, standards, and guidelines.
- CO3: Evaluate interactive systems using usability testing, heuristic evaluation, and user-centred assessment methods.

Syllabus

Foundations of HCI: History of HCI - Evolution of Interactive Systems - Concepts of Interaction Design - Scientific Foundations of HCI - Human Factors in Design - Cognitive, Social and Emotional Aspects of Interaction - Conceptual Models - Usability Principles.

HCI Models and User Research: HCI Framework - Introduction to Interaction Models - Goals, Operators, Methods and Selection Rules - Fitts's Law - Hick-Hyman Law - Shneiderman's Eight Golden Rules - Norman's Seven Principles - Gestalt Laws of Design - Nielsen's Ten Heuristics with Examples - Garrett's Elements of User Experience - User Data Gathering Techniques.

Usability Engineering and Prototyping: Usability Life Cycle Model - Needs Analysis - Systems Analysis - User Profiling - Rapid Prototyping - Interactive Design - Usability Testing - Standards and Guidelines in Interface Design - Case Studies and Projects.

Interaction Design and User Interfaces: Interface Types and Interaction Styles - Screen Design and Layout - Command Line Interface - Windows, Icons, Menus and Pointing Devices - Form Fill-ins - Navigation Design - Dialog Notations and Dialog Design - 2D and 3D Interfaces - Mobile Interfaces - Wearable Interfaces - Multimodal Interfaces.

Evaluation and Emerging Trends in HCI: Evaluation Techniques - Wizard of Oz Method - Universal Design - Web Usability Guidelines - Accessibility Considerations - Recent Trends in HCI - VR-AR Design Guidelines - CSCW - Conversational Agents - Inclusive Design.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	2	1	1		2				1	2	1
CO2	3	3	3	2	3	1	1	2		1		1	3	3
CO3	2	3	2	3	2	1	1	2	1	2		2	3	2

Textbooks / References

1. Alan Dix, J Finlay, G D Abowd, R Beale Human Computer Interaction, Prentice Hall, 2003.
2. Jenny Preece, Helen Sharp, Yvonne Rogers. "Interaction Design: Beyond Human-Computer Interaction". 2019. (5th Edition) (Wiley)
3. Norman, Donald A. "The design of everyday things." New York City, NY, USA: Doubleday (1988).
4. B. Shneiderman, Designing the User Interface: Strategies for Effective Human-Computer Interaction, 3rd Ed., Addison Wesley, 2000.
5. W.O. Galitz, The Essential Guide to User Interface Design of Interaction Design, John Wiley & Sons, 2002.
6. Krug, Steve (2000). Don't Make Me Think: A Common Sense Approach to Web Usability. USA: New Riders Press.

7. Kuniavsky, Mike (2003). Observing The User Experience: A Practitioner's Guide to User Research. USA: Morgan Kaufmann.
8. C. Stephanidis (ed.), User Interface for All: Concepts, Methods and Tools. Lawrence Erlbaum Associates, 2001.
9. J. M. Caroll (ed.), HCI Models, Theories and Frameworks: Towards a Multidisciplinary Science (Interactive Technologies), Morgan Kauffman, 2003.

CSO002 Digital Signal Processing [3-0-0-3]

Course Objectives

To impart knowledge on

- Theoretical principles underlying DSP.
- Digital signal processing systems in time and frequency domain.
- Sampling and reconstruction of analog signals.
- Design FIR and IIR filters to meet specific magnitude and phase requirements.

Course Outcomes

After successful completion of the course, students should be able to:

- CO1: Explain the basic concepts of signal processing.
- CO2: Characterise signals and systems in discrete time, including use of the z-transform.
- CO3: Use Fourier transform and convolution to filter signals.
- CO4: Design FIR and IIR filters to satisfy a desired frequency response.
- CO5: Explain the role of the window function and describe its influence on FIR filters.
- CO6: Design IIR filters on the basis of an analogue design.
- CO7: Design and evaluate a real-time DSP system.

Syllabus

Introduction, signals, systems, examples, classification of signals, basic operations on signals; folding, shifting, time scaling, classification of systems, linear and nonlinear, time variant and invariant, causal and noncausal, stable and unstable systems. DSP Advantages and disadvantages; Quantisation and sampling; Aliasing; Anti-aliasing and reconstruction filters.

Time domain analysis of discrete signals and systems: Impulse response, Linear Time Invariant (LTI) systems, Convolution sum.

Frequency domain analysis of systems using Fourier transform. Difference equations and z transforms.

Finite and infinite impulse response filters, FIR and IIR, Poles, zeros and frequency response. Design of Digital Filters-Design of non-recursive and recursive digital filters. Hardware & software implementations.

Some Typical DSP Applications Speech recognition; Control; Image recognition; Radar; Room response analysis, signal compression. DSP Hardware: A/Ds, D/As and over-sampling.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2			1							1	2	1
CO2	3	3	1	2	2							1	3	2
CO3	3	3	2	2	2							1	3	2
CO4	3	3	3	2	2							1	3	3
CO5	2	2	2	1	2							1	2	2
CO6	3	3	3	2	2							1	3	3
CO7	3	3	3	3	3	1		1	1	2		2	3	3

Textbooks / References

1. Proakis John G and Manolakis Dimitris G, Digital Signal Processing: Principles, Algorithms and Applications, 4th Edition, Prentice-Hall, 2006.
2. Oppenheim Alan V, Schafer Ronald W and Buck John R, Discrete Time Signal Processing, 3rd Edition, Pearson Education, 2009.
3. Prandoni Paolo and Vetterli Martin, Signal Processing for Communication, 1st Edition, EPFL Press.
4. Mitra Sanjit K, Digital Signal Processing: A Computer Based Approach, 4th Edition, McGraw Hill, 2011.
5. Kuo Sen M, Lee Bob H and Tian Wenshun, Real-Time Digital Signal Processing: Implementations and Applications, 2nd Edition, John Wiley, 2006.
6. Lapsley Phil, DSP Processor Fundamentals: Architectures and Features, IEEE Press, 1997.
7. Ackenhusen John G, Real Time Signal Processing: Design and Implementation of Signal Processing Systems, PrenticeHall,1999.

CAO001 Soft Computing [3-0-0-3]

Course Objectives

- To understand the fundamental concepts, architectures, and learning mechanisms of neural networks, fuzzy systems, and genetic algorithms.
- To analyze fuzzy reasoning, inference mechanisms, and optimization techniques for solving problems involving uncertainty and imprecision.
- To explore neuro-fuzzy modeling approaches and their applications in intelligent systems for prediction, classification, and decision support.

Course Outcomes

- CO1: Explain the principles, architectures, and learning methods of neural networks, fuzzy systems, and genetic algorithms used in soft computing.
- CO2: Apply fuzzy logic, neural network models, and evolutionary computation techniques to solve complex optimization and decision-making problems.
- CO3: Design and evaluate neuro-fuzzy models for real-world applications such as healthcare, forecasting, transportation, and intelligent IoT systems.

Syllabus

Foundations of Soft computing: Artificial neural network: Introduction, characteristics - learning methods - taxonomy - Evolution of neural networks -basic models - applications. Fuzzy logic: Introduction - crisp sets and fuzzy sets - crisp relations and fuzzy relations: cartesian product of relation - classical relation, tolerance and equivalence relations. Genetic algorithm - Introduction - biological background - traditional optimization and search techniques - Encoding schemes - Fitness functions - Selection - Crossover - Mutation - Applications.

Neural Networks: McCulloch-Pitts neuron - linear separability - Hebb network - supervised learning network: perceptron networks - adaptive linear neuron, multiple adaptive linear neuron, BPN, RBF, associative memory network: auto-associative memory network, hetero-associative memory network, hopfield networks, unsupervised learning networks: Kohonen Self Organizing Feature Maps.

Fuzzy Logic and Fuzzy Systems: Membership functions - Linguistic variables - fuzzification, methods of membership value assignments -Defuzzification: lambda cuts - methods - fuzzy arithmetic - extension principle - fuzzy measures - measures of fuzziness - fuzzy rule base and approximate reasoning : truth values and tables, fuzzy propositions, formation of rules -decomposition of rules, aggregation of fuzzy rules, fuzzy reasoning - fuzzy inference systems - overview of fuzzy expert system - fuzzy decision making.

Neuro-Fuzzy Modelling: Adaptive Neuro-Fuzzy Inference Systems (ANFIS) - Architecture, Hybrid learning algorithm, ANFIS as a universal approximator. Applications – Medical diagnosis - Energy load forecasting - Financial prediction - Smart transportation - IoT-based intelligent systems.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2		1	2							1	2	1
CO2	3	3	2	2	2							1	3	2
CO3	2	3	3	3	2		1	1		1		1	3	2

Textbooks / References

1. S.R.Jang, C.T. Sun and E.Mizutani, “Neuro-Fuzzy and Soft Computing”, PHI / Pearson Education 2004.
2. S.N.Sivanandam , S.N.Deepa, “Principles of Soft Computing”, Wiley India Pvt. Ltd., Paperback, 2018.
3. S.Rajasekaran, G.A.Vijayalakshmi Pai, “Neural Networks, Fuzzy Logic and Genetic Algorithm, Synthesis and Appli-cations“, PHI Learning Pvt. Ltd., 2017.
4. George J. Klir and Bo Yuan, Fuzzy Sets and Fuzzy Logic-Theory and Applications, Prentice Hall, Paperback – 2015.
5. Timothy J. Ross, Fuzzy Logic with Engineering Applications, 4th Edition, CRC Press, 2023.
6. S. Haykin, Neural Networks and Learning Machines, 3rd Edition, Pearson Education, 2008.

CSO003 Quantum Computing for Engineers [3-0-0-3]

Course Objectives

- Introduce the requisite mathematical concepts that are relevant to computing.
- Explore the core concepts that distinguish quantum computation from classical computation.
- Examine various quantum computing models and algorithms.
- Develop knowledge of quantum gates, circuits, and error correction techniques.
- Familiarize students with quantum programming using modern tools for practical implementation.

Course Outcomes

After successful completion of the course, students should be able to:

- CO1: Understand the fundamental principles of mathematics relevant to computing.
- CO2: Explain the basic concepts of quantum computing
- CO3: Demonstrate quantum computing models and algorithms.
- CO4: Examine quantum gates, circuits, and error correction techniques.
- CO5: Formulate quantum inspired algorithms to solve computationally challenging problems.

Syllabus

Complex numbers, Tensor products, Unitary and Hermitian matrices, Introduction to Quantum Computing: Classical vs. Quantum Computing, Mathematical representations for Quantum Computing, Quantum Bits (Qubits) and Quantum Registers, Quantum Parallelism and Quantum Speedup, No-Cloning Theorem, Measurement, Quantum Decoherence and Noise, Platforms for Exploring Quantum Computing (Classiq, IBM Quantum, Google Quantum)

Quantum Gates and Circuits: Quantum Gates: Pauli, Hadamard, CNOT, Phase, and Toffoli Gates, Quantum Circuits and Their Representations, Reversible and Universal Gate Sets, Quantum Error Correction Techniques, Quantum Teleportation and Superdense Coding

Quantum Algorithms and Programming: Introduction to Quantum Algorithms, Deutsch Josza algorithm, Quantum Fourier Transform, Phase Estimation, Shor's Algorithm for Factorization, Grover's Algorithm, Amplitude Amplification, Quantum Random walk, Variational Quantum Algorithms

Introduction to Quantum Programming Languages (Qiskit, Cirq, PennyLane, etc.), Quantum Software Engineering and Applications: Quantum Software Development Life Cycle, Hardware Implementations and Current Limitations, Future Trends in Quantum Computing.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2										2	2	
CO2	3	2	1		1							2	2	1
CO3	2	3	2	2	2							2	3	2
CO4	2	3	2	2	3							2	3	2
CO5	2	3	3	2	3	1		1		1		3	3	3

Textbooks / References

1. Yanofsky, Noson S., and Mirco A. Mannucci. Quantum computing for computer scientists. Cambridge University Press, 2008.
2. Hiu Yung Wong, Introduction to quantum computing: from a layperson to a programmer in 30 steps. Springer Nature, 2023.
3. Thomas G. Wong, Introduction to classical and quantum computing. Omaha, NE, USA: Rooted Grove, 2022.
4. Robert S. Sutor, Dancing with Qubits: How quantum computing works and how it can change the world. Packt Publishing Ltd, 2019.
5. Nielsen, Michael A., and Isaac L. Chuang. Quantum computation and quantum information. Cambridge university press, 2010.
6. Robert Loredo, Learn Quantum Computing with Python and IBM Quantum Experience: A hands-on introduction to quantum computing and writing your own quantum programs with Python. Packt Publishing Ltd, 2020.
7. Willi-Hans Steeb, and Yorick Hardy. Problems and solutions in quantum computing and quantum information. World Scientific Publishing Company, 2018.
8. Anshul Saxena, Javier Mancilla, Iraitz Montalban, and Christophe Pere. Financial Modeling Using Quantum Computing: Design and manage quantum machine learning solutions for financial analysis and decision making. Packt Publishing Ltd, 2023.

CSO004 Introduction to DevOps and Microservices [3-0-0-3]

Course Objectives

- To understand and explain the concepts of business needs for DevOps and Microservices.
- To gain knowledge in DevOps design issues and adopting DevOps in business use cases.
- To develop and integrate Microservices using DevOps.
- To deploy the DevOps automation for different use cases.

Course Outcomes

Upon successful completion of the course, the students will be able to:

- CO1: Understand the key concepts and principles of DevOps and Microservices.
- CO2: Familiar with DevOps automation tools.
- CO3: Apply Microservices in DevOps.
- CO4: Apply DevOps for Business Analytics.

Syllabus

Introduction to DevOps: DevOps: An overview - Understanding the Business Needs for DevOps - History of DevOps - DevOps and Software Development Life Cycle – Waterfall Model - Agile Model – DevOps Lifecycle - Process and Technology in DevOps - DevOps Myths – Barriers.

DevOps Adoption and Tools: Plan and Measure, Develop and Test (collaborative and continuous), Release and Deploy Monitor and Optimize. Tools: distributed version control tool Git, Containerization tool Docker, Docker compose.

Microservices: Introduction to Microservices, Architecture: Monolith and Microservices- Benefits, Challenges, Characteristics: Coordination model - Building and testing - Deployment pipeline - Development and Precommit Testing - Build and Integration Testing - Continuous integration – Monitoring. CI/CD Tools: Github Actions, Jenkins. Container Orchestration – Kubernetes.

Integration of Microservices and DevOps Build a DevOps platform - Benefits of combining DevOps and Microservices- Working of DevOps and Microservices in Cloud environment (Case studies: Openstack, GCP and AWS).

Devops Automation: Infrastructure Automation- Configuration Management – Infrastructure as Code - Terraform. Test Automation. Deployment Automation - Performance Management - Log Management -Monitoring.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2			2			1		1		2	2	1
CO2	2	2	2	2	3			1	1	1	1	2	3	2
CO3	2	3	3	2	3	1		1	2	2	1	2	3	3
CO4	2	3	3	3	3	2	1	2	2	2	2	2	3	3

Textbooks / References

1. Joyner Joseph, “Devops for Beginners”, 1st Edition, Mihails Konoplovs publisher, 2015.
2. Stephen Flemming, “DevOps And Microservices Handbook: Non-Programmer’s Guide to DevOps and Microservices”, 2018
3. Michael Hüttermann, DevOps for Developers, 1st Edition, APress, e-book, 2012.
4. James A Scott, A Practical Guide to Microservices and Containers, MapR Data Technologies
5. Ebert, C., Gallardo, G., Hernantes, J., & Serrano, N. (2016). DevOps. Ieee, 33(3), 94-100.

CAO004 Large Language Models [3-0-0-3]

Course Objectives

- To understand the transformer architecture in depth, including self-attention, positional encoding, and encoder-decoder design.
- To distinguish encoder-only, decoder-only, and encoder-decoder model families based on their training objectives.
- To gain understanding of tokenisation algorithms including BPE, WordPiece, and SentencePiece.
- To analyze data pipelines and quality filtering strategies used in major LLM pre-training efforts.

Course Outcomes

- CO1: Derive and implement the self-attention mechanism and explain its computational complexity.
- CO2: Trace a complete pre-training pipeline from raw data to a deployable language model.
- CO3: Compare positional encoding schemes such as sinusoidal embeddings, ALiBi, RoPE, and NoPE based on length-generalisation trade-offs.
- CO4: Configure a distributed training setup using data, tensor, and pipeline parallelism with ZeRO optimisation.

Syllabus

Transformer Architecture: Introduction to the transformer architecture, Self-attention mechanism, Scaled dot-product attention, Multi-head attention, Encoder layers and encoder stack, Decoder layers, Teacher forcing and masked (causal) attention, Position encoding using sinusoidal embeddings, Feed-forward sublayers, Residual connections, Batch normalisation, Layer normalisation.

LLM Families and Training Objectives: Language modelling objectives, Causal language modelling, Decoder-only Large Language Models (LLMs), GPT architecture and training, Pre-training and fine-tuning, Decoding strategies including greedy search, beam search, top-k sampling, and top-p sampling, Encoder-only LLMs, BERT architecture, Masked Language Model (MLM) and Next Sentence Prediction objectives, Encoder-decoder models, BART, T5 text-to-text framework, GPT-2 zero-shot learning.

Tokenisation and Data Pipelines: Word-level and character-level tokenisation challenges, Sub-word tokenisation, Byte Pair Encoding (BPE), WordPiece, SentencePiece, Vocabulary size trade-offs, Data sources including C4, mC4, Pile, RedPajama, SlimPajama, The Stack, and Sangraha, Data pipelines used in Gopher, OPT, LLaMA, Falcon, and SETU, Quality filtering, Deduplication, Clean data and training effectiveness.

Efficient Attention and Positional Encodings: Time and space complexity of standard attention mechanisms, Sub-quadratic attention methods including local, dilated, random, block-sparse, low-rank, and kernel approaches, FlashAttention, KV-caching, Multi-Query Attention (MQA), Grouped-Query Attention (GQA), Paged attention, Limitations of sinusoidal positional embeddings, ALiBi, RoPE, NoPE, Length generalisation, Pre-normalisation and post-normalisation.

Training and Scaling Large Models: Scaling laws and Chinchilla compute-optimal training, Adam and LION optimisers, Learning rate schedules, Gradient clipping, Training failure modes, Mixed-precision training using FP16 and BF16, Activation checkpointing, CPU offloading, Data parallelism, Tensor parallelism, Pipeline parallelism, ZeRO (Zero Redundancy Optimizer) stages 1–3.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	2	2	3			1		1		2	3	2
CO2	2	3	3	2	3			1		1		2	3	3
CO3	2	3	2	2	2	2	2	3		1		2	2	2
CO4	2	3	3	3	3	1		2	1	2	1	2	3	3
CO5	2	3	3	3	3	2	2	3	2	2	1	3	3	3

Textbooks / References

1. Vaswani, A. et al., "Attention Is All You Need," *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
2. Devlin, J. et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *NAACL*, 2019.
3. Brown, T. et al., "Language Models are Few-Shot Learners," *NeurIPS*, 2020.
4. Raffel, C. et al., "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer," *Journal of Machine Learning Research (JMLR)*, 2020.
5. Touvron, H. et al., "LLaMA: Open and Efficient Foundation Language Models," *arXiv*, 2023.
6. Hoffmann, J. et al., "Training Compute-Optimal Large Language Models," *NeurIPS*, 2022.
7. Dao, T. et al., "FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness," *NeurIPS*, 2022.
8. Rajbhandari, S. et al., "ZeRO: Memory Optimizations Toward Training Trillion Parameter Models," *SC20*, 2020.

CAO005 Prompt Engineering & LLM Application Development [3-0-0-3]

Course Objectives

- Master a range of prompting strategies — from zero-shot and chain-of-thought to structured output control.
- Understand and implement parameter-efficient fine-tuning methods including LoRA, QLoRA, and prefix tuning.
- Gain a thorough understanding of the RLHF pipeline and alignment techniques including DPO and Constitutional AI.
- Design and build retrieval-augmented generation systems and tool-using LLM agents.

- Evaluate LLMs rigorously using standard benchmarks and evaluation frameworks.
- Understand the challenges of multilingual LLMs and the responsible deployment of AI systems.

Course Outcomes

- CO1: Design and systematically evaluate prompts using few-shot examples, CoT reasoning, and format constraints.
- CO2: Implement LoRA fine-tuning on a pre-trained LLM and choose appropriately between PEFT methods.
- CO3: Describe the full InstructGPT training pipeline (SFT ⟶ reward model ⟶ PPO) and compare it with DPO.
- CO4: Build an end-to-end RAG pipeline covering chunking, embedding, retrieval, reranking, and generation.
- CO5: Implement a tool-using LLM agent and identify common agentic failure modes.

Syllabus

Prompting Strategies: Zero-shot and few-shot in-context learning, Chain-of-thought (CoT) prompting and self-consistency, Tree-of-thoughts and plan-and-solve prompting, Instruction following — templates, delimiters, and role prompting, Output format control — JSON and structured outputs, Prompt injection and adversarial prompt failure modes, GPT-2 zero-shot learning as a foundation for prompting.

Fine-tuning & Parameter-Efficient Adaptation: When prompting is insufficient — motivation for fine-tuning, Full supervised fine-tuning and multi-task fine-tuning, Prompt tuning and prefix tuning, LoRA — low-rank adaptation of large language models, QLoRA — quantised LoRA for memory-efficient fine-tuning, Adapter layers, Catastrophic forgetting and mitigation strategies, Choosing between prompting, PEFT, and full fine-tuning.

Alignment & RLHF: The alignment problem — helpfulness, harmlessness, and honesty, Supervised instruction tuning — InstructGPT pipeline, Human preference data collection and reward model training, Reinforcement learning from human feedback (RLHF) using PPO, Direct preference optimisation (DPO) — RLHF without a reward model, Constitutional AI, Toxicity, bias, fairness, and red-teaming methods.

Retrieval-Augmented Generation & Agentic Systems: Retrieval-augmented generation (RAG) — motivation and architecture, Dense passage retrieval, vector databases, and embedding models, Chunking, indexing, reranking, and hybrid retrieval, Tool use and function calling, ReAct and plan-and-execute agent frameworks, Multi-agent systems and memory management, Failure modes — hallucinated tool calls, context overflow, infinite loops.

Evaluation & Responsible Deployment: Standard benchmarks — MMLU, HellaSwag, TruthfulQA, GSM8K, Evaluation frameworks — lm-evaluation-harness and HELM, Open LLM leaderboards and their limitations, Human vs automatic evaluation and hallucination detection, Multilingual evaluation and low-resource language challenges, deployment on edge devices, IndicLLMs — Anudesh platform and data contribution for Indian languages, Safety guardrails, red-teaming, content moderation, and model cards.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3							2	3	2
CO2	3	3	3	2	3							2	3	3
CO3	3	3	2	2	2	1		2				2	2	2
CO4	3	3	3	3	3					1		2	3	3
CO5	2	3	3	3	3	2	1	3	1	2		3	3	3

Textbooks / References

1. Wei, J. et al. (2022). Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. NeurIPS.
2. Hu, E. et al. (2022). LoRA: Low-Rank Adaptation of Large Language Models. ICLR.
3. Dettmers, T. et al. (2023). QLoRA: Efficient Finetuning of Quantized LLMs. NeurIPS.
4. Ouyang, L. et al. (2022). Training language models to follow instructions with human feedback. NeurIPS.
5. Rafailov, R. et al. (2023). Direct Preference Optimization. NeurIPS.
6. Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS.
7. Yao, S. et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. ICLR.
8. Liang, P. et al. (2022). Holistic Evaluation of Language Models (HELM). arXiv.

CAO006 Introduction to AyurInformatics [3-0-0-3]

Course Objectives

- To introduce the fundamental biomedical principles of Ayurveda and their structural translation into digitized health data.
- To analyze multi-omic, clinical, and sensor-based datasets using machine learning, deep learning, and network pharmacology models.
- To design computational solutions and IoT-driven frameworks for personalized health profiling, quality control, and automated pulse/tongue diagnostics.

Course Outcomes

- CO1: Explain traditional Ayurvedic concepts (Prakriti, Dhatus, Agni) using modern physiological, genomic, and phenotypic data representations.
- CO2: Implement machine learning classifiers and deep learning architectures for automated clinical sub-typing and medical image diagnostics.

- CO3: Evaluate computational chemistry and bioinformatics pipelines to map polyherbal formulations onto modern molecular target networks.
- CO4: Design IoT-based smart sensing architectures for non-invasive digital health tracking and physiological data acquisition.

Syllabus

Foundations of Ayurveda and Data Digitization: Introduction to Ayurinformatics: Scope, motivation, and challenges. Core concepts of Ayurveda: Tridosha (Vata, Pitta, Kapha), Sapta Dhatu (tissues), and Agni (metabolism). Concept of Prakriti (human constitution) and genomic correlations. Methods for converting traditional clinical texts into digitized, structured clinical ontologies and Electronic Health Records (EHR).

Digital Prakriti assessment: Rule-based systems vs. statistical classification. Applying machine learning (Decision Trees, SVMs, Random Forests) to questionnaire-based phenotype mapping. Classifying multi-spectral physiological profiles. Handling high-dimensional, imbalanced data in traditional healthcare datasets.

Computer vision applications in traditional diagnostics. Automated Jivha Pariksha (Tongue Analysis): Segmentation, color-texture classification, and feature extraction using Convolutional Neural Networks (CNNs). Deep learning for Netra Pariksha (Eye Examination) and skin lesion analysis for personalized cosmetic and therapeutic recommendations.

Ayurgenomics: Bridging HLA typing, SNPs, and gene expression profiles with Prakriti classification. Network Pharmacology: Computational mapping of polyherbal formulations (Rasayana). Molecular docking and virtual screening of active phytochemical compounds against modern disease targets. Databases in focus: Bio-Informatics databases, KNApSAcK, and IMPPAT.

Natural Language Processing (NLP) for mining classical Sanskrit texts (Charaka Samhita, Sushruta Samhita). Standardization of raw material quality using machine vision. IoT, Edge Computing & Biosensors in Ayurinformatics:Digitizing Nadi Pariksha (Pulse Diagnosis): Real-time health monitoring architectures using IoT and Edge AI.

Regulatory, ethical, and privacy challenges in integrating traditional medicine with mainstream digital healthcare.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2				2		2		1		1	1	
CO2	2	3	3	2	2	1		1		1		2	3	2
CO3	2	3	2	3	2	2	1	2		1		2	3	2
CO4	2	3	3	3	3	2	2	2	1	2	1	2	3	3

Textbooks / References

1. Pradeep Kumar, G., et al. Ayurinformatics: An Emerging Trend in Digital Health and Traditional Medicine. Academic Press, 2024.
2. Valiathan, M. S. The Legacy of Caraka. Orient Blackswan, 2003.
3. Aggarwal, B. B. Molecular Targets of Ayurvedic Medicinal Plants. CRC Press, 2011.
4. Selected Research Papers from the Journal of Ayurveda and Integrative Medicine (JAIM) and Briefings in Bioinformatics on Ayurgenomics and Network Pharmacology.
5. Patwardhan, Bhushan. Drug Discovery and Development: Traditional Medicine and Ethnopharmacology. New India Publishing Agency, 2007.

CAO007 Introduction to Culture Computing [3-0-0-3]

Course Objectives

- Introduce Interdisciplinary Foundations: Connect human philosophy, anthropology, and computer science to analyze how global cultures shape algorithm designs.
- Examine Computers as Social Artifacts: Investigate the historical development of computation, automata, and localized user behavior using critical theoretical frameworks.
- Critique Algorithmic Neutrality: Expose how automated systems, search engines, and data mining reinforce data colonialism and structural discrimination.
- Teach Culturally Situated Methodologies: Provide empirical human-computer interaction (HCI) and ethnographic design frameworks for diverse global populations.

Course Outcomes

- CO1: Define the core philosophical and anthropological intersections between human cultural practices and computational structures.
- CO2: Identify and articulate how modern machine learning and search algorithms perpetuate systemic biases and social inequities.
- CO3: Apply standard ethnographic research methods to map the specific technology needs of localized regional communities.
- CO4: Design user interfaces and digital media architectures that successfully respect regional aesthetics, color systems, and local demographic patterns.

Syllabus

Foundations of Cultural Computing: Defining the intersection of culture, human philosophy, and computation. The Myth of Tech Neutrality: How cultural values, ethics, and human biases shape algorithm designs and localized interfaces. Philosophical & Historical Frameworks: Historical evolution of computing, comparing binary logic against traditional global philosophies. Computers as Social Artifacts: Anthropological history of automata, language syntax biases, and localized user behaviors.

Algorithmic Bias and Representation: Investigating data colonialism, machine learning ethics, and tech equity. The "New Jim Code": How automated systems and machine learning perpetuate structural discrimination and data colonialism.

Search Engine Bias: Investigating how search indexes, data mining, and ad optimization reinforce racism and social inequities. Global Tech Power Asymmetry: Digital extraction, platform monopoly, and the exploitation of global tech labor patterns.

Culturally Situated Design: UI/UX localization, ethnography, and inclusive digital media architectures. UI/UX Localization: Empirical frameworks for adapting interfaces, layout structures, and text direction to respect global user demographies.

Ethnographic HCI Methods: Field research guidelines, community co-design, and converting qualitative data into user personas.

Visual & Structural Aesthetics: Engineering grid setups, localized color psychology, and adaptive architectures for varying digital literacy.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2				2		2		2		1	1	
CO2	2	3		2		3	2	3		2		2	1	
CO3	1	2	2	3		2	2	2	2	2		2	2	1
CO4	2	3	3	2	2	2	2	2	3	3	1	2	2	2

Textbooks / References

1. Culturally Responsive Computing by Devan J. Walton.
2. Philosophical and Cultural Aspects of Computing by Gerard O'Regan.
3. MIT OpenCourseWare: Cultures of Computing.

CAO008 Indian Theories of Meaning [3-0-0-3]

Course Objectives

- To introduce the fundamental concepts of ancient Indian semantics, logic, and grammatical traditions.
- To analyze the cognitive and structural mechanics of language comprehension through Sphota, Dhwani, and Karaka frameworks.
- To contextualize traditional Indian philosophies of meaning by evaluating them alongside modern Western linguistic theories.

Course Outcomes

- CO1: Explain the classical Indian systems of logic, inference, and semantic relations as defined by the Nyaya and Vyakarana schools.
- CO2: Deconstruct the layers of speech perception, conceptualization, and aesthetic suggestion using Sphota and Dhwani theories.
- CO3: Map structural dependency and semantic roles in sentences using the Paninian Karaka framework for computational processing.
- CO4: Contrast Indian philosophies of language with Western semantic paradigms, specifically analyzing the structural intersections with Wittgenstein's philosophy.

Syllabus

Foundations of Indian Logic & Semantics, Introduction to Indian linguistic philosophy; pramāṇas (sources of knowledge); Epistemology of language; Theories of Inference and semantic competency according to the Nyāya school. The Sphota Theory of Language: The concept of Sphota (indivisible linguistic unit); Bhartṛhari's philosophy of Vākyapadīya; The speech act: conceptualization (Paśyantī), performance (Madhyamā), and comprehension (Vaikharī). The Dhwani Theory of Suggestion: Introduction to the Dhwani framework (poetic/suggested meaning); Nature and classification of Dhwani; Literal vs. implied meaning; Historical and philosophical debates surrounding Dhwani.

Structural Semantics: Dhwani vs. Sphota-Comparative analysis of Dhwani and Sphota; Distinction between primary sounds (prākṛta-dhvani) and secondary acoustic realizations (vaikṛta-dhvani); Word-meaning vs. sentence-meaning (Anvitābhidhāna-vāda vs. Abhihitānvaya-vāda). The Karaka Theory & Dependency Grammar Pāṇini's structural framework; The Kāraka system (semantic relations between nouns and verbs); Mapping case-endings (Vibhakti) to semantic roles; Modern applications of Karaka in Paninian computational linguistics. Comparative Philosophies: Indian vs. Western Perspectives Cross-cultural evaluation of meaning; Intersections between Indian grammarians (Bhartṛhari) and Western analytical philosophers (Ludwig Wittgenstein); Language games, picture theory, and usage-based meaning.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2		1		1		2				2	1	
CO2	2	3		2		1		2		1		2	1	
CO3	2	3	1	2	2			1				2	3	2
CO4	2	3		2		2		3		2	1	2	1	1

Textbooks / References

1. Kunjunni Raja, K. Indian Theories of Meaning. The Theosophical Publishing House, 2000.
2. Coward, Harold G. The Sphota Theory of Language: A Philosophical Analysis. Motilal Banarsidass Publishers, 1980.
3. Matilal, Bimal K. The Word and the World: India's Contribution to the Study of Language. Oxford University Press, 1990.
4. Shah, K.J. "Bhartrihari and Wittgenstein." Perspectives on the Philosophy of Meaning, Vol. I, No. 1, New Delhi.
5. Akshara Bharathi, Vineet Chaitanya, and Rajeev Sangal. Natural Language Processing: A Paninian Perspective. PHI Learning, 1996.

6. Patnaik, Tandra. Śabda: A Study of Bhartrhari's Philosophy of Language. DK Printworld, 1994.

CAO009 Health Informatics & Digital Health Systems [3-0-0-3]

Course Objectives

- To introduce the fundamental concepts, components, and applications of health informatics and digital health systems.
- To familiarize students with healthcare information systems, electronic health records, and healthcare data standards.
- To develop skills in health data management, analytics, and clinical decision support technologies.
- To understand the role of telemedicine, mobile health, wearable devices, and IoT in modern healthcare.
- To evaluate privacy, security, ethical, and regulatory issues associated with digital healthcare systems.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Explain the principles, architecture, and applications of health informatics and digital health technologies.
- CO2: Analyze healthcare data and utilize health information systems for effective healthcare management.
- CO3: Apply healthcare data standards, interoperability frameworks, and electronic health record systems in healthcare environments.
- CO4: Design and evaluate digital healthcare solutions incorporating telemedicine, IoT, AI, and mobile health technologies.
- CO5: Assess ethical, legal, privacy, and cybersecurity challenges associated with digital health systems.

Syllabus

Introduction to Health Informatics and Healthcare Information Systems: Evolution of Healthcare Information Technology , Introduction to Health Informatics, Healthcare Ecosystem and Stakeholders , Components of Healthcare Information Systems, Clinical Information Systems, Hospital Information Systems (HIS), Electronic Medical Records (EMR), Electronic Health Records (EHR), Digital Transformation in Healthcare.

Healthcare Data Management and Standards: Healthcare Data Types and Sources, Clinical Data Collection and Management, Healthcare Databases and Data Warehousing , Medical Coding Systems (ICD, CPT, SNOMED CT), Healthcare Data Standards , HL7 and FHIR Frameworks , Interoperability in Healthcare Systems, Data Quality and Governance , Health Information Exchange (HIE).

Digital Health Technologies and Telemedicine: Digital Health Concepts and Frameworks, Telemedicine and Telehealth Systems, Mobile Health (mHealth) Applications, Wearable Health Monitoring Devices, Internet of Medical Things (IoMT), Remote Patient Monitoring Systems, Cloud Computing in Healthcare, Smart Hospitals and Connected Healthcare, Case Studies in Digital Healthcare.

Clinical Decision Support and Healthcare Analytics: Clinical Decision Support Systems (CDSS), Health Data Analytics, Predictive Analytics in Healthcare, Artificial Intelligence and machine Learning in Healthcare, Natural Language Processing for Clinical Data, Medical Image Informatics, Population Health Analytics, Personalized and Precision Medicine, Applications of Big Data in Healthcare.

Security, Ethics, and Emerging Trends in Digital Health: Healthcare Data Privacy and Confidentiality, Cybersecurity in Healthcare Systems, Risk Management and Data Protection, Ethical Issues in Health Informatics, Regulatory Frameworks and compliance, Digital Health Policies, Blockchain for Healthcare, Digital Twins in Healthcare, Future Trends: Generative AI, Robotics, and Intelligent Healthcare Systems.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2		1	2	1						1	2	
CO2	2	3	1	2	3	1						1	3	1
CO3	2	3	2	2	3	2		1				1	3	2
CO4	2	3	3	2	3	2	1	2	1	1		2	3	2
CO5	2	3	1	2	2	3	1	3		1	1	2	2	2

Textbooks / References

1. Health Informatics: Practical Guide for Healthcare and Information Technology Professionals. Robert E. Hoyt, Ann Yoshihashi and Brendan M. Thyvalikakath.
2. Biomedical Informatics: Computer Applications in Health Care and Biomedicine Springer. Edward H. Shortliffe and James.
3. Guide to Health Informatics (formerly Introduction to Health Informatics): Enrico Coiera CRC Press / Taylor & Francis.
4. Digital Health: Scaling Healthcare to the World. Homa Javahery
5. Healthcare Analytics for Quality and Performance Improvement. Trevor L. Strome
6. Clinical Decision Support Systems. Theory and Practice, Eta S. Berner
7. Artificial Intelligence in Healthcare. Adam Bohr and Kaveh Memarzadeh

CSO006 Internet of Things: Concepts and Applications [3-0-0-3]

Course Objectives

- To understand the architecture, ecosystem, and enabling technologies of IoT systems.
- To study sensing, communication, and service integration mechanisms in IoT environments.

- To understand IoT protocols, middleware platforms, and data management frameworks.
- To analyze security, privacy, and interoperability challenges in IoT deployments.
- To examine real-world IoT applications and deployment strategies.
- To develop the ability to design and evaluate IoT-enabled solutions.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Explain IoT architectures, enabling technologies, and ecosystem components.
- CO2: Analyze sensing, communication, and interoperability requirements in IoT systems.
- CO3: Evaluate IoT communication protocols and platform technologies.
- CO4: Analyze data management, security, and privacy challenges in IoT deployments.
- CO5: Design IoT-based solutions for application-specific environments.
- CO6: Evaluate emerging trends and future directions of IoT technologies.

Syllabus

Foundations of the Internet of Things: Evolution of Internet of Things, Characteristics of IoT systems, IoT ecosystem and stakeholders, Enabling technologies for IoT, Machine-to-Machine (M2M) communication, IoT World Forum (IoTWF) Reference Architecture, Simplified IoT Architecture, Core IoT Functional Stack, Functional blocks of an IoT ecosystem, Smart objects and connected environments, Sensors and actuators, Introduction to Fog, Edge, and Cloud paradigms in IoT.

IoT Connectivity and Communication Protocols: Communication requirements in IoT systems, Constrained devices and constrained networks, IEEE 802.15.4 overview, IEEE 802.11ah overview, LoRaWAN architecture and applications, IPv6 for IoT, 6LoWPAN concepts and architecture, Application layer protocols: MQTT and CoAP, SCADA systems in industrial environments, Protocol selection considerations for IoT deployments.

IoT System Design and Development: IoT design methodology, IoT system building blocks, Device integration and deployment workflow, Fundamentals of embedded computing for IoT, Overview of microcontrollers and System-on-Chip (SoC) platforms, IoT hardware ecosystem, Raspberry Pi platform overview, Arduino platform overview, Design considerations for scalable IoT systems.

IoT Data Management, Security and Supporting Services: IoT data lifecycle, Structured and unstructured data, Data in motion and data at rest, Data acquisition and organization in IoT environments, IoT data analytics challenges, Data visualization concepts, Middleware and supporting services, Cloud-enabled IoT applications, Security and privacy challenges in IoT environments, Identity and trust management, Sustainable IoT deployment considerations, Responsible and ethical IoT systems.

IoT Applications and Emerging Trends: Smart homes and smart appliances, Smart buildings and infrastructure, Smart healthcare systems, Smart agriculture, Environmental monitoring, Industrial IoT (IIoT), Industry 4.0 and 5.0 concepts, Smart energy systems, Digital twins, Human-centered IoT applications, Sustainable IoT systems, Future trends and research challenges in IoT.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2		1	2							1	2	
CO2	2	3	1	2	2							1	2	1
CO3	2	3	2	2	3			1				1	3	2
CO4	2	3	2	2	2	2	1	2			1	2	2	1
CO5	2	3	3	2	3	1	1	2	1	1	1	2	3	2
CO6	2	3	3	2	3	1	2	2	2	1	1	2	3	2

Textbooks / References

1. Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Introduction to IoT, Cambridge University Press, 2021.
2. Arshdeep Bahga and Vijay Madisetti, Internet of Things: A Hands-on Approach, Universities Press, 2014.
3. Olivier Hersent, David Boswarthick, and Omar Elloumi, The Internet of Things: Key Applications and Protocols, Wiley, 2011.
4. Adrian McEwen and Hakim Cassimally, Designing the Internet of Things, Wiley, 2013.
5. Pethuru Raj and Anupama Raman, The Internet of Things: Enabling Technologies, Platforms and Use Cases, CRC Press, 2017.
6. Honbo Zhou, The Internet of Things in the Cloud: A Middleware Perspective, CRC Press, 2013.

CSO007 Project Management and Agile Practices [3-0-0-3]

Course Objectives

- To understand the fundamentals of project management and project life cycles.
- To apply project planning, scheduling, budgeting, and risk management techniques.
- To analyze project performance using standard project management tools and metrics.
- To understand Agile values, principles, and frameworks used in modern software development.
- To implement Scrum, Kanban, and Lean practices for iterative project delivery.
- To develop project management and teamwork skills for industry-oriented projects.

Course Outcomes

Upon successful completion of the course, the students will be able to:

- CO1: Explain project management concepts, project life cycles, and organizational structures.

- CO2: Apply project planning, scheduling, estimation, and resource management techniques.
- CO3: Analyze project risks, quality requirements, and project performance metrics..
- CO4: Implement Agile methodologies and frameworks such as Scrum and Kanban in project execution.
- CO5: Evaluate Agile project outcomes using appropriate metrics and continuous improvement practices.
- CO6: Manage software and technology projects collaboratively using Agile tools and project management practices.

Syllabus

Fundamentals of Project Management: Introduction to project management- Characteristics of projects-Project management framework-Project life cycle-Project stakeholders and governance-Organizational structures- Functional-Matrix-Projectized-Project Management Knowledge Areas-Role of project managers-Ethics and professional responsibility.

Project Planning and Scheduling: Project initiation and feasibility analysis-Project scope management-Work Breakdown Structure (WBS)-Effort and cost estimation-Project scheduling techniques-Gantt Charts-Network Diagrams-Critical Path Method (CPM)-Program Evaluation and Review Technique (PERT)-Resource allocation and levelling-Project documentation and planning tools.

Risk, Quality, and Performance Management: Project risk management process-Risk identification and assessment-Risk response planning-Project quality management-Quality assurance and quality control-Earned Value Management (EVM)-Performance monitoring and reporting-Change management-Project closure and lessons learned.

Introduction to Agile Development: Limitations of traditional project management-Agile Manifesto and principles-Agile project life cycle-Iterative and incremental development-Agile roles and responsibilities-Agile planning and estimation-User stories and acceptance criteria-Agile metrics and tracking.

Agile Frameworks and Practices: Scrum framework-Scrum roles-Scrum events-Scrum artifacts-Sprint planning and reviews-Product backlog management-Kanban methodology-Lean software development-Extreme Programming (XP)-Agile testing practices-Continuous feedback and improvement.

Agile Project Execution and Industry Practices: Agile project management tools-DevOps and Agile integration-Continuous Integration and Continuous Delivery (CI/CD)-Agile scaling frameworks-Agile project governance-Case studies in software and IT project management-Emerging trends in project management.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	1			1		2	1	1	2	1	1	
CO2	2	3	2	1	2	1		2	2	1	3	1	2	1
CO3	2	3	2	2	2	2		2	1	1	3	1	2	1
CO4	2	2	3	1	2	1		2	3	2	3	2	2	2
CO5	2	3	3	2	2	1	1	2	2	2	3	2	2	2
CO6	2	2	3	1	3	2	1	2	3	3	3	2	3	2

Textbooks / References

1. J. A. Schwalbe, Information Technology Project Management, 9th ed. Boston, MA, USA: Cengage Learning, 2019.
2. K. Schwaber and J. Sutherland, The Scrum Guide: The Definitive Guide to Scrum-The Rules of the Game, 2020.
3. R. C. Martin, Agile Software Development: Principles, Patterns, and Practices. Upper Saddle River, NJ, USA: Prentice Hall, 2002.
4. H. Kerzner, Project Management: A Systems Approach to Planning, Scheduling, and Controlling, 13th ed. Hoboken, NJ, USA: Wiley, 2022.
5. M. Cohn, Agile Estimating and Planning. Upper Saddle River, NJ, USA: Prentice Hall, 2005.
6. K. Beck, Extreme Programming Explained: Embrace Change, 2nd ed. Boston, MA, USA: Addison-Wesley, 2004.
7. E. Ries, The Lean Startup. New York, NY, USA: Crown Business, 2011.

CSO008 Introduction to Indian Knowledge Systems [3-0-0-3]

Course Objectives

- To introduce and explain the foundational concepts, classifications, and historical development of Indian Knowledge Systems.
- To examine major Indian traditions in philosophy, mathematics, science, astronomy, medicine, and governance.
- To analyze key concepts related to Indian epistemology, ethics, sustainability, and holistic approaches to knowledge.
- To evaluate the relevance and application of Indian Knowledge Systems in contemporary interdisciplinary and sustainable practices.

Course Outcomes

- CO1: Explain the historical foundations, classification, and scope of Indian Knowledge Systems.
- CO2: Describe and examine important contributions of Indian traditions in philosophy, mathematics, science, astronomy, and medicine.
- CO3: Analyze key concepts related to Indian epistemology, ethics, governance, and holistic knowledge traditions.

- CO4: Evaluate the contemporary relevance of Indian Knowledge Systems in interdisciplinary studies, sustainability, and global knowledge practices.

Syllabus

Foundations of Indian Knowledge Systems: Introduction to Indian Knowledge Systems, Nature and scope of traditional knowledge, Vedas, Vedangas, Upanishads, Itihasas, and Puranas, Overview of Indian philosophical systems (Darshanas), Oral traditions and knowledge transmission, Concepts of Dharma, Artha, Kama, and Moksha, Gurukula system.

Science and Knowledge Traditions in India: Indian mathematics and numeral systems, Contributions of Aryabhata and other scholars, Astronomy and Indian calendar systems, Traditional medicine and wellness systems, Indian approaches to logic and reasoning (Pramana), Sanskrit and scientific knowledge traditions, Indian contributions to metallurgy, architecture, and engineering.

Society, Culture, and Contemporary Relevance: Indian ethics and governance traditions, Environmental consciousness and sustainability in Indian Knowledge Systems (IKS), Yoga and holistic well-being, Indian arts, aesthetics, and performance traditions, Indian Knowledge Systems and modern interdisciplinary studies, Influence of IKS on global thought and culture, Contemporary applications and relevance of IKS.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1				2		2		1		1		
CO2	2	2				2		2		1		1	1	
CO3	2	3		1		2	2	3		1		2		
CO4	2	3	1	1	1	2	3	3		2		2	1	1

Textbooks / References

1. Shibayan Chattopadhyay, "Reviving Timeless Insights: A Deep Dive into the Indian Knowledge System and its Modern Significance," *International Journal of Law Management and Humanities*, Vol. 7, Issue 6, 2024.
2. Trina Das, "Relevance of Indian Knowledge System in the Present," in *Indian Knowledge Systems: Journey from Vedic Wisdom to Digital Repositories*, 2026.
3. Rimmi Datta and Jayanta Mete, "Exploring the Foundations and Modern Relevance of Indian Knowledge Systems."
4. Kapil Kapoor and Avadhesh K. Singh (Eds.), *Indian Knowledge Systems*, Vol. 1, Indian Institute of Advanced Study, New Delhi, 2005.
5. Salim Khan and Meeta Sharma, "An Overview on Indian Knowledge System," *Integrated Journal for Research in Arts and Humanities*, Vol. 4, No. 4, 2024, pp. 42–46.
6. Pallavi Kumari and Anjan Niyogi, "The Indian Knowledge System and Indigenous Pedagogies: A Historical and Contemporary Review," *IUJ Journal of Management*, Vol. 13, No. 1, 2025, pp. 9–27.
7. B. Mahadevan, Vinayak Rajat Bhat, and Pavana R. N. Nagendra, *Introduction to Indian Knowledge System: Concepts and Applications*, 2022.
8. Uddipta Nayan Medhi, "Epistemic Foundation of Indian Knowledge System: A Study in Primary Narratives and Secondary Exegesis," *International Journal of Social Science Research*, Vol. 2, No. 4, 2025, pp. 339–349.
9. Tanmay Mehta and Nrapendra Vir Singh, "An Exploration of the Indian Knowledge System: Roots, Significance, and Contemporary Relevance," in *Bhartiya Knowledge Systems*, 2024, pp. 33–39.
10. B. Mohanapriya and M. Suriya, *Indian Knowledge Systems: Principles and Practices*, SSS Publications, 2025.

CSO009 Fundamentals of Sanskrit [3-0-0-3]

Course Objectives

- To introduce and explain the structure, pronunciation system, and linguistic features of Sanskrit language.
- To develop foundational skills in Sanskrit reading, writing, transliteration, and simple communication.
- To examine and apply basic Sanskrit grammatical concepts including Sandhi, Samasa, nouns, and verb forms in sentence construction.
- To interpret and appreciate selected Sanskrit literary and philosophical traditions and their contribution to Indian Knowledge Systems.

Course Outcomes

- CO1: Explain the basic linguistic structure and phonetic system of Sanskrit language.
- CO2: Demonstrate elementary skills in Sanskrit pronunciation, transliteration, reading, writing, and communication.
- CO3: Apply fundamental Sanskrit grammatical concepts such as Sandhi, Samasa, and sentence formation in simple language exercises.
- CO4: Analyze and interpret selected Sanskrit literary and philosophical texts in the context of Indian intellectual traditions.

Syllabus

Introduction to Sanskrit Language and Tradition: Historical overview of Sanskrit, Sanskrit as a language of knowledge and culture, Introduction to sounds (Dhvani) and alphabets (Varna), Transliteration using IAST system, Basic vocabulary and pronunciation, Difference between Bhasa, Vak, and Vani, Oral traditions and preservation of knowledge, Sanskrit and Indian intellectual traditions.

Fundamentals of Sanskrit Grammar and Communication: Introduction to Sanskrit grammar traditions, Basic sentence structure, Introduction to nouns, gender, and number, Basic verb forms and sentence making, Introduction to Sandhi and Samasa, Conversational Sanskrit expressions, Reading and writing simple Sanskrit sentences, Introduction to Panini and grammatical systems.

Sanskrit Literature and Contemporary Relevance: Overview of Sanskrit literary traditions, Introduction to Vedas, Upanishads, and classical literature, Selected Subhashitas and simple verses, Sanskrit and Indian philosophy, Sanskrit in science and Natural Language Processing (NLP), Introduction to epics and poetic traditions, Reading and interpretation of selected passages, Relevance of Sanskrit in contemporary interdisciplinary studies, Introduction to the software BARAHA.

Course Outcome – PO/PSO Mapping

CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	2	1				2		2		1		1		
CO2	2	2				2		2		1		1	1	
CO3	2	3		1		2	2	3		1		2		
CO4	2	3	1	1	1	2	3	3		2		2	1	1

Textbooks / References

1. Anil K. Biltoo, *First Steps Towards Sanskrit: Language, Linguistics and Culture*, Routledge, 2021.
2. Sourav Chakraborty, *The Impact of Sanskrit on the World and Science: A Comprehensive Study of its Influence on Global Languages and Science*, Chyren Publication, 2026.
3. Robert P. Goldman and Sally J. Sutherland Goldman, *Devavanipravesika: An Introduction to the Sanskrit Language*, 2011.
4. Jan Houben, *Ideology and Status of Sanskrit: Contributions to the History of the Sanskrit Language*, Vol. 13, Brill, 2023.
5. Gopa Kar and Niranjan Manna, "Role of the Sanskrit Language in the Indian Knowledge System and Its Integration into Modern Education: A Critical Study."
6. Laxman Majhi, "Exploring the Significance of Sanskrit Literature in Shaping of the Indian Knowledge System," *Partners Universal International Innovation Journal*, Vol. 2, No. 3, 2024, pp. 103–114.
7. Antonia Ruppel, *The Cambridge Introduction to Sanskrit*, Cambridge University Press, 2017.
8. Richard Salomon, *The Sanskrit Language: An Introductory Grammar and Reader*, 2000.
9. Samiran Samanta, "Indian Knowledge System in NEP-2020: A New Era of Sanskrit Study," in *Indian Knowledge System (IKS)*, 2025.
10. Sukanta Sarkar, "Implication of the Sanskrit Language in the 21st Century: An Exploratory Study," *Journal of Global Economy*, Vol. 21, No. 3, 2025.
11. Malati J. Shendge, "The Interdisciplinary Approach in Indian Studies," *Annals of the Bhandarkar Oriental Research Institute*, Vol. 63, No. 1–4, 1982, pp. 63–98.